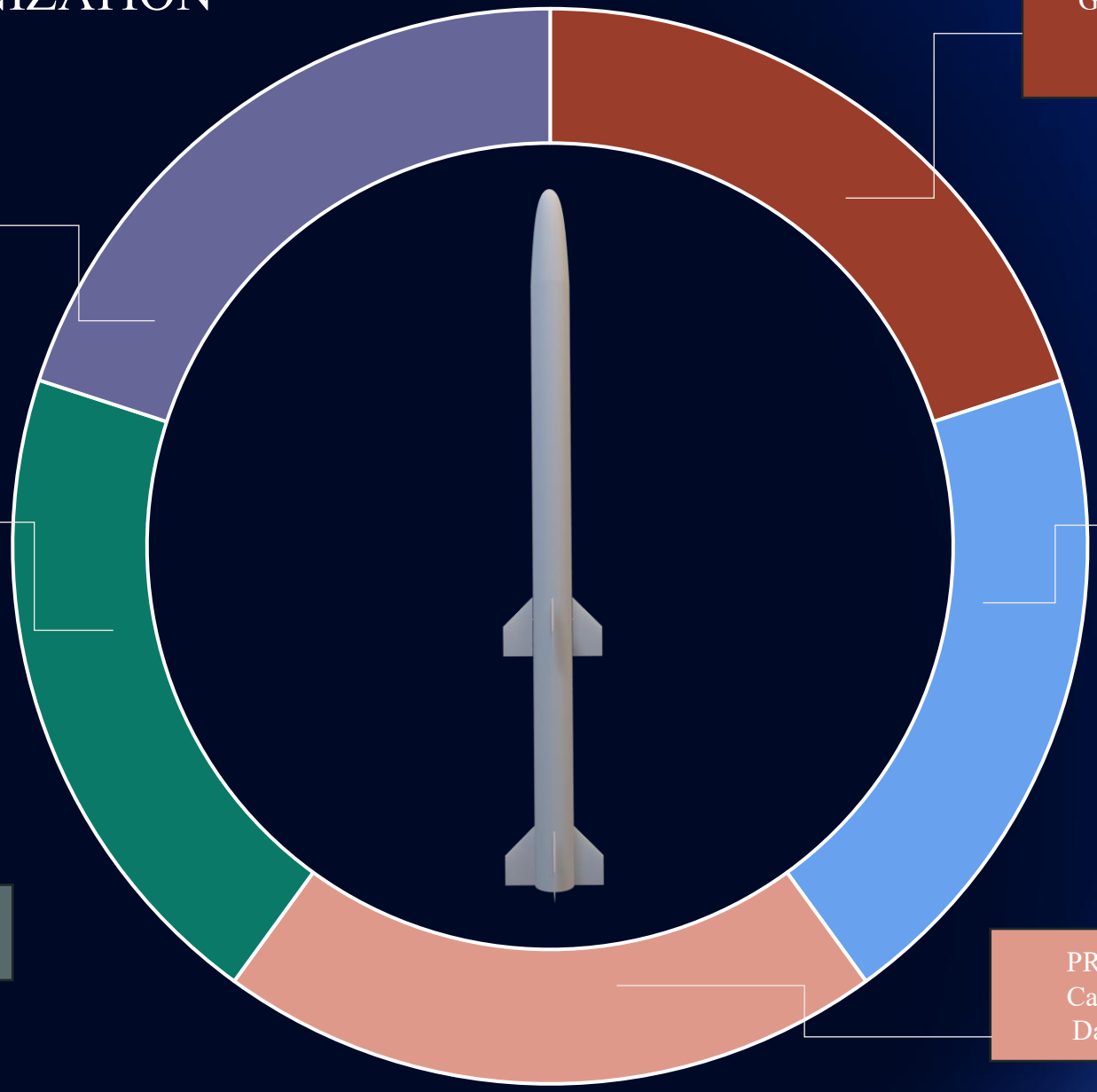
SAGITTARIUS

MBDA

MISSILE SYSTEMS

ROLES AND ORGANIZATION

ROLES AND ORGANIZATION



GUIDANCE AND CONTROL
Antonio Niro

AERODYNAMIC
Gabriele Foggiano

WARHEAD
Matteo Giannitti

STRUCTURE
Francesco Gasperini

SYSTEM ENGINEER
Vincenzo Junior Di Rosa

PROPULSION
Carmine Marra
Davide Sergio

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- Reference Scenario
- Simulation Trade-Off
- Guidance Law
- Warhead
- Missile Overview
- Second Stage
- First Stage
- Off Design Scenarios

CONCEPT OF OPERATION

Mission Characteristics

- Ground-based Air defence
- Radious to defend: 3000 m
- Maximum interception altitude: 5500 m

Targets characteristics

- Speed range: 30 – 60 $^{m}/_{s}$
- Weight range: 100 – 300 kg
- Maximum load factor: 2





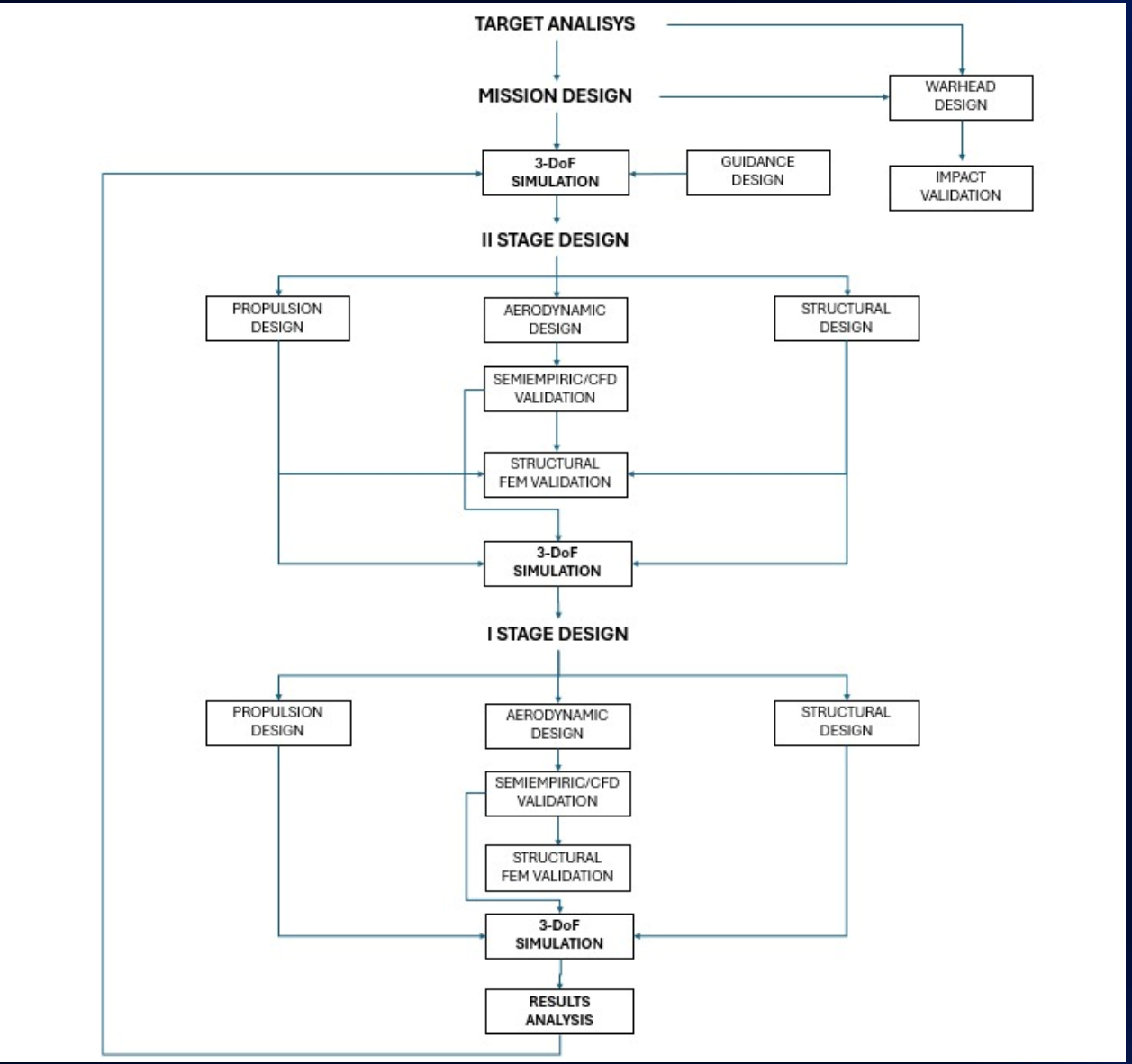
Interceptor characteristics

- Vehicle mounted
- Two stages
- Expected weight: 15 kg
- Expected lenght: 1.20 m
- Expected diameter: 0.08 m
- Cruise Velocity ≈ 200-275 $^{m}/_{s}$



FLOW CHART

FLOW CHART


TARGET ANALISYS
MISSION DESIGN
WARHEAD DESIGN
3-DoF SIMULATION
GUIDANCE DESIGN
IMPACT VALIDATION
II STAGE DESIGN
PROPULSION DESIGN
AERODYNAMIC DESIGN
STRUCTURAL DESIGN
SEMIEMPIRIC/CFD VALIDATION
STRUCTURAL FEM VALIDATION
3-DoF SIMULATION
I STAGE DESIGN
PROPULSION DESIGN
AERODYNAMIC DESIGN
STRUCTURAL DESIGN
SEMIEMPIRIC/CFD VALIDATION
STRUCTURAL FEM VALIDATION
3-DoF SIMULATION
RESULTS ANALYSIS

REFERENCE SCENARIO

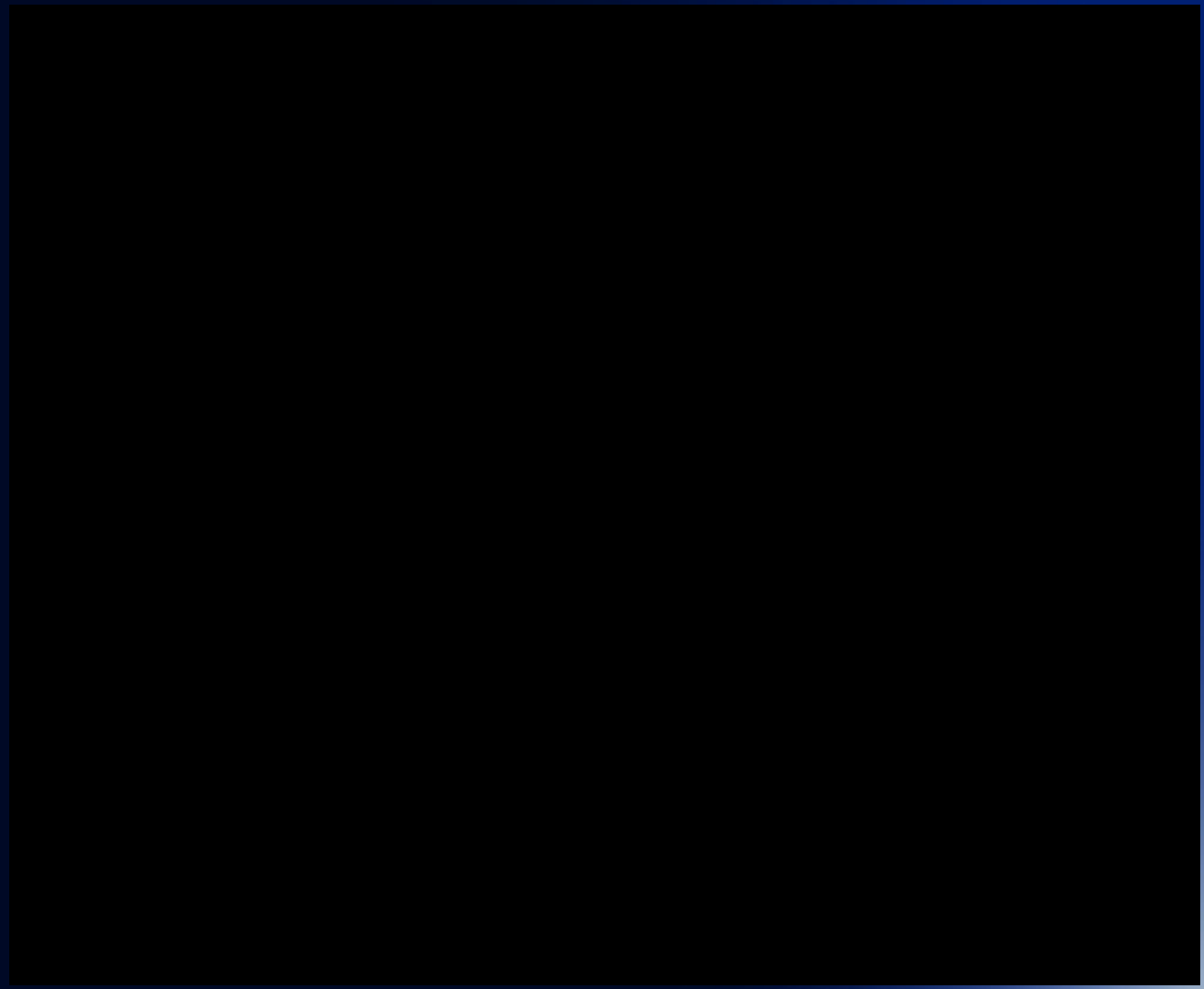
REFERENCE SCENARIO

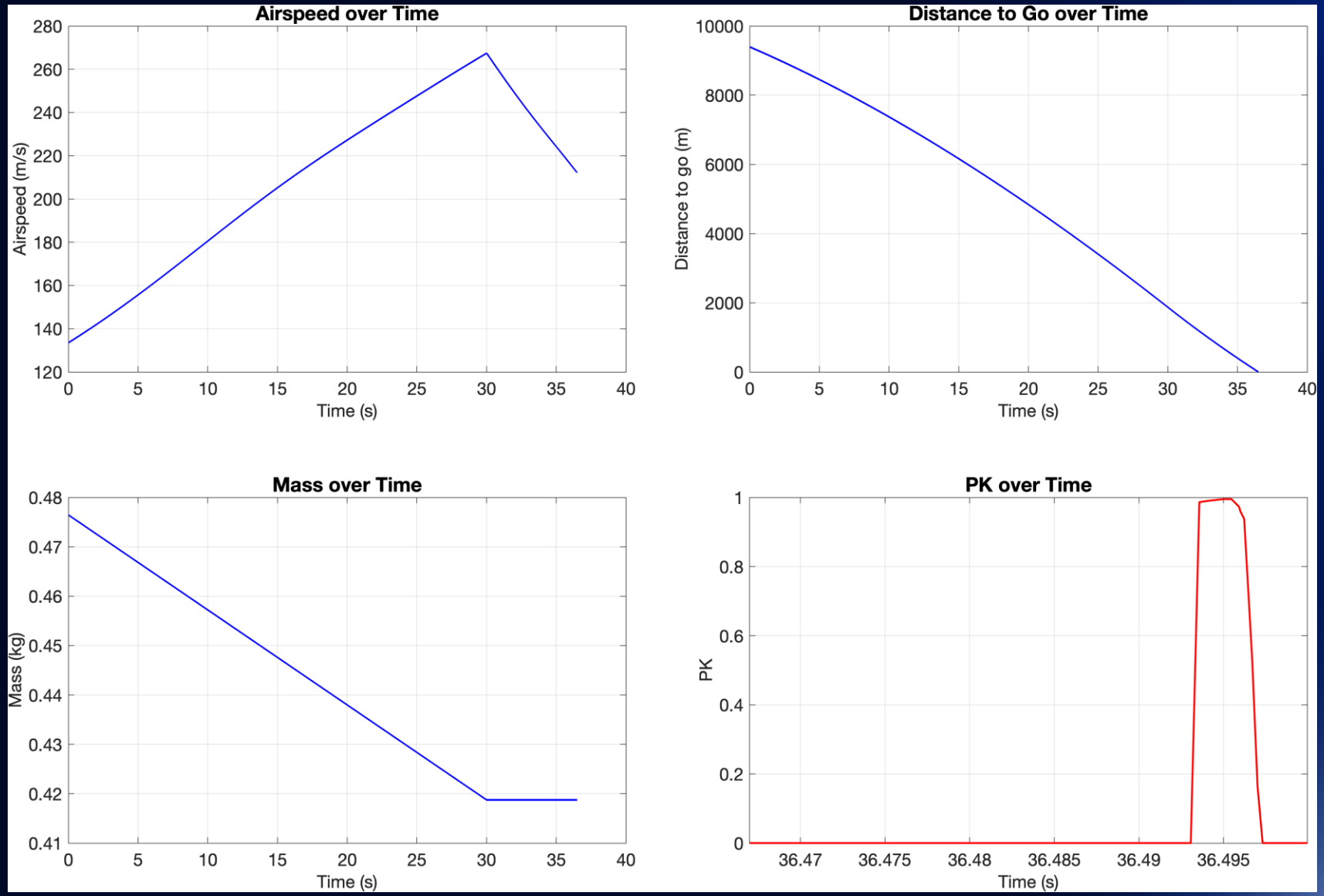
Target conditions:

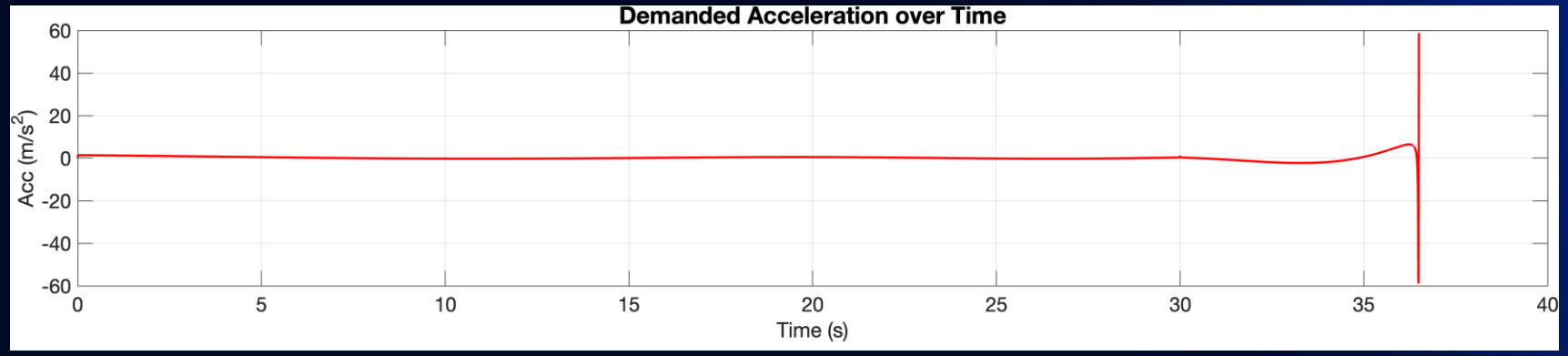
X(0) = 8000 m

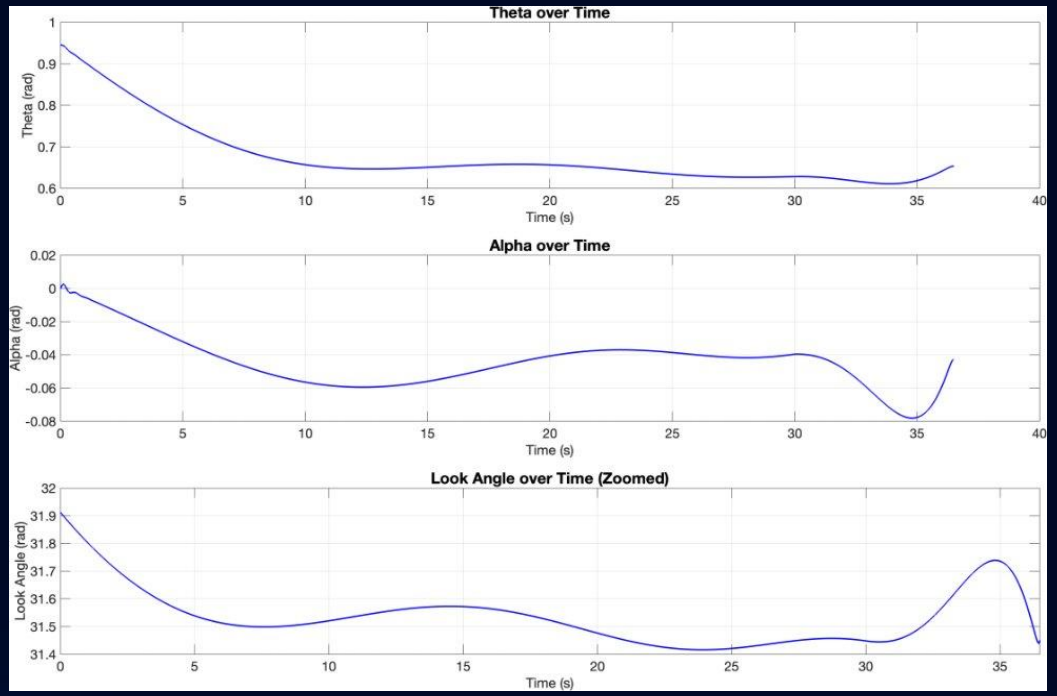
Y(0) = 5000 m

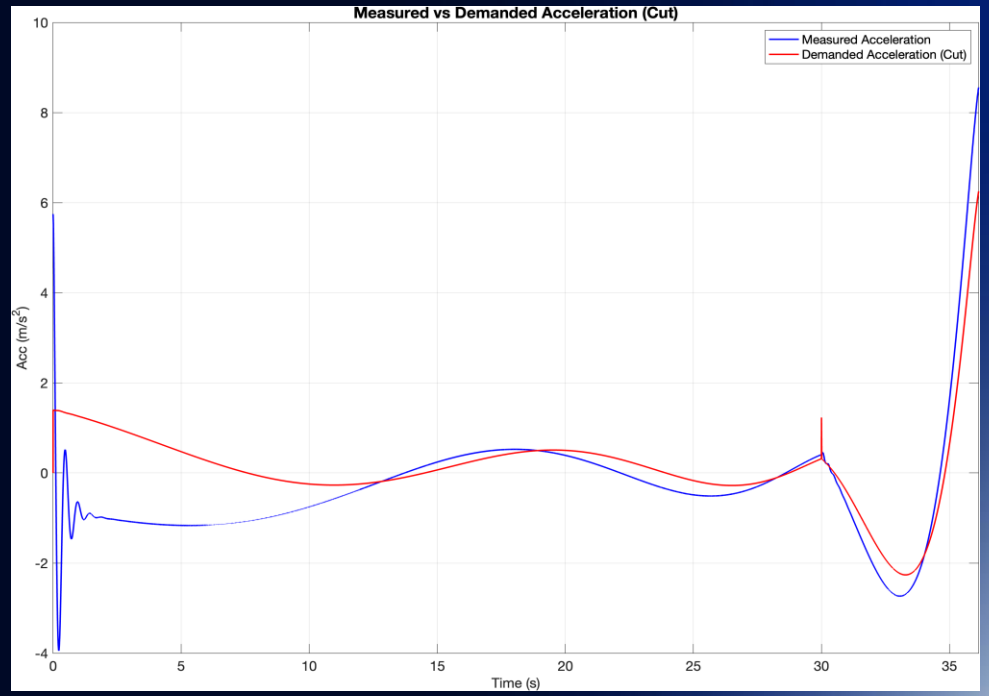
v = 60 $^{m}/_{s}$




Airspeed over Time
Airspeed (m/s)
Time (s)
Distance to Go over Time
Distance to go (m)
Time (s)
Mass over Time
Mass (kg)
Time (s)
PK over Time
PK
Time (s)

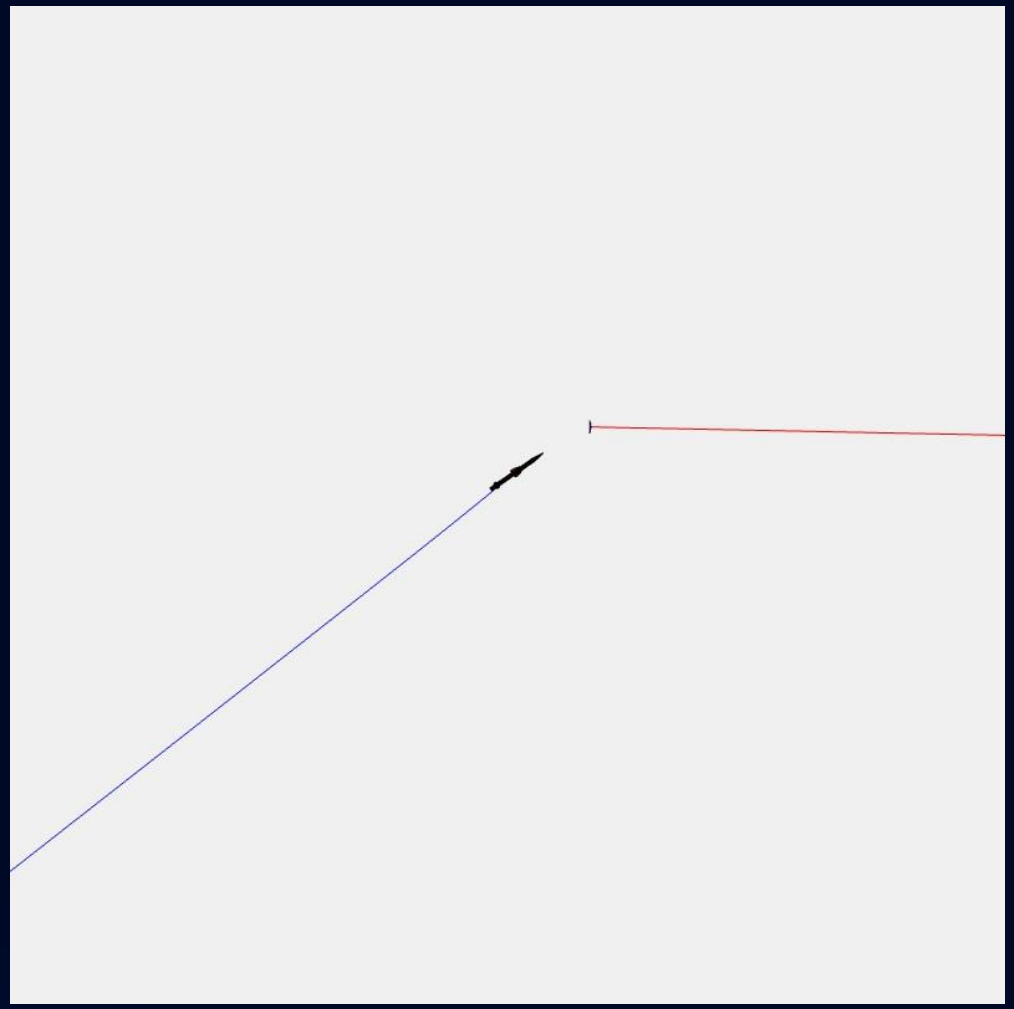

Demanded Acceleration over Time
Acc (m/s^2)
Time (s)


Theta over Time
Theta (rad)
Time (s)
Alpha over Time
Alpha (rad)
Time (s)
Look Angle over Time (Zoomed)
Look Angle (rad)
Time (s)


Measured vs Demanded Acceleration (Cut)
Measured Acceleration
Demanded Acceleration (Cut)
Acc (m/s^2)
Time (s)

SIMULATION AND TRADE - OFF

SIMULATION AND TRADE - OFF



The objective is to simulate a trajectory and find the right balance between thrust, aerodynamics, and guidance, so it was necessary to run different simulations and perform different iteration .

It is fully customizable and takes the input data from a Matlab file and gives in output the trajectory and all the useful dimensions for a scenario.

Input data

Second Stage:

- Masses:
  - With loaded propellant
  - Propellant mass
- Inertia I_{YY} for full and empty configuration
- Reference surface and diameter
- Center of gravity that changes with time
- Center of Pressure
- Actuators data
  - $w_n,\ \zeta,\ fin_{min},\ fin_{max}$
- Sensors position with respect to CG
- n_{max}
- Aerodynamics coefficients:
  - $C_D, C_{N,\alpha}, C_{X,fin}, C_{M,q}, C_{z,fin}$

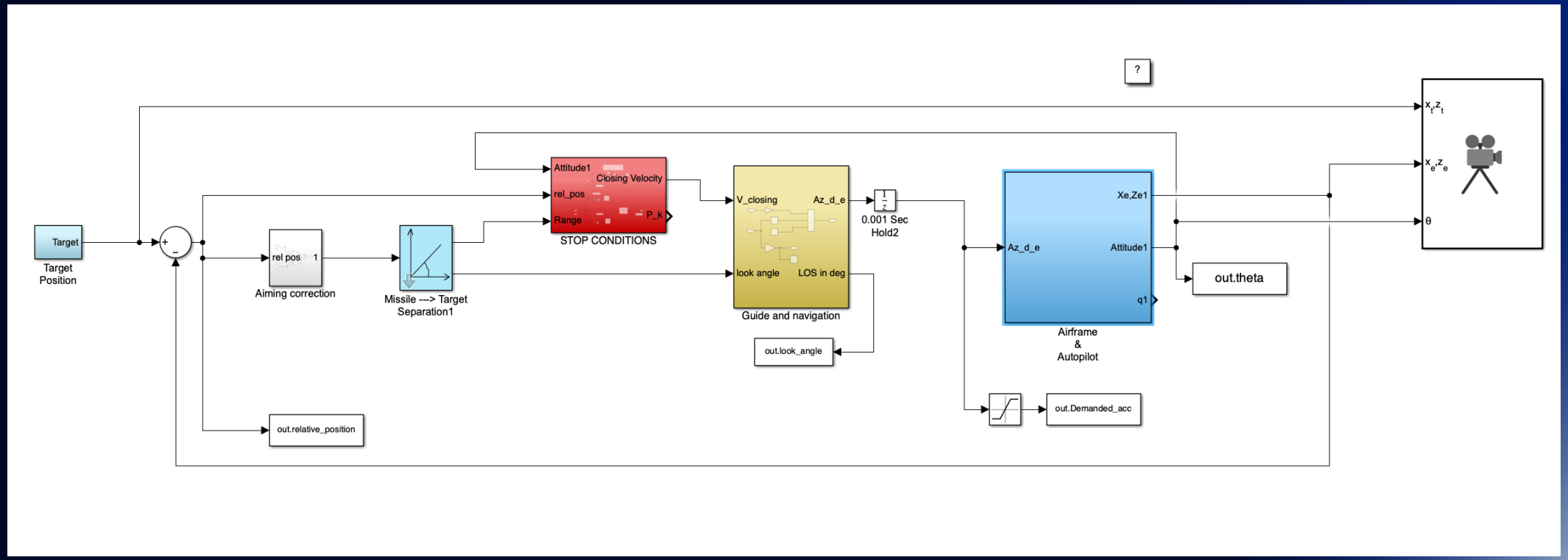
Target:

- Starting position on X and Z, correspond to where the defence systems have identified the target
- Speed $V_{tgt} = 60\ ^m/_s$
- Trajectory angle $\theta_{tgt} = 180\ deg$

Additionally:

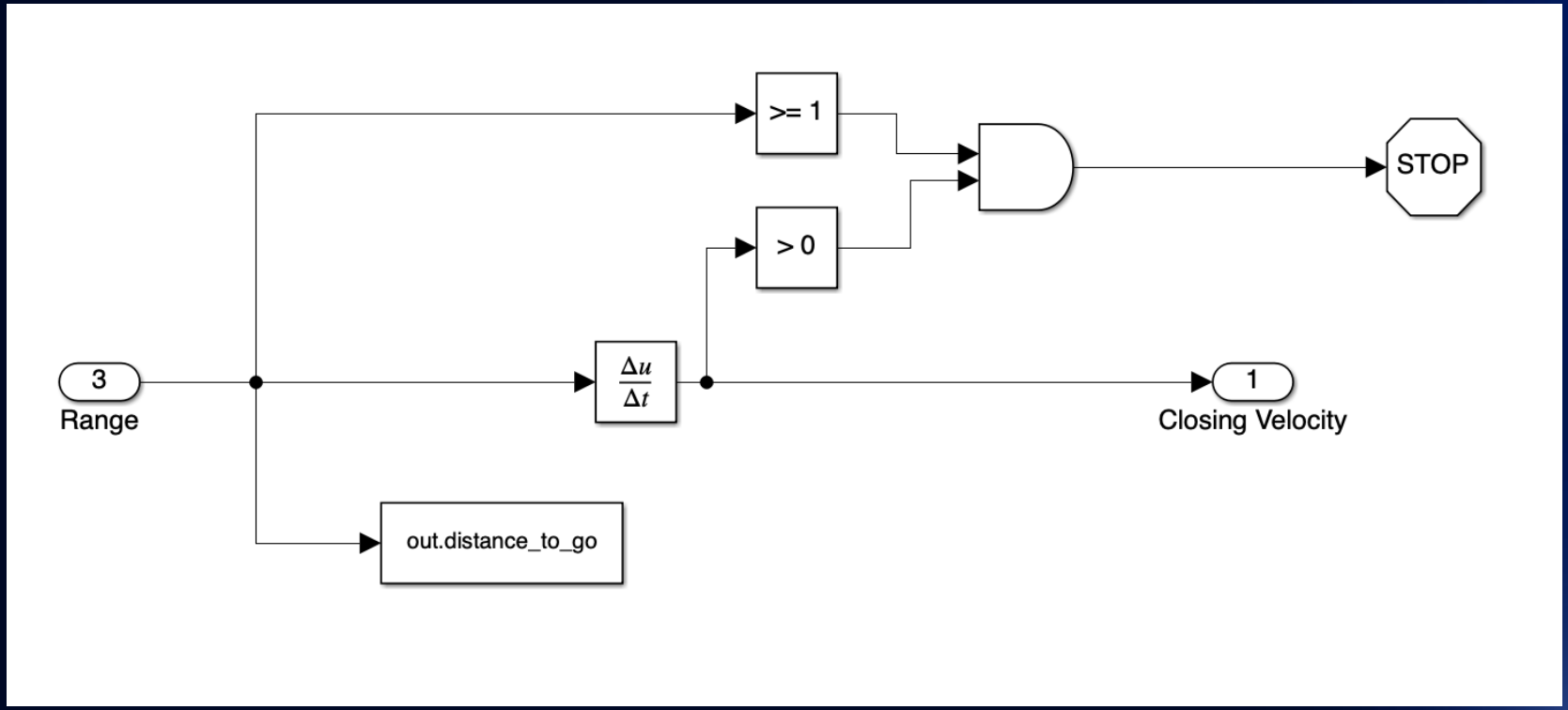
- Second stage starting position is given by the end of booster phase, calculated by a dedicated matlab code
- Equipped with an *auto-aiming* function

Simulink model overview

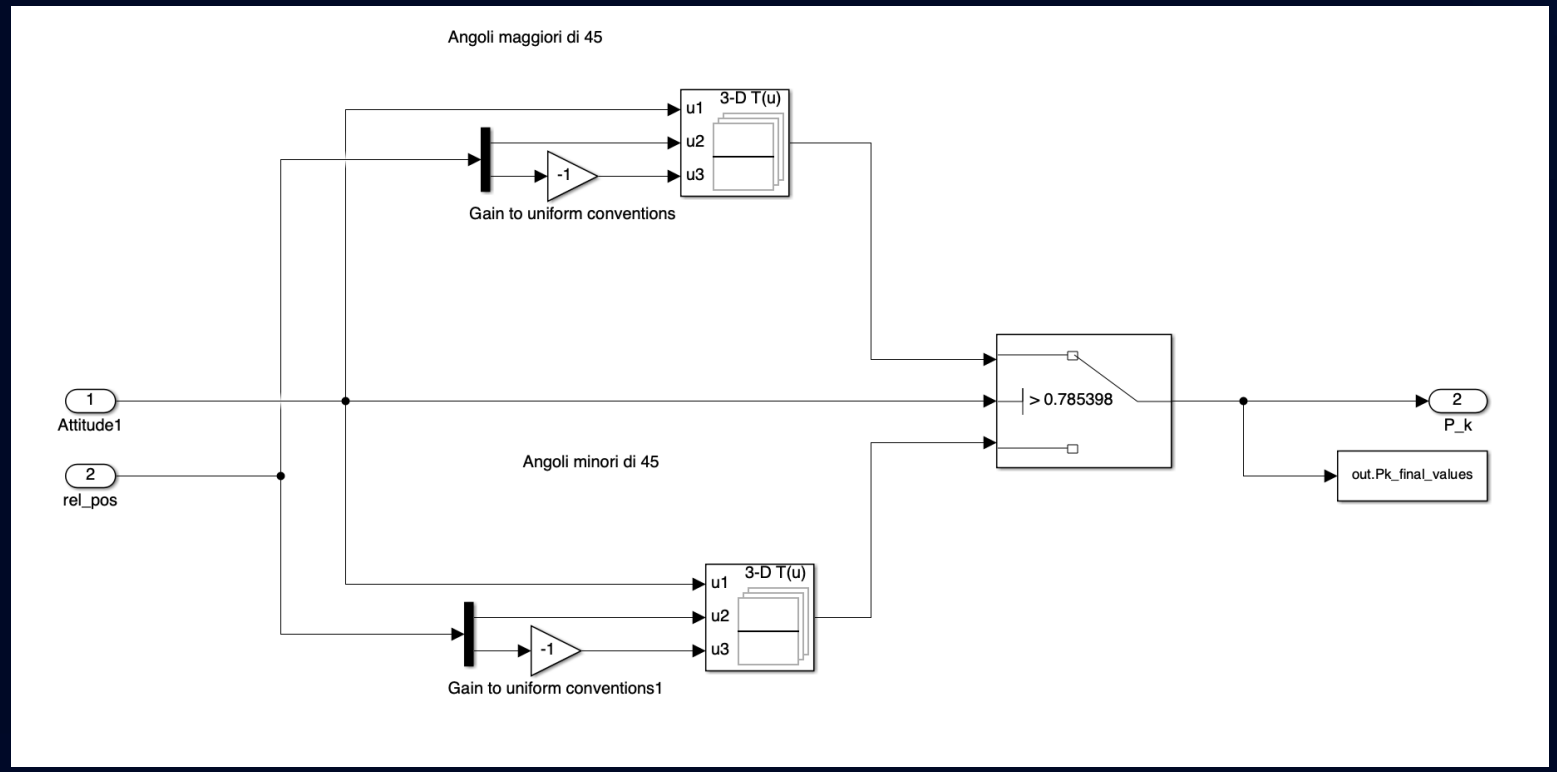


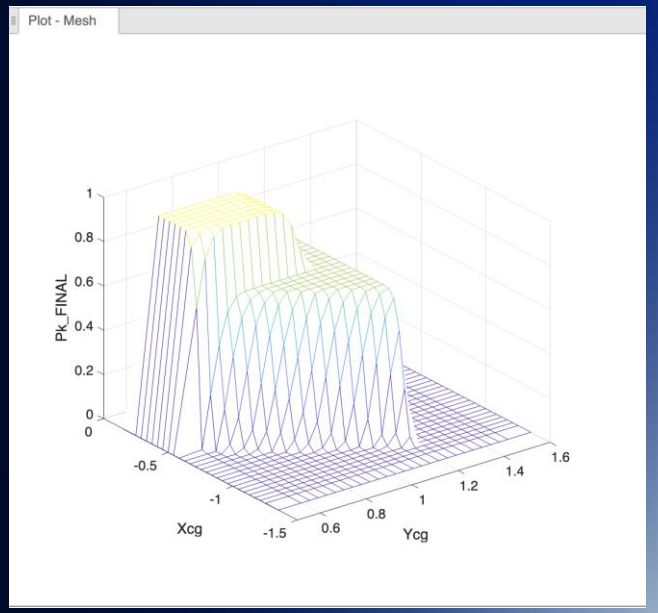
Stop conditions

The simulation stops when two conditions are met at the same time:

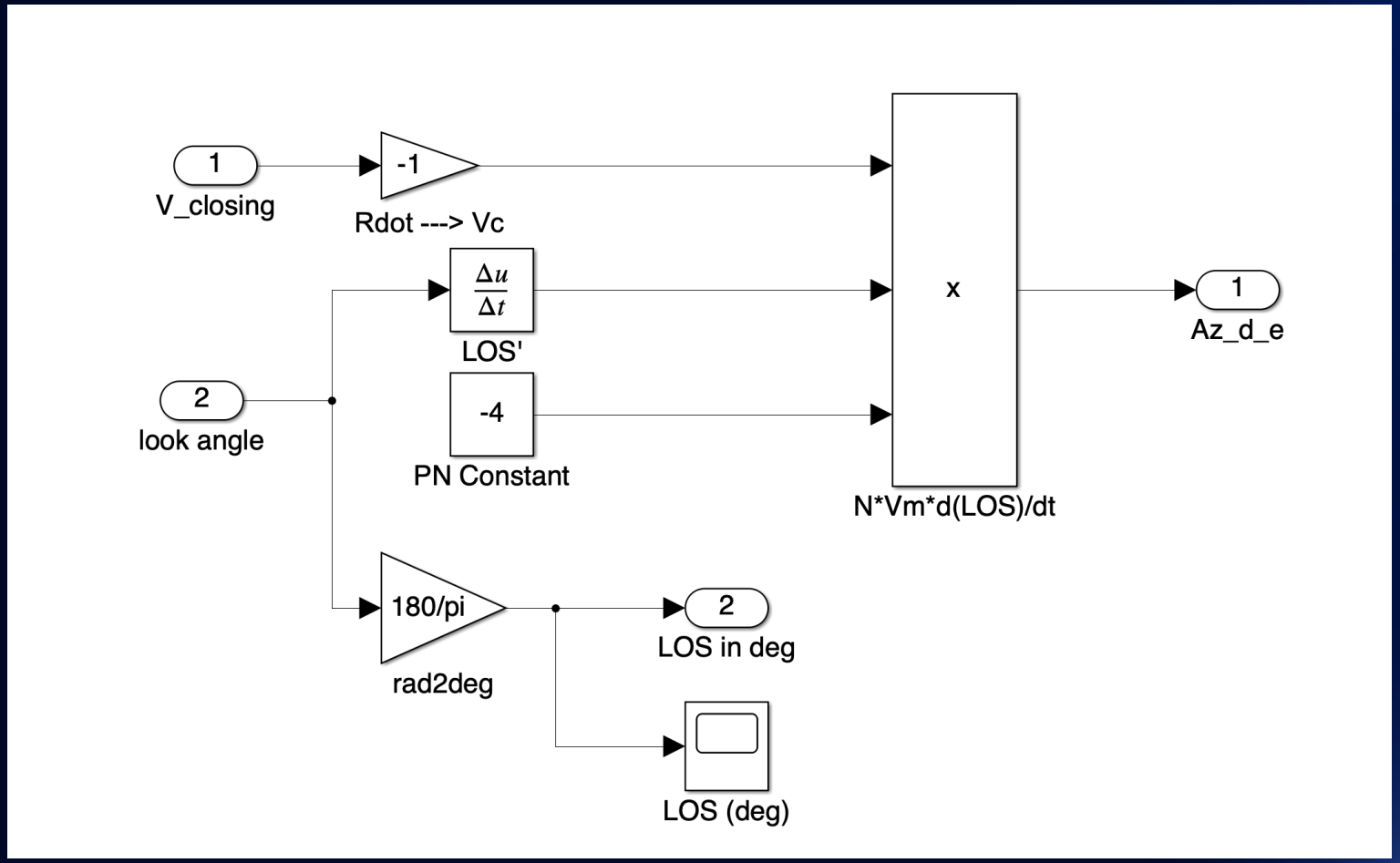


PK (*Probability to Kill*) Calculation

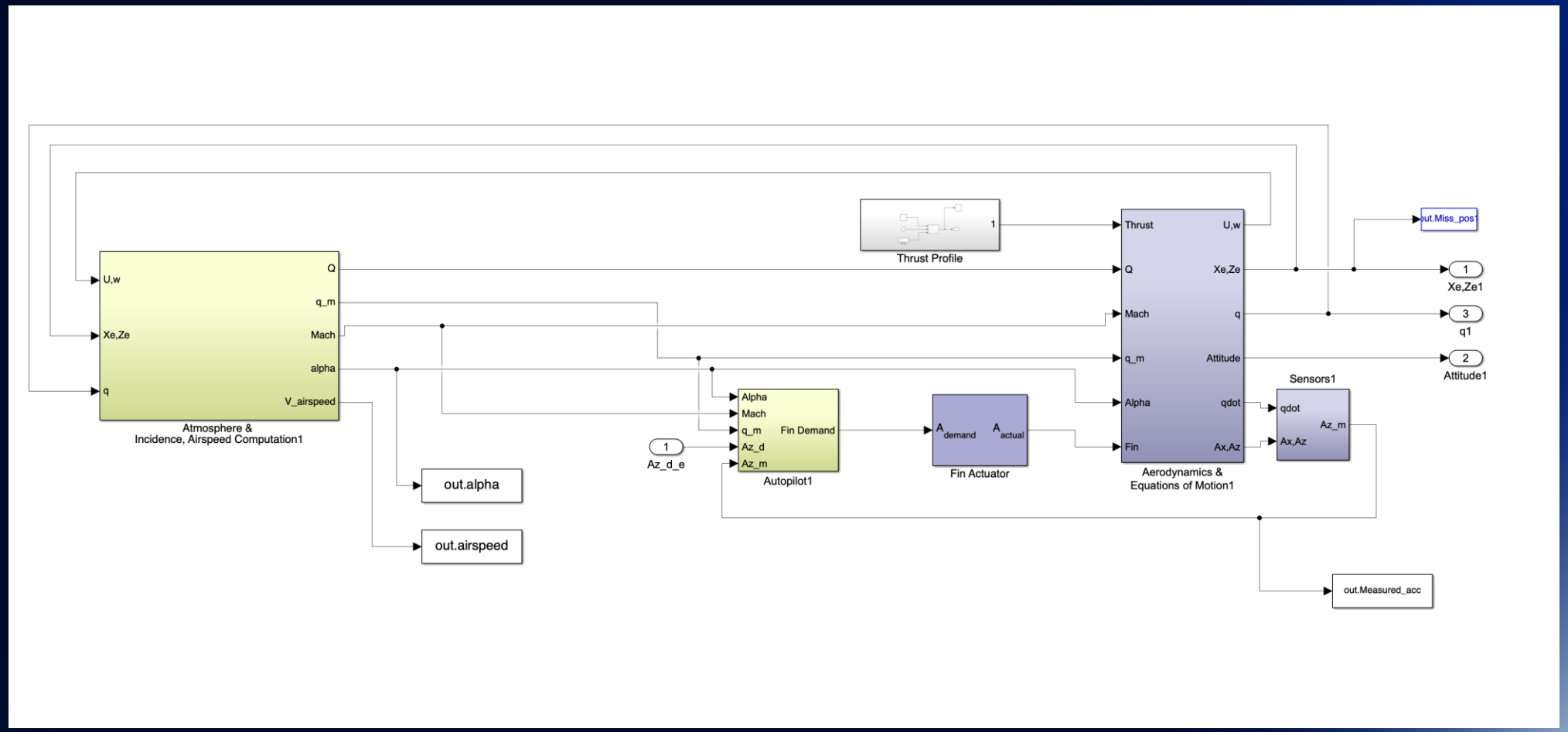




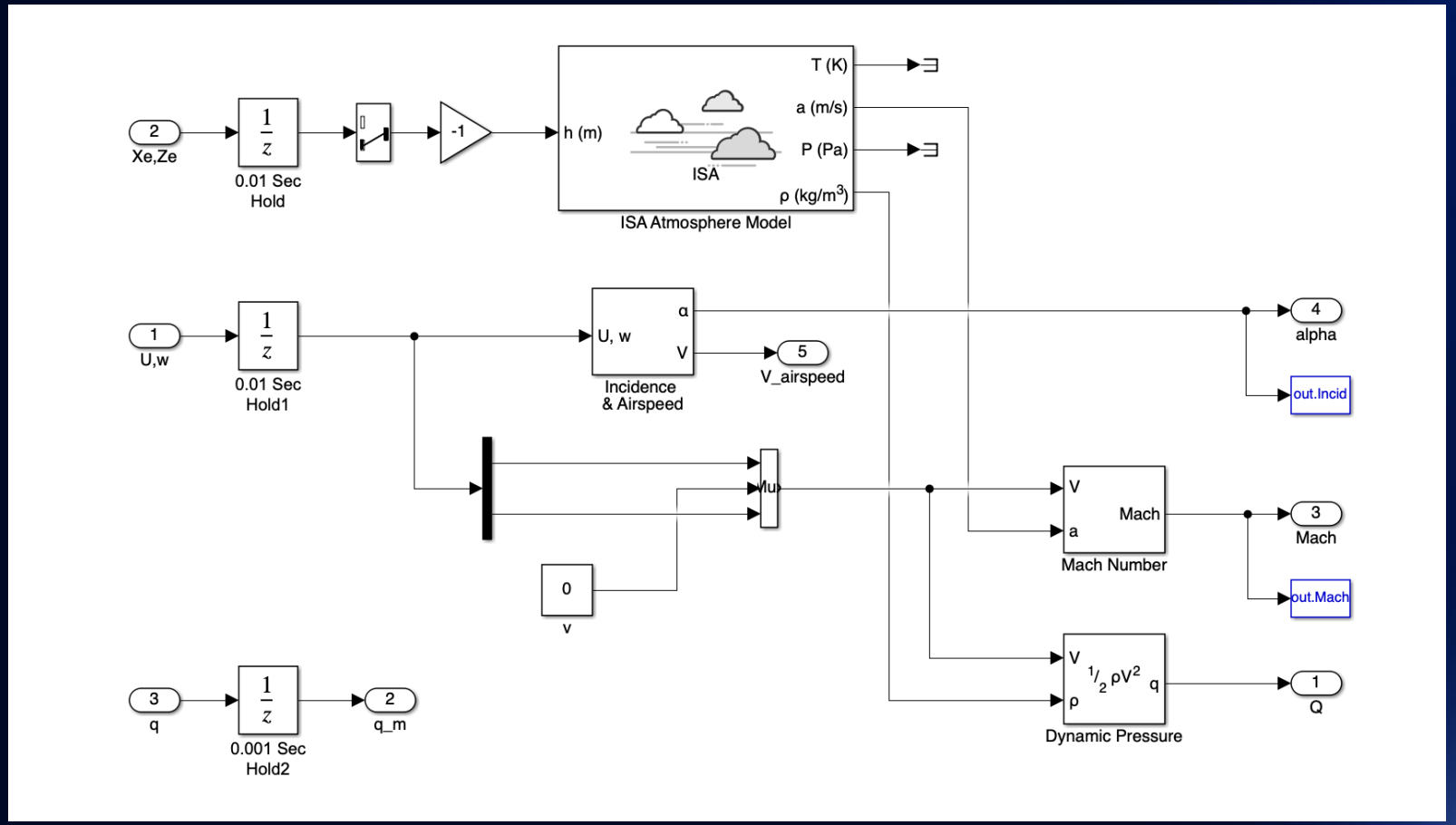
Guide and Navigation block



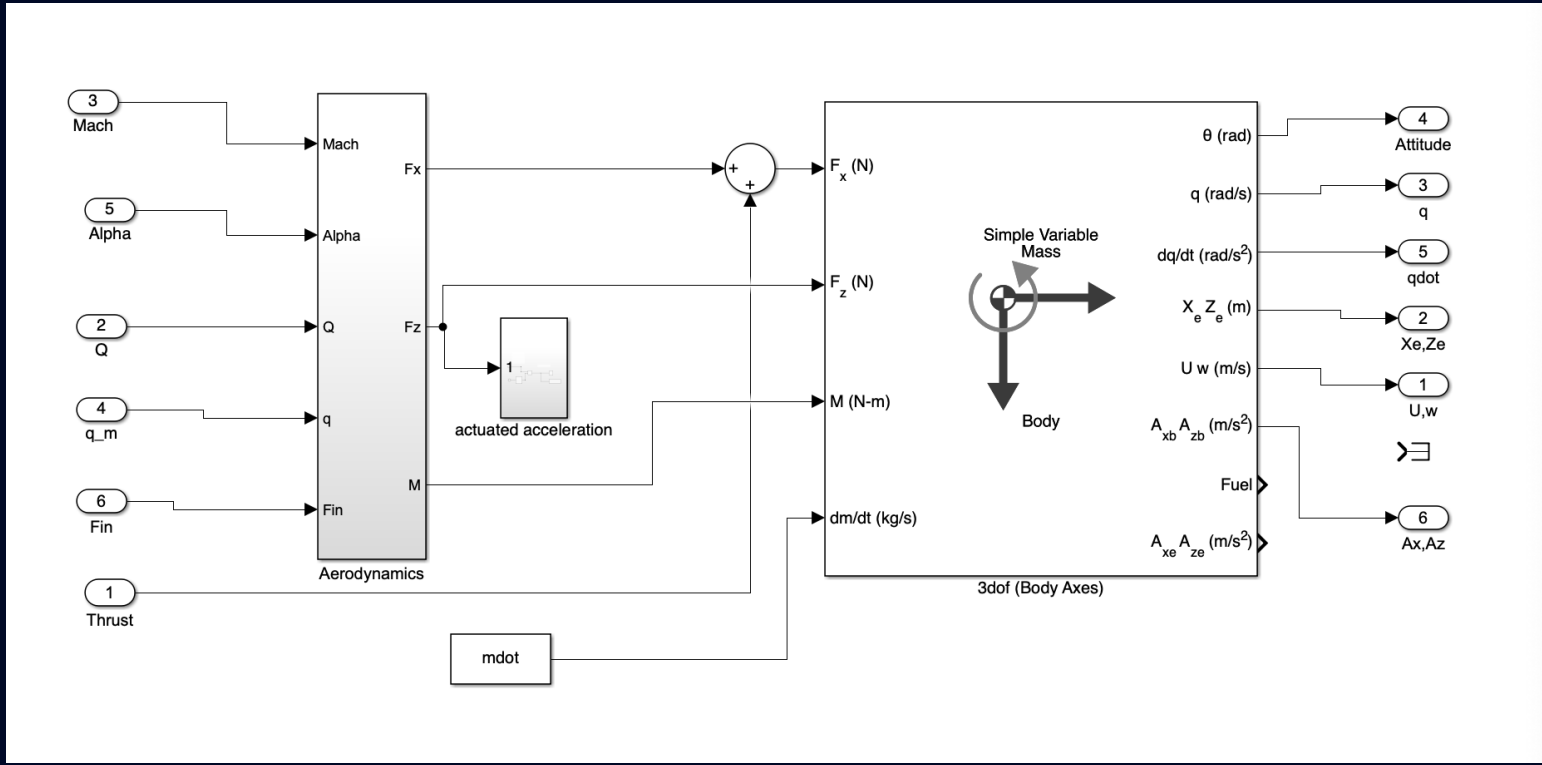
Airframe and Autopilot

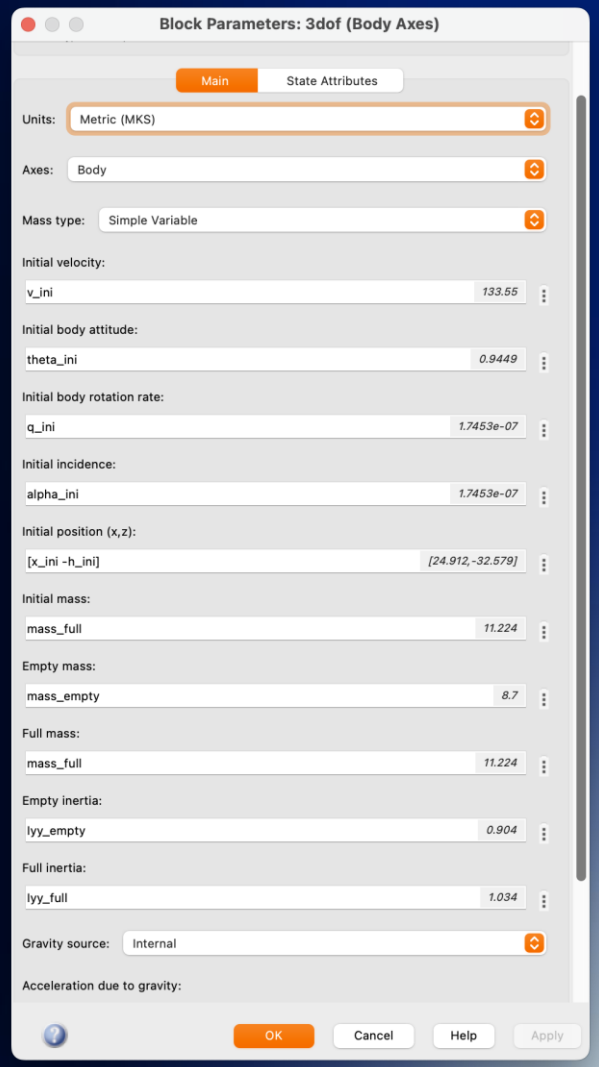


Athmosphere, speed and incidence block

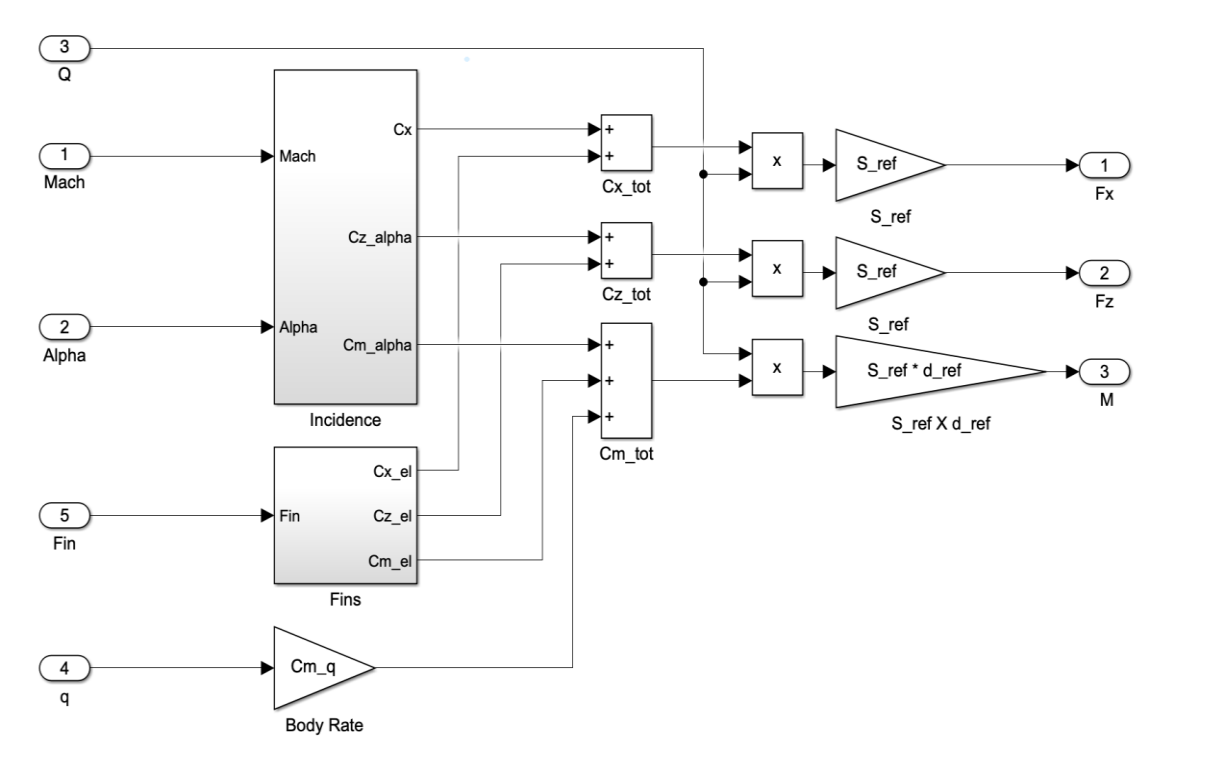


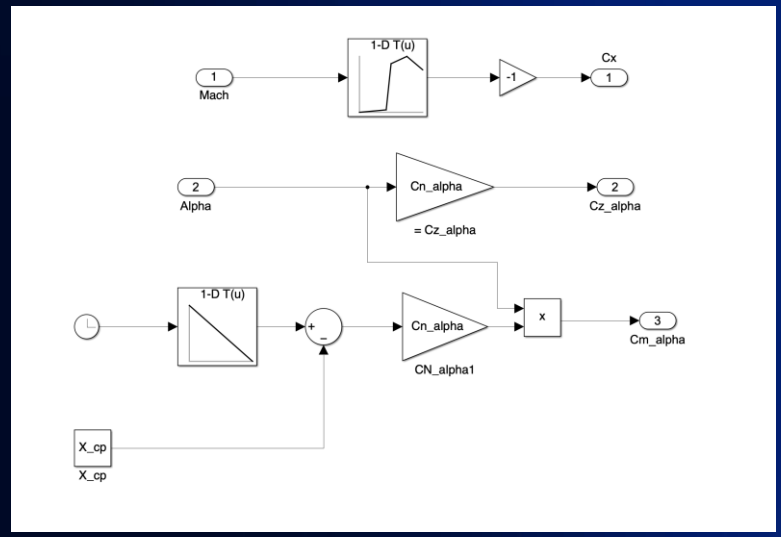
Aerodynamics and Equation of motion



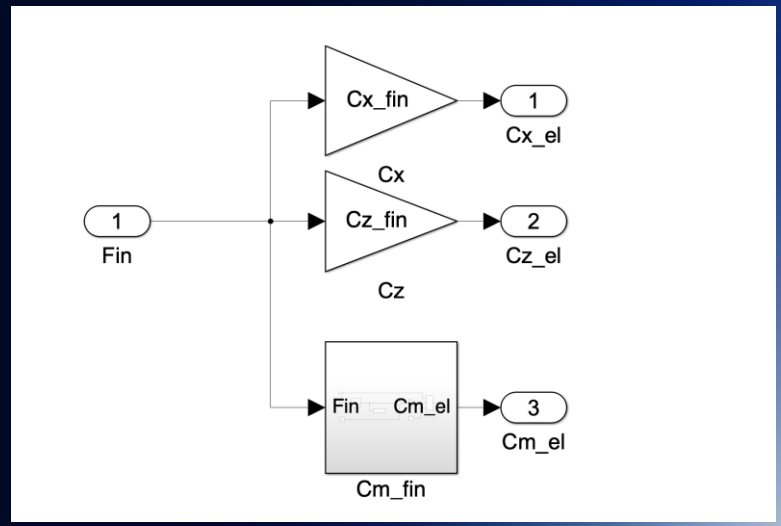


Aerodynamics Block





Incidence Block

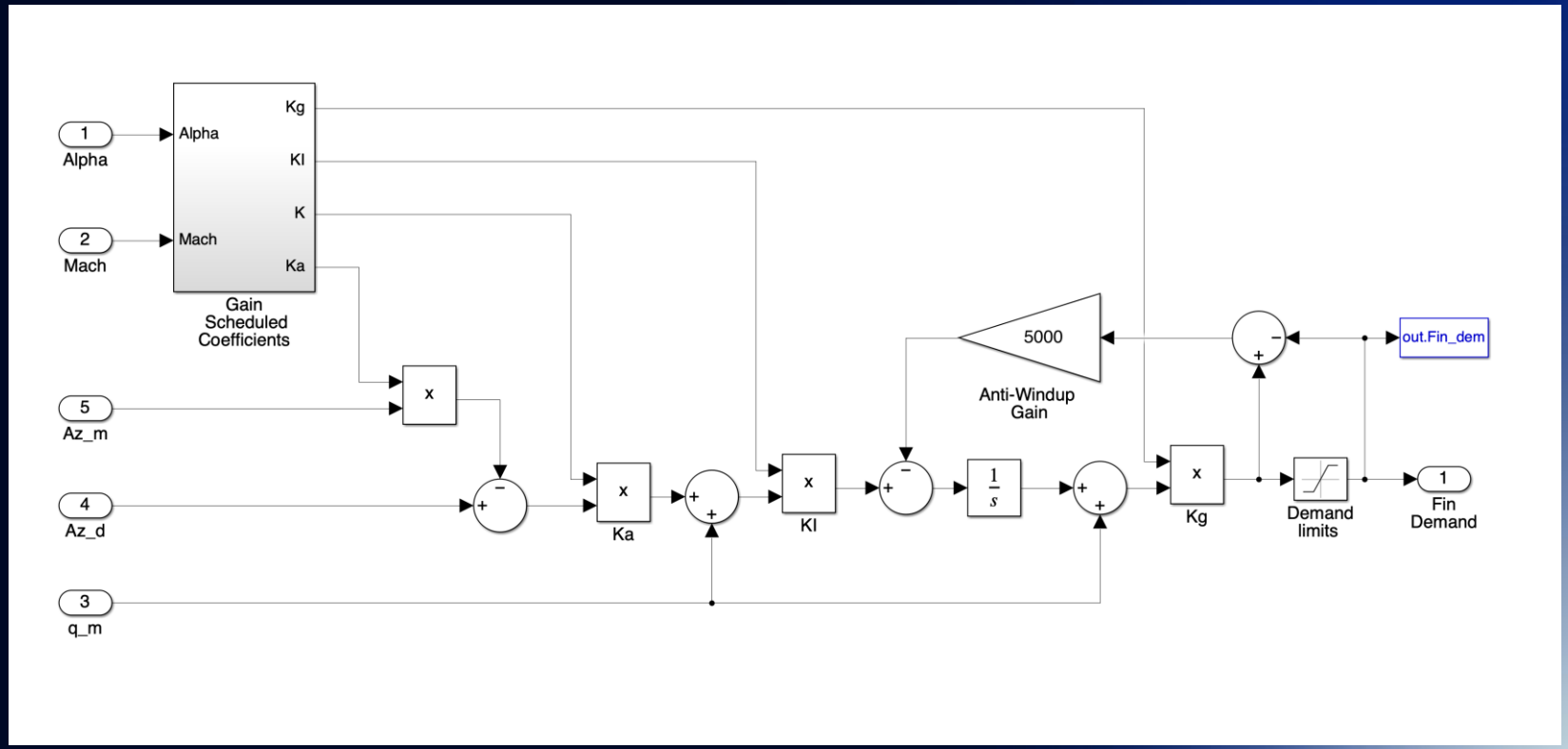


Fins Block

Non linear PID Autopilot Block

Inputs:

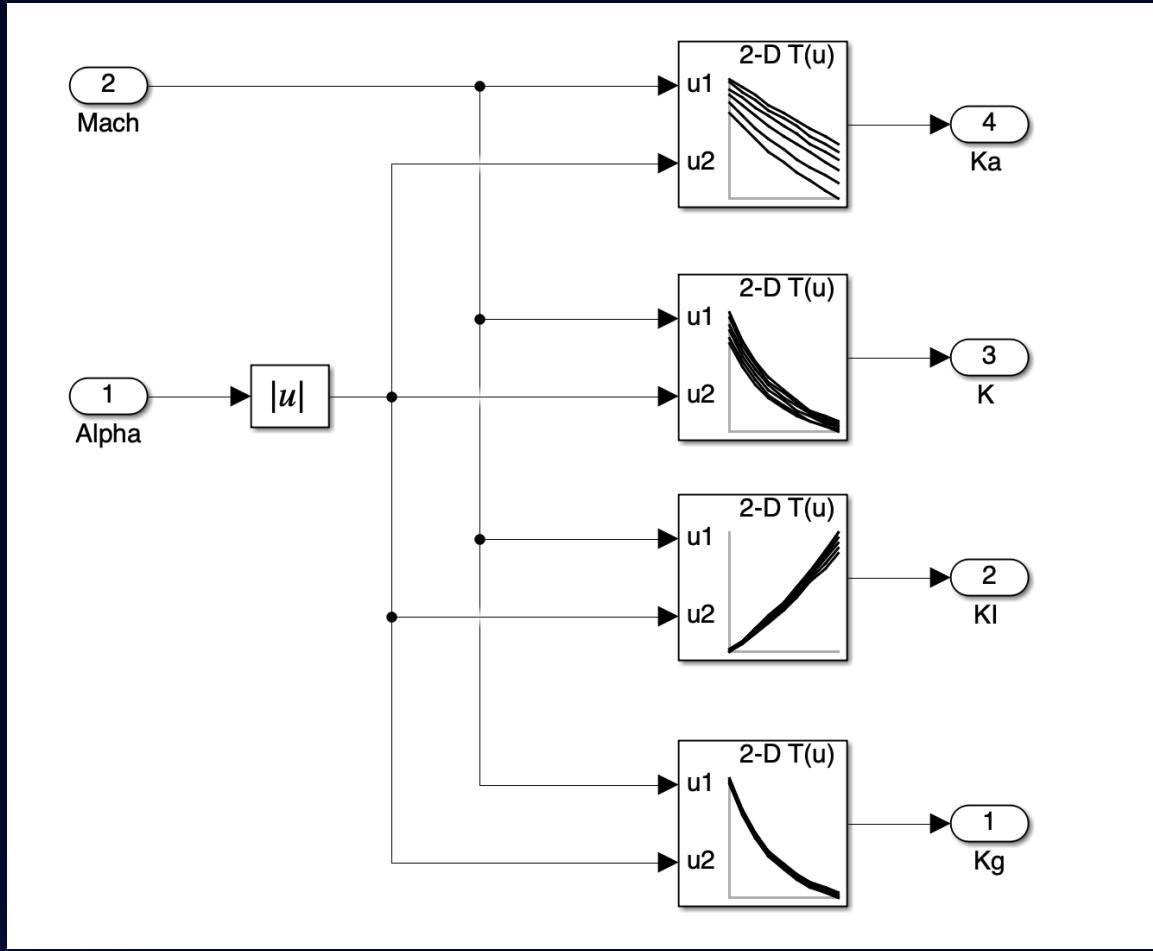
- Attitude (α) and Mach number (M)
- Demanded Normal Acceleration $A_{Z,d}$
- Measured Normal Acceleration $A_{Z,m}$
- Pitch Rate q_m



Output:

Demanded fin deflection

Gain scheduled coefficients block



4 different 2D lookup table to compute the **control gains**

$$K, K_a, K_I, Kg$$

Based on the current Mach number and the angle of attack.

Values of gain coefficients were carefully selected through a manual tuning process so that the system response was as accurate as possible.

Non Linear Second Order Actuators

The ”Fin demand” value is then passed to the actuator system.

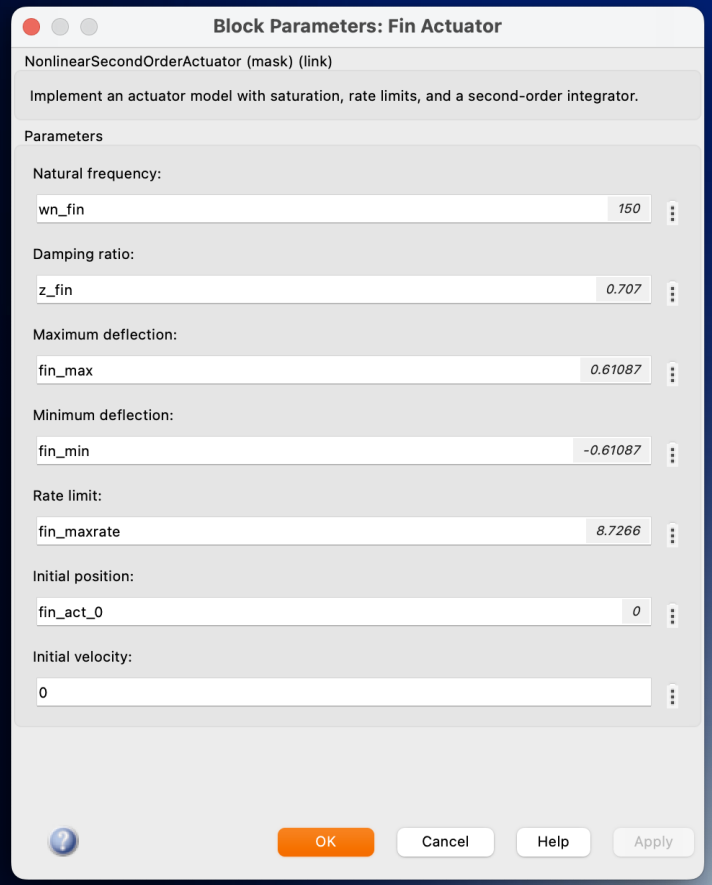
The Aerospace Block-set’s ”Non-Linear Second order actuator” is used with the following parameters:

- «natural frequency» is $\omega_n = \frac{k}{m}$.

  For the porpuse of the simulation a value of $150\ ^{rad}/_{s}$ has been chosen

- «dumping ratio» is $\zeta = \frac{b}{2\sqrt{mk}}$.

  For the porpuse of the simulation a value of $\zeta = \frac{1}{\sqrt{2}} = 0.707$ has been chosen



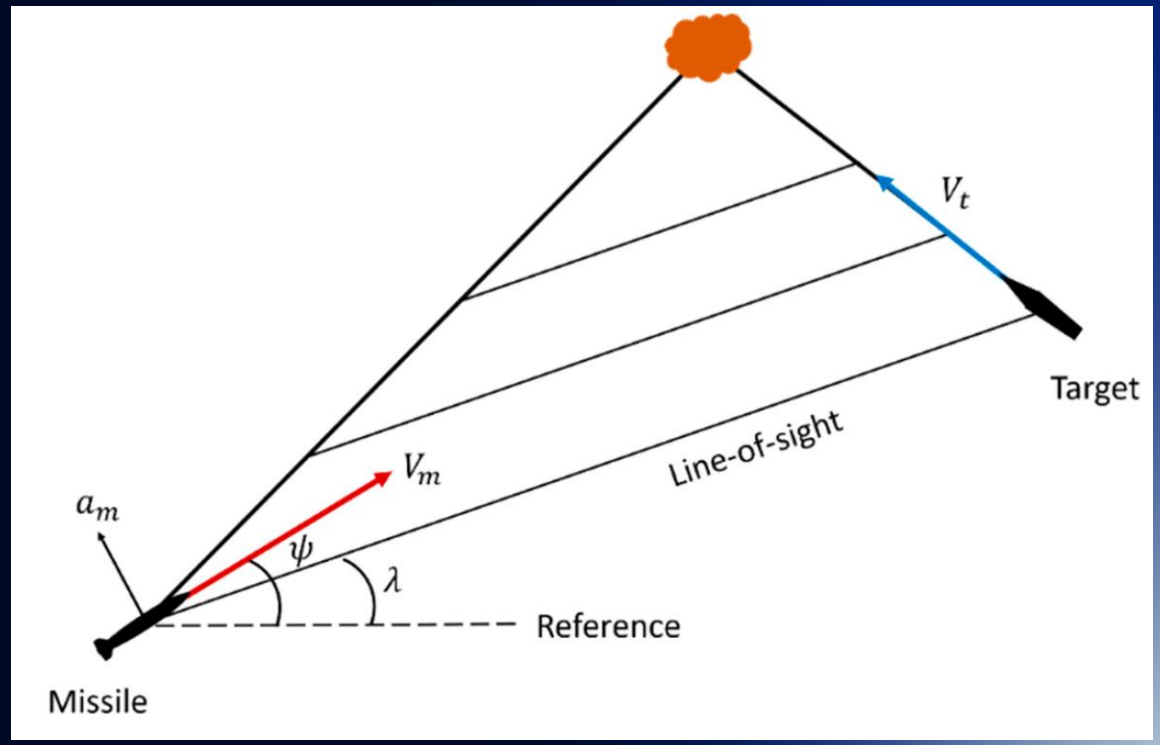
GUIDANCE

Proportional Navigation

The core principle is that the missile's commanded acceleration A_Z is proportional to the rate of change of the line-of-sight (LOS) angle between the missile and the targe.

Key parameters:

- LOS (Line of sight)
- LOS Angle λ
- Angular Rate of LOS Angle $\dot{\lambda}$
- Closing Velocity V_C
- PN Navigation Constant N
- Commanded Vertical Acceleration A_C



Proportional Navigation

The core equation is the following:

$$A_C = N \cdot V_C \cdot \dot{\lambda}$$

where:

$$V_C = -\frac{\delta R}{\delta t}$$

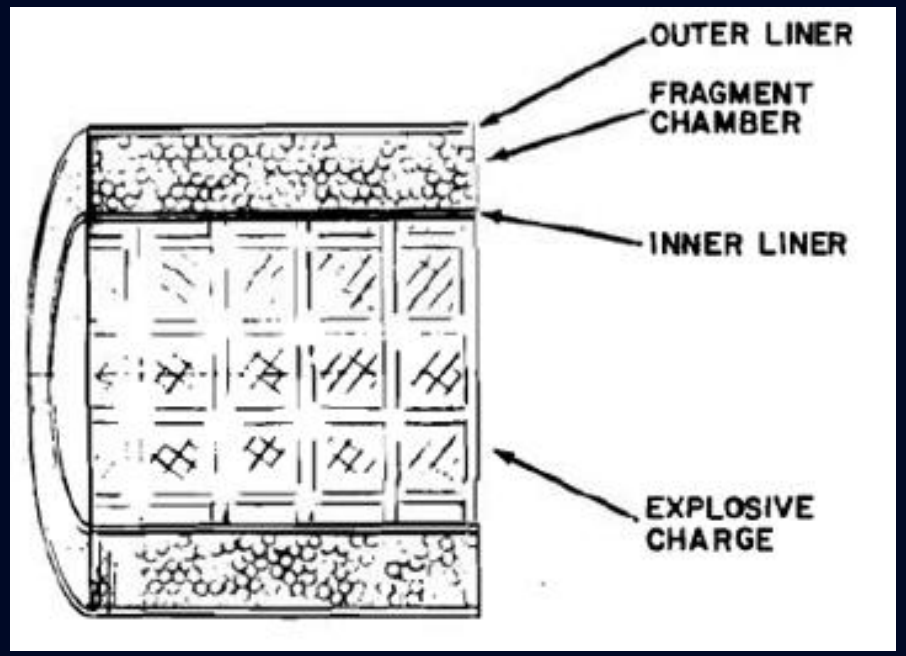
$$\dot{\lambda} = \frac{\delta \lambda}{\delta t}$$

$$N \in (3,5)$$

WARHEAD

Design

Fragmentation warhead with pre-cut spherical fragments:



Design

Using Garney formula it is possible to calculate the fragments' velocity (V_r) in radial direction:

$$V_r = \sqrt{2E}\sqrt{\frac{{}^{c}/_{m}}{1 + 0.5\ {}^{c}/_{m}}}$$

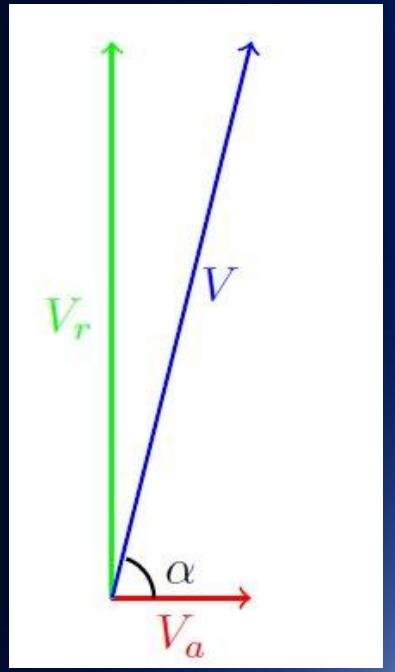
Where:

- $\sqrt{2E} = 2682.24\ {}^{m}/_{s}$ (Composition B)
- ${}^{c}/_{m} = 0.5$

The result is: $V_r = 1696\ {}^{m}/_{s}$

The fragments' velocity in axial (V_a) direction is equal to missile velocity: $V_a = 250\ {}^{m}/_{s}$

The fragments' velocity (V) is: $V = 1715\ {}^{m}/_{s}$



Design

It's possible to calculate the slowdown caused by the aerodynamic drag: $V_{new} = V_{old} \cdot e^{-C_D \rho A m_{sf} d}$

If the distance traveled is less than $\sqrt{5}$ m it involves a reduction of less than 2%

For this reason it is possible to approximate the impact speed (V_{imp}) to V: $V_{imp} \approx V = 1714\,^{m}/_{s}$

Design

Considering:

- Mass of Steel Fragments: 4.1 g
- Thickness of Target Aluminum plate: 10 mm

It is possible to analyze the fragments penetration using the Thor Formula:

$$V_{res}=V_{imp}-0.3048\cdot 10^{c11}\cdot(61023.75\cdot t\cdot A)^{c12}\cdot\left(915432.1\cdot m_{sf}\right)^{c13}\cdot\left(\frac{1}{\cos(\alpha)}\right)^{c14}\cdot(3.28084)^{c15}\cdot V_{imp}=763.965\ {}^{m}/_{s}$$

$$M_{res}=m_{sf}-6.48\cdot 10^{c31-5}\cdot(61023.75\cdot t\cdot A)^{c32}\cdot\left(915432.1\cdot m_{sf}\right)^{c33}\cdot\left(\frac{1}{\cos(\alpha)}\right)^{c34}\cdot(3.28084)^{c35}\cdot V_{imp}=2.6\ g$$

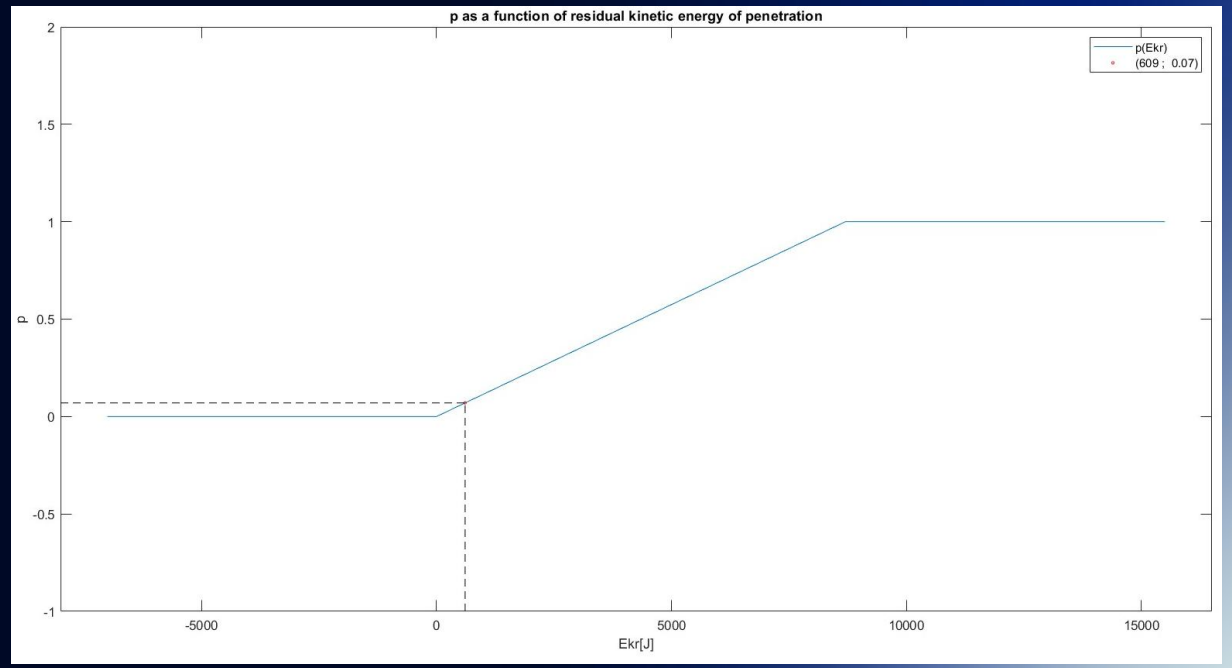
The Residual Kinetic Energy ($E_{K\,res}$) is: $E_{K\,res}=761\ J$

P_K Estimation

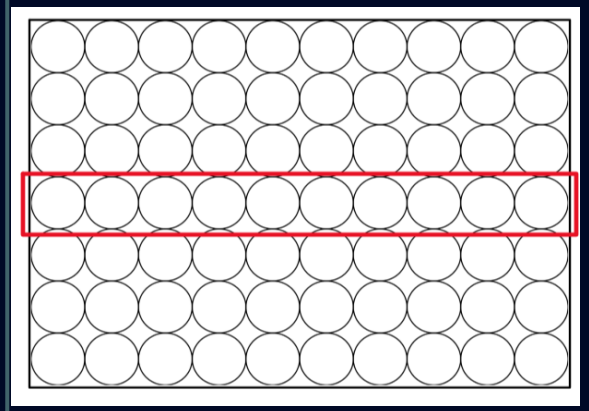
For each triplet of values (X, Y, θ) → n → $P_K = 1 - (1 - p)^n$

Where «p» is the success probability of Bernoulli Experiment

Supposing a realistic value of 0.07 for «p» it has been obteined the following ramp function $p = p(E_{k\,res})$

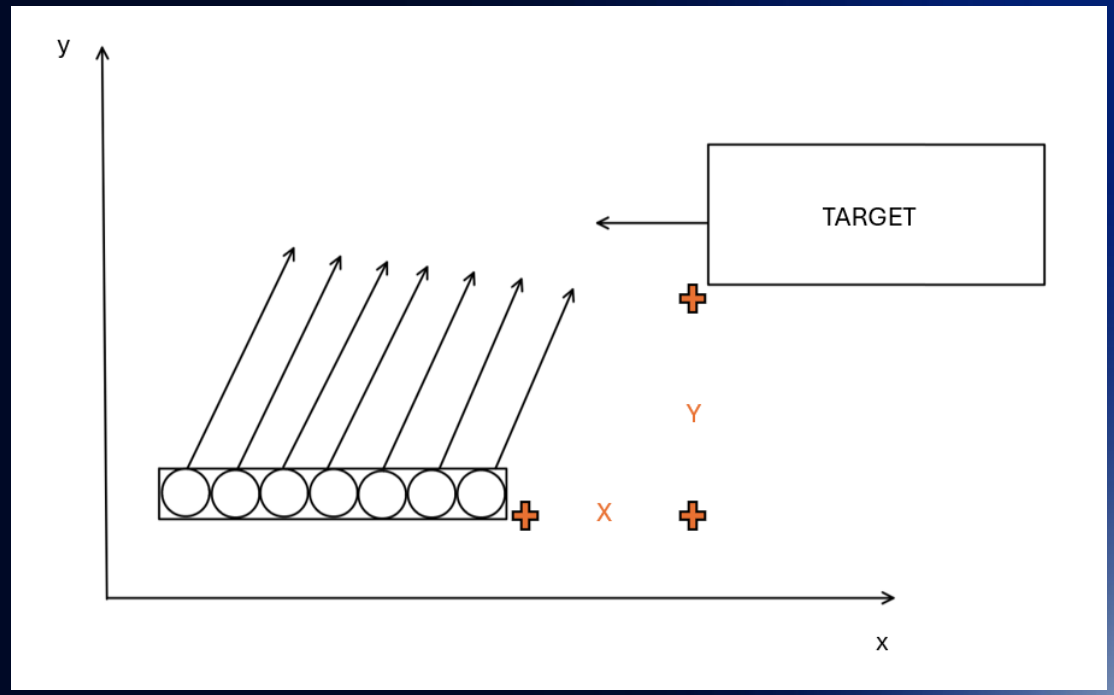


P_K Estimation



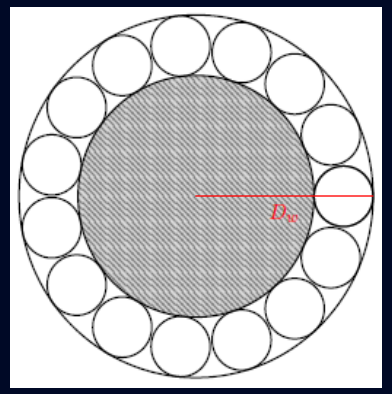
Top view of the warhead

2D analysis to understand how many rings hit



Front view of a hypothetical scenario ($\theta = 0$)

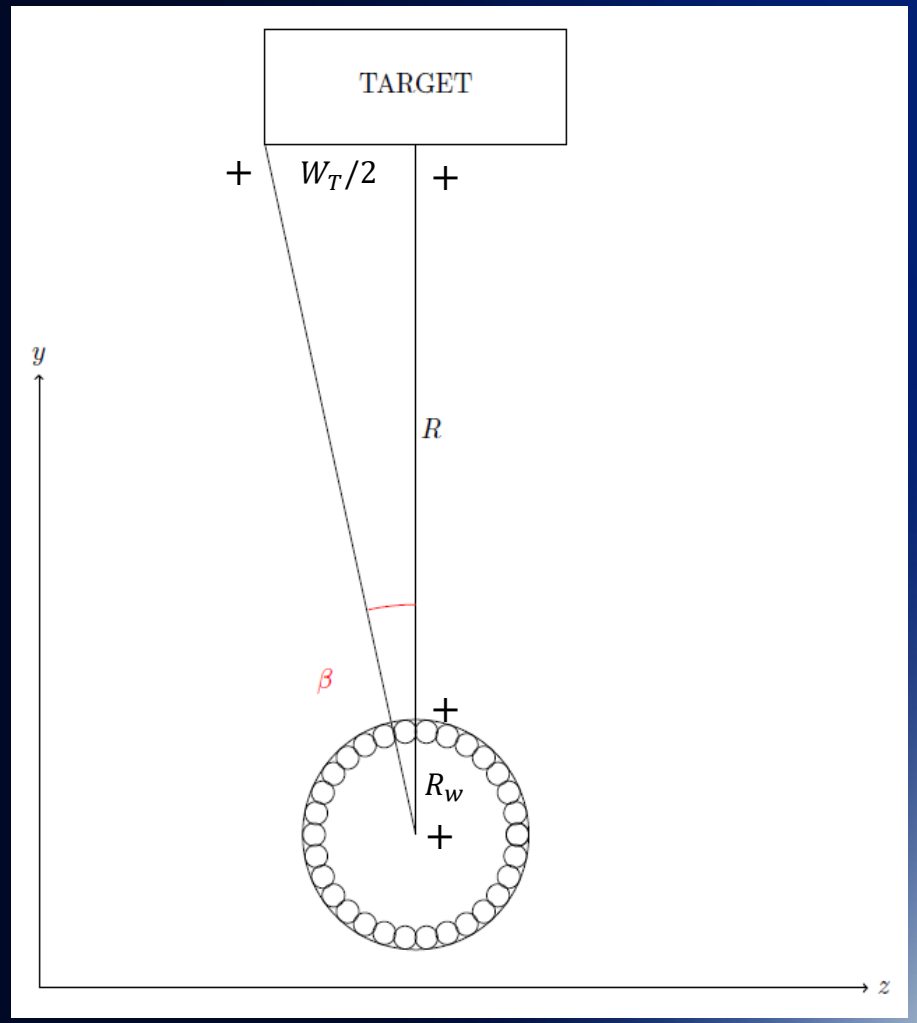
P_K Estimation



3D analysis to understand for each ring how many fragments hit

$$\beta = \tan^{-1}\left(\frac{W_T/2}{R + R_W}\right)$$

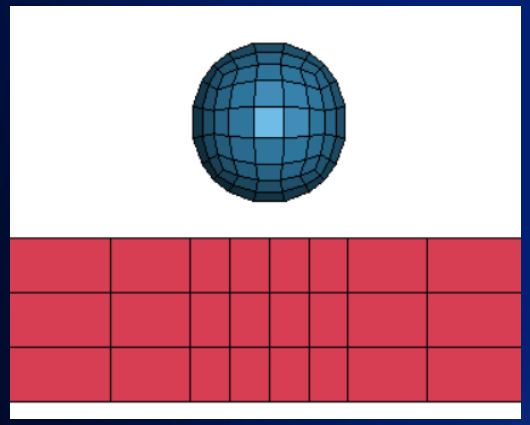
$$2 \cdot (\beta \cdot R_w) = \pi \cdot r_f \cdot n_{pr} \Rightarrow n_{pr} = \frac{2 \cdot (\beta \cdot R_w)}{\pi \cdot r_f}$$



LS-Dyna

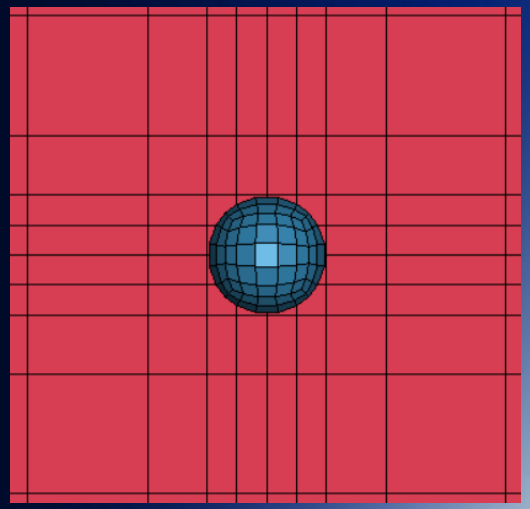
The fragment, whose mesh was modeled using sphere solid, has the following characteristics:

- it is a deformable sphere with a radius of 5 mm and a mass of 4.1 g
- its velocity is perpendicular to the target surface and it is 1714 $^{m}/_{s}$
- it is made of steel



The target, whose mesh was modeled using box solid, has the following characteristics:

- it is a plate with a width and length of 1 m and a thickness of 10 mm
- it is made of aluminum



LS-Dyna

Materials:

MAT_003_PLASTIC_KINEMATIC for the Target, while MAT_024_PICEWISE_LINEAR_PLASTICITY for the Fragment:

	ρ	E	PR	SIGY	ETAN	FS
Fragment	$8.300 \cdot 10^{-9} \frac{t}{mm^3}$	210 GPa	0.284	1000 MPa	2000 MPa	
Target	$2.8 \cdot 10^{-9} \frac{t}{mm^3}$	72 GPa	0.33	576 MPa	0	0.3

Contact:

ERODING CONTACT SURFACE TO SURFACE

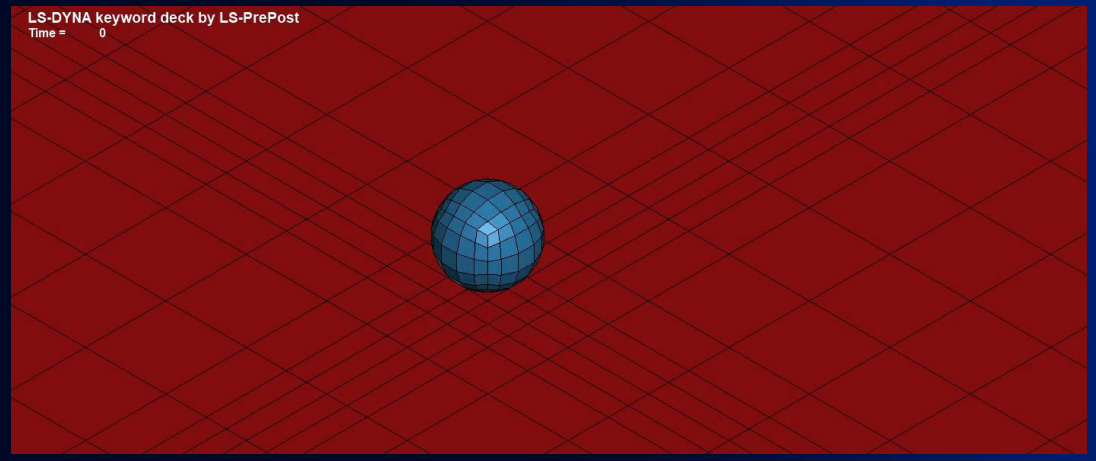
Selecting the fragment as the master (due to its higher stiffness) and the target as the slave

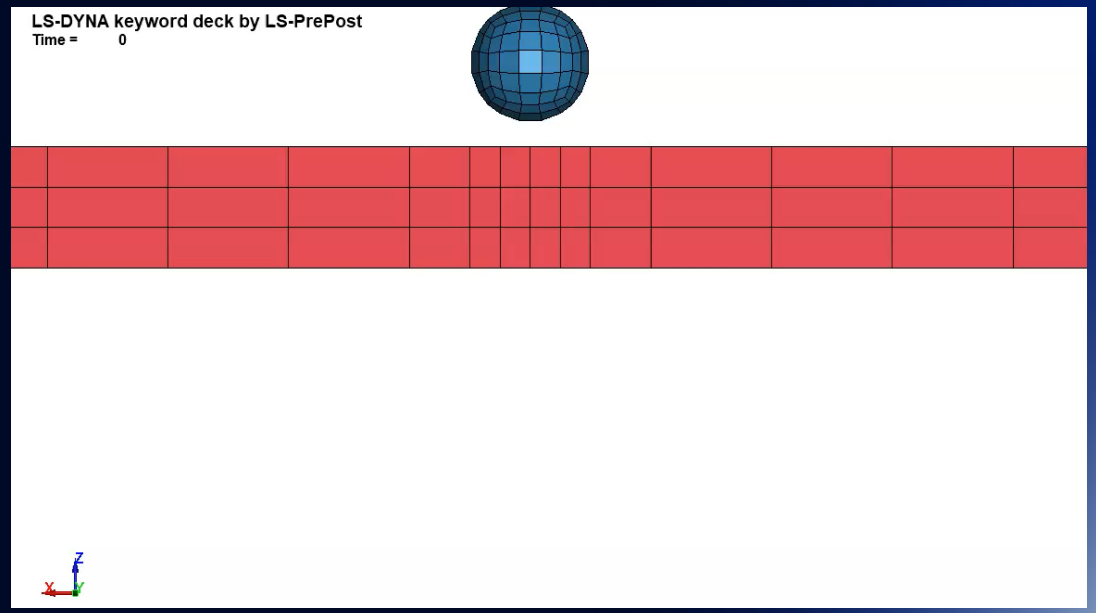
Boundary Conditions:

All degrees of freedom for the edge nodes of the target were constrained

LS-Dyna

Analysis of Penetration:

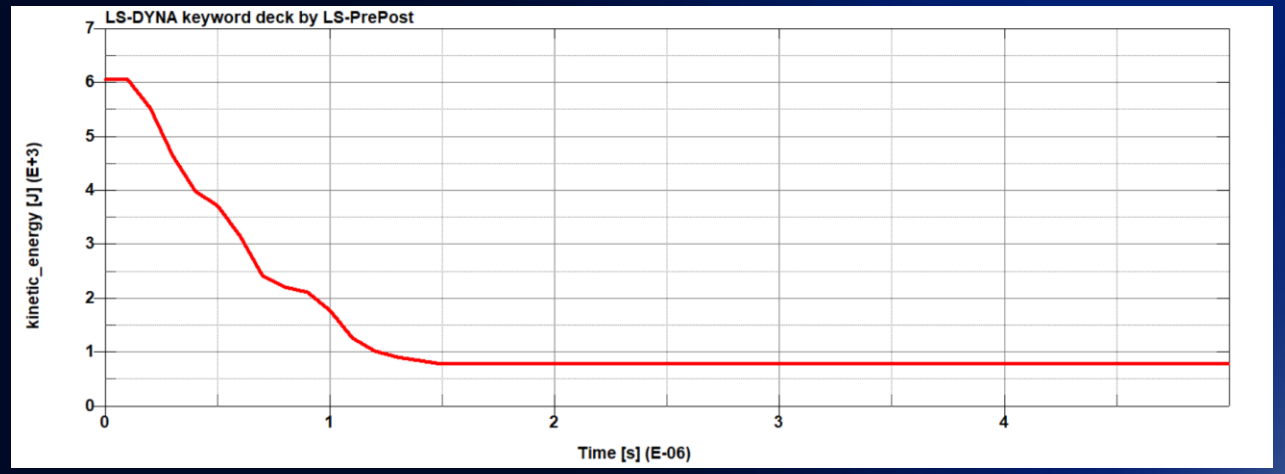


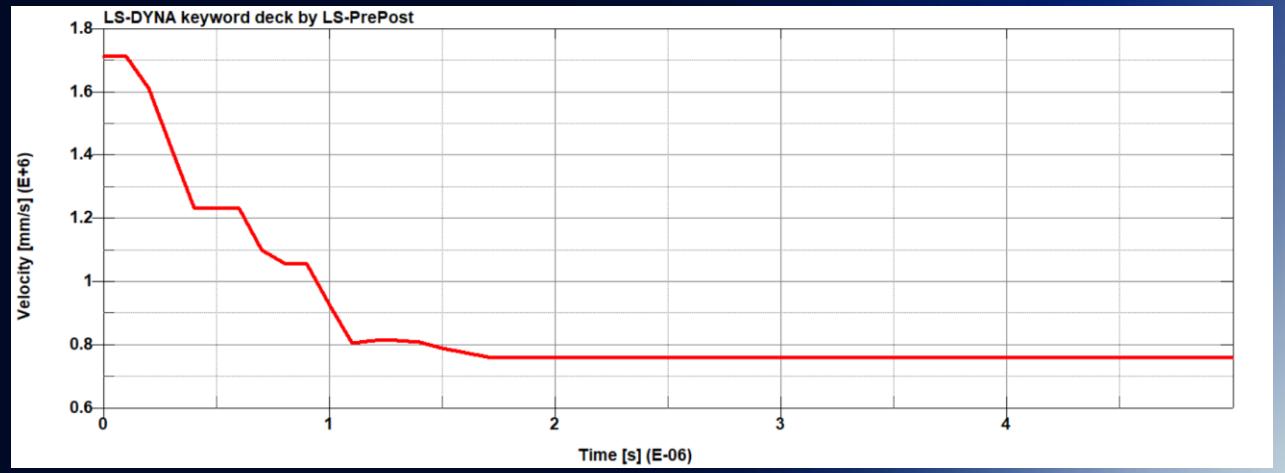


LS-Dyna

Results:

	LS-DYNA	THOR
residual kinetic energy	730 J	761 J
residual velocity	760 $\frac{m}{s}$	764 $\frac{m}{s}$
residual mass	2.6 g	2.6 g

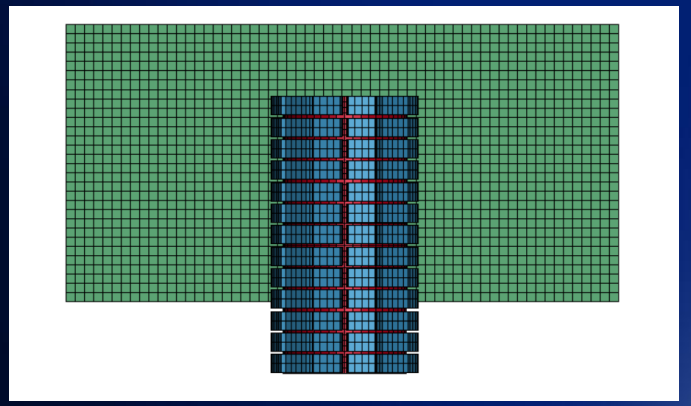




Explosion Simulation

Three main components:

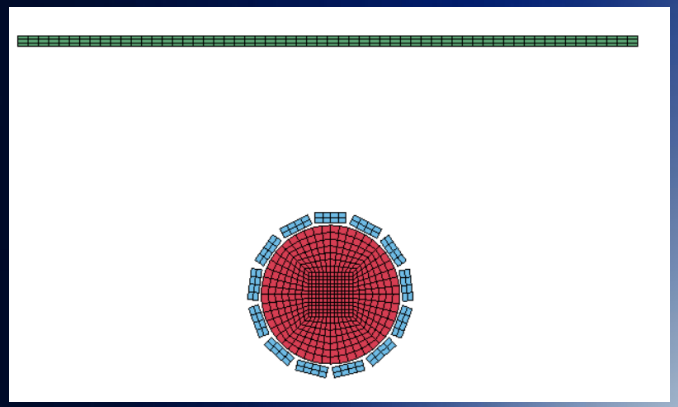
- Explosive, that represents the Component B;
- 225 Fragments, they are collocated around the Explosive;
- Plate, that represents the Target.



Materials:

- HIGH_EXPLOSIVE_BURN_008 + ADD_EROSION_00 for the explosive:

1	MID	RO	D	PCJ	BETA	K	G	SIGY
	1	1.600e-09	6.780e+06	2.100e+04	0.0	0.0	0.0	0.0

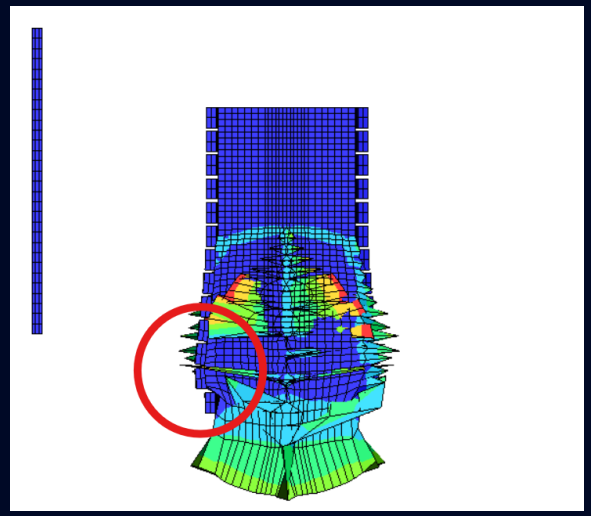


- MAT_003_PLASTIC_KINEMATIC for the Target, same parameters shown in the previous Simulation;
- MAT_024_PIECEWISE_LINEAR_PLASTICITY, same parameters shown in the previous Simulation;

LS-Dyna

Contacts:

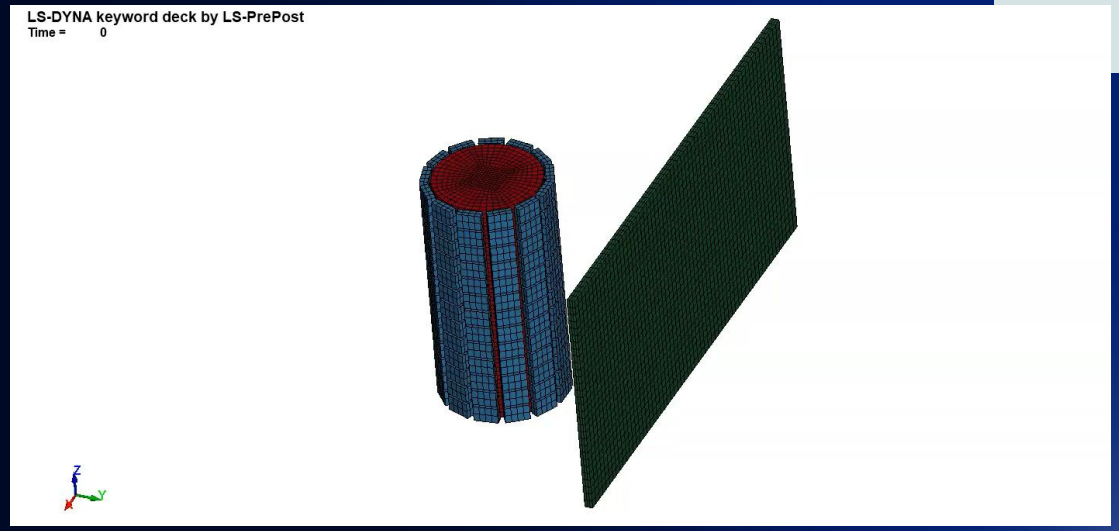
- AUTOMATIC_SURFACE_TO_SURFACE between Explosive and fragments, cause the easy of the contact. To ensure it the parameters SFSA e SFSB were modified:

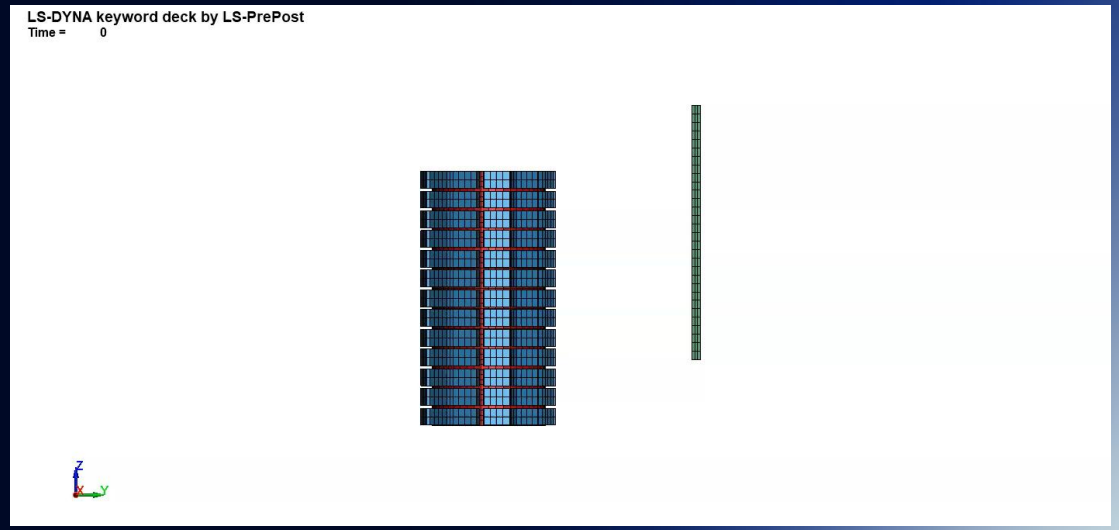


- ERODING_SURFACE_TO_SURFACE between fragments and target, due the nature of the problem.

LS-Dyna

Analysis of Penetration:

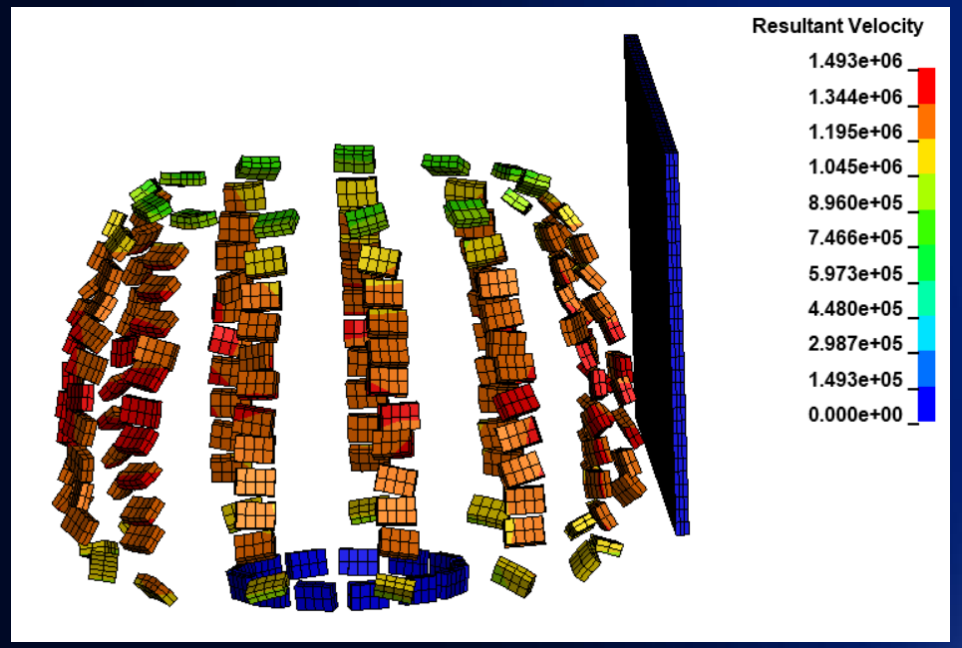




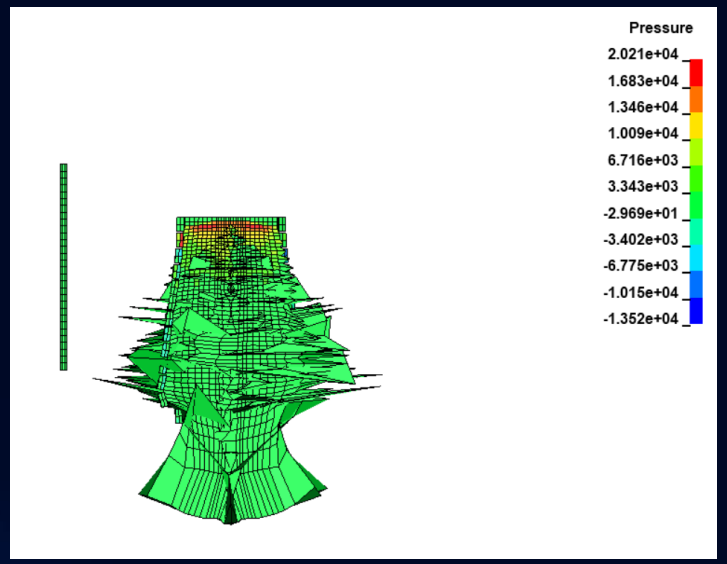
LS-Dyna

Results:

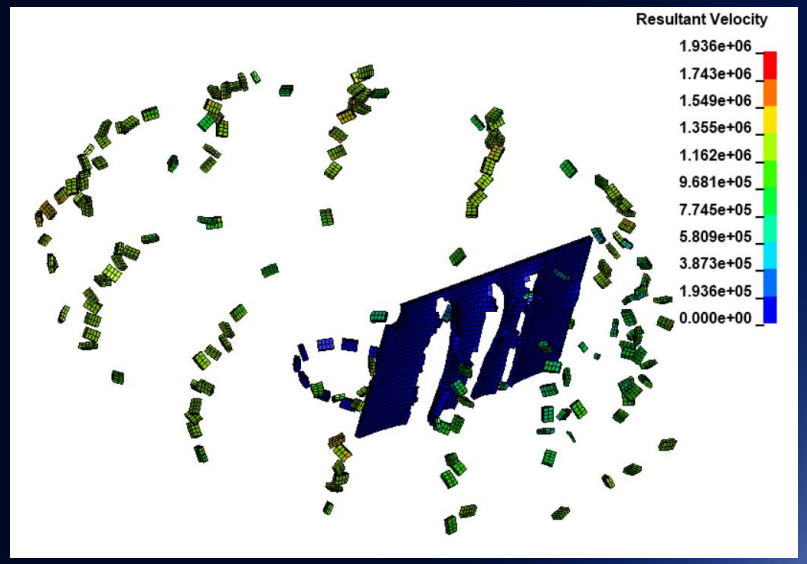
- Ejection Velocity 1500 m/s
- Residual Velocity 800 m/s
- Pressure Peak 21 GPa



(mm/s)



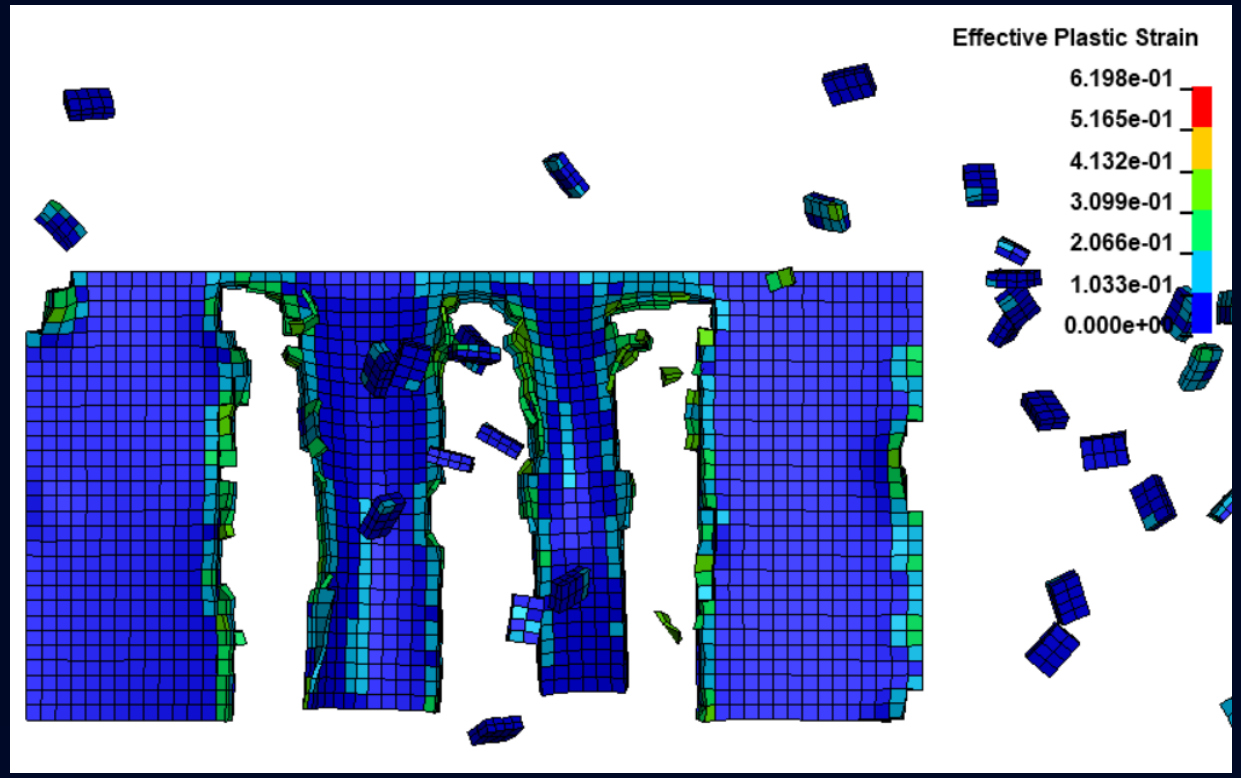
(MPa)

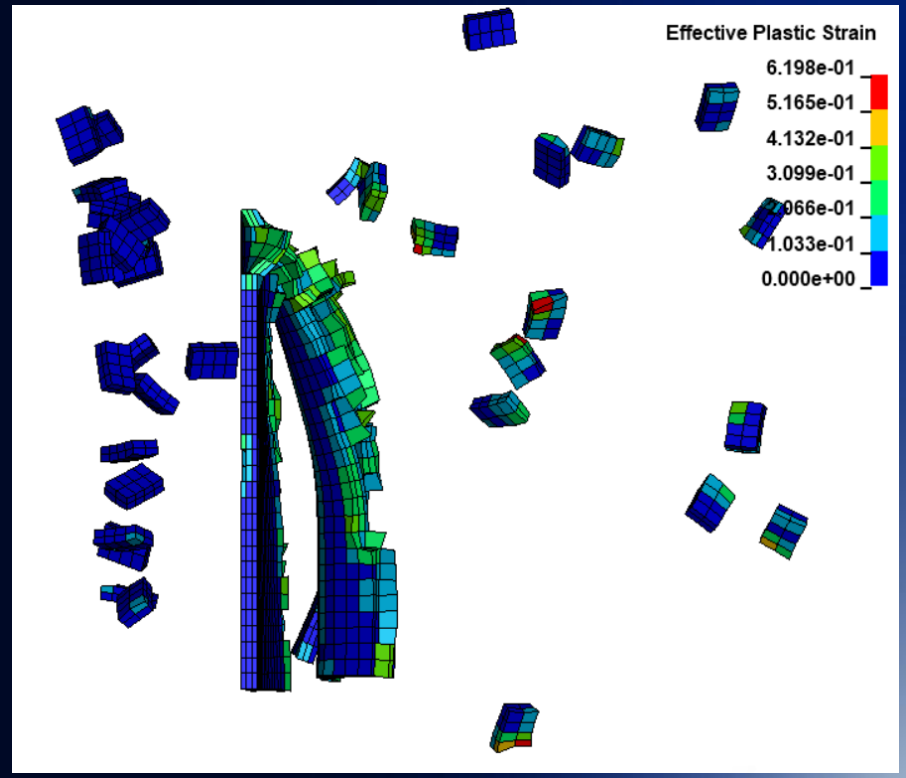


(mm/s)

LS-Dyna

Damage Results:

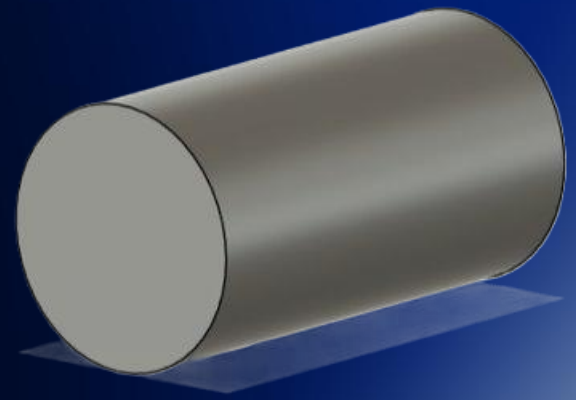




Recap

- Type: Fragmentation
- Shape: cylindrical
- Fragment material: Steel
- Shape of fragments: Spherical
- Radius of fragments: 5 mm
- Number of rings: 15
- Fragment numbers per ring: 15
- Total number of fragments : 225
- Mass of the single fragment: 4.1 g
- Mass of fragments: 918.9 g

- Type of explosive: composition B
- c/m: 0.5
- Explosive mass: 459.5 g
- Diameter of the explosive: 48.6 mm
- Dead mass: 590.7 g



Warhead dimentions:

- Total Warhead Mass: 1969.1 g
- Total Warhead Length: 150 mm
- Total Warhead Diameter: 77 mm

SAGITTARIUS OVERVIEW

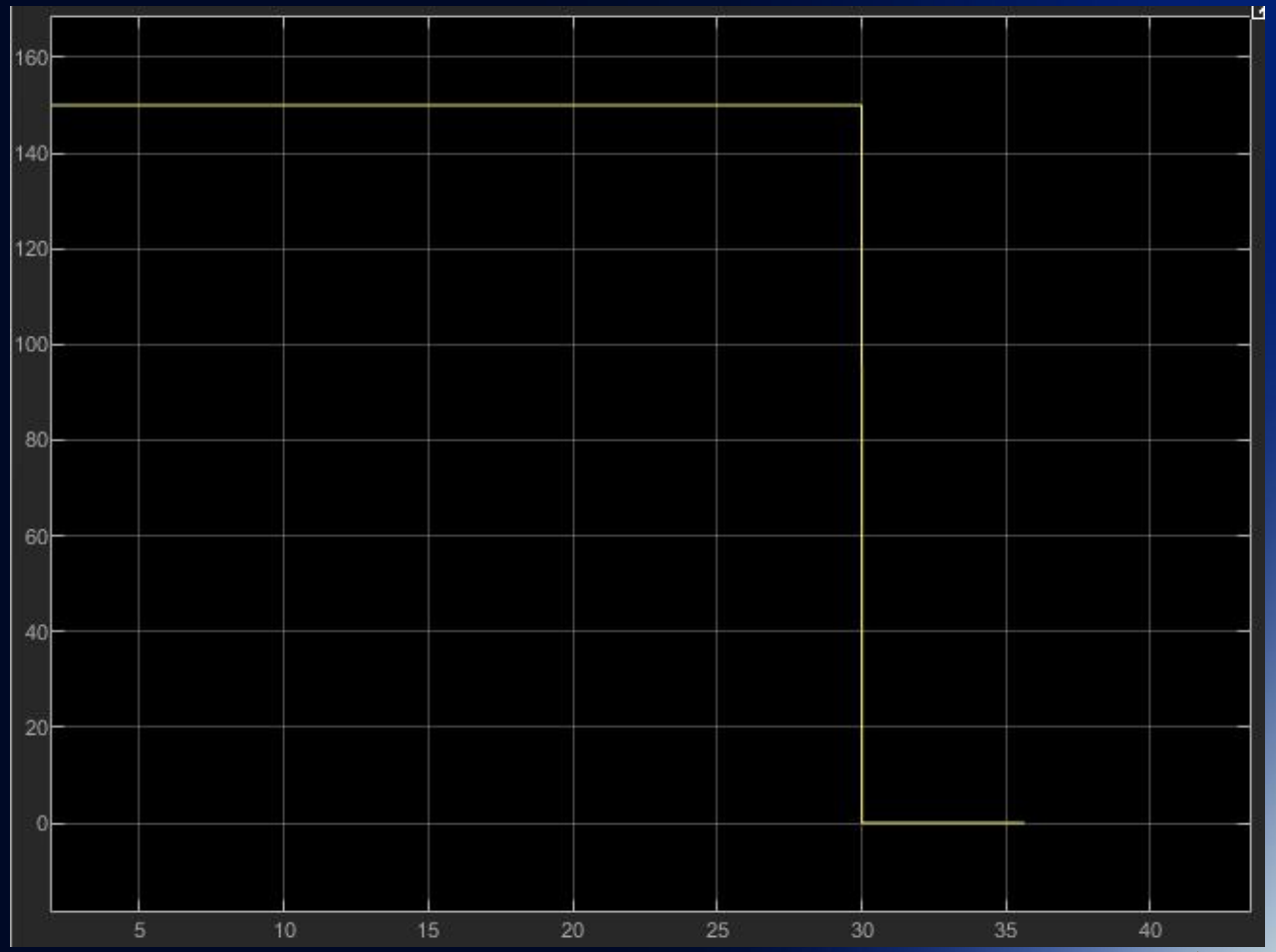

SAGITTARIUS
SAGITTARIUS

PROPULSION

Thrust Profile

Engine requirements:

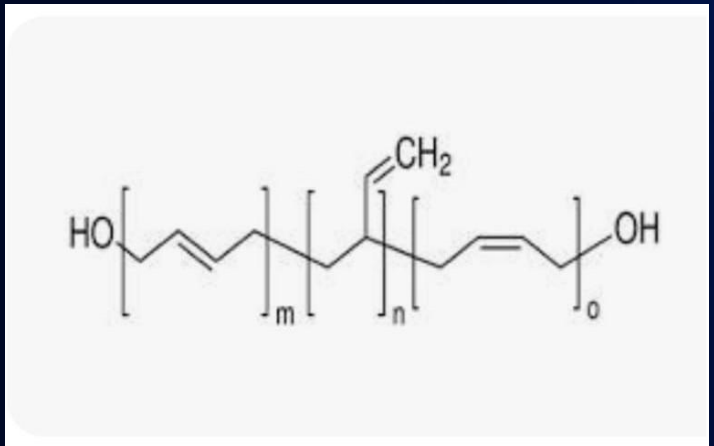
- Thrust: 150N
- Burning time: 30s
- Thrust Profile shape: Step



Propellant

Propellant charcteristics:

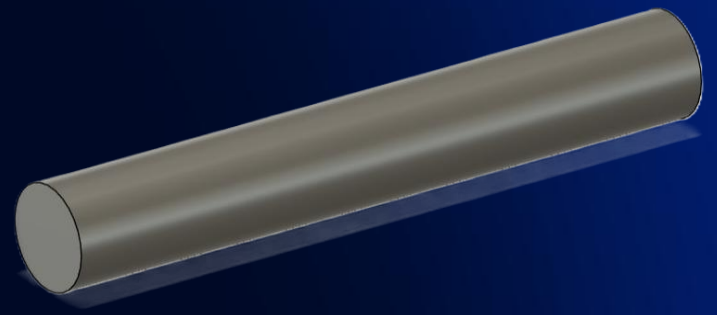
- Propellant: HTPB 1912
- Density: 1854.55 $^{kg}/_{m^3}$
- Burning rate: 0.0141 $^{m}/_{s}$
- $\dot{m} = 0.0841\ ^{kg}/_{s}$
- $\dot{l} = 0.0141\ ^{m}/_{s}$



Chamber Design

Chamber charcteristics:

- Propellant: HTPB 1912
- End-burning (or «cigarette grain»)
- Burning rate: 0.0141 $^{m}/_{s}$
- $\dot{m} = 0.0841\ ^{kg}/_{s}$
- $p_{chamber}$: 7 bar
- p_{exit} : 1 bar (adapted to sea level)
- Exit diameter: 22 mm



Input from CEA (Chemical equilibrium application):

- Expansion ratio: 1.9
- Specific Impulse: 182.1s (Corrected with 0.9)
- Characteristic velocity (c^*): 1666 $^{m}/_{s}$

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 101.5 PSIA
CASE = _____

REACTANT		WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	AL	0.6129000	0.000	0.000
FUEL	C4H6,butadiene	0.3871000	0.000	0.000
OXIDANT	NH4CLO4(I)	1.0000000	0.000	0.000

O/F= 2.23000 %FUEL= 30.959752 R,EQ.RATIO= 1.765042 PHI,EQ.RATIO= 2.377076

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7170	7.0000
P, BAR	7.0000	4.0770	1.0000
T, K	3528.08	3387.56	3044.36
RHO, KG/CU M	6.3091-1	3.8788-1	1.0943-1
H, KJ/KG	0.00000	-583.80	-1962.23
U, KJ/KG	-1109.50	-1634.88	-2876.02
G, KJ/KG	-37249.1	-36349.3	-34104.2
S, KJ/(KG)(K)	10.5579	10.5579	10.5579
M, (1/n)	26.439	26.797	27.700
MW, MOL WT	24.770	24.951	25.464
(dLV/dLP)t	-1.06630	-1.05865	-1.04135
(dLV/dLT)p	2.2177	2.1048	1.8296

SON VEL,M/SEC	1110.5	1080.6	999.9
MACH NUMBER	0.000	1.000	1.950

PERFORMANCE PARAMETERS

Ae/At	1.0000	1.7861
CSTAR, M/SEC	1670.1	1670.1
CF	0.6470	1.1674
Ivac, M/SEC	2053.3	2375.8
Isp, M/SEC	1080.6	1949.7

MASS FRACTIONS

*AL	0.00150	ALCL	0.03863	ALCL2	0.00118
ALCL3	0.00014	ALH	0.00011	ALHCL	0.00004
ALHCL2	0.00003	*ALO	0.00372	ALOCL	0.00374
ALOCL2	0.00001	ALOH	0.02525	ALOHCL	0.00160
ALOHCL2	0.00067	ALO2	0.00007	AL(OH)2	0.00041
AL(OH)2CL	0.00021	AL(OH)3	0.00006	AL2O	0.00190
AL2O2	0.00079	*CO	0.23611	*CO2	0.01904
*CL	0.04147	CLO	0.00003	CL2	0.00004
*H	0.00429	HALO	0.00002	HALO2	0.00010
HCO	0.00001	HCL	0.14497	HOCL	0.00001
*H2	0.01930	H2O	0.07178	*N	0.00001
*NO	0.00163	*N2	0.08153	*O	0.00318
*OH	0.01396	*O2	0.00102	AL2O3(L)	0.28143

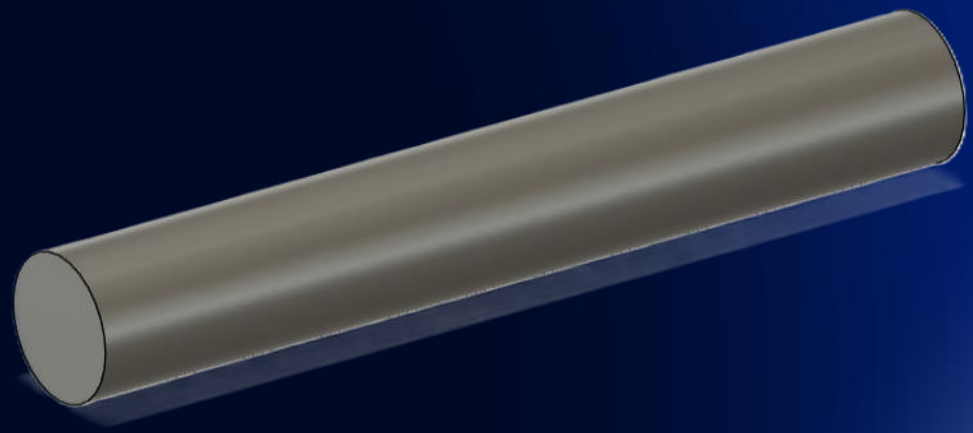
* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

Grain Sizing

Grain characteristic:

- End-burning (or «cigarette grain»)
- Grain diameter: 64 mm
- Grain length: 423.1 mm
- Propellant mass: 2522 g
- $p_{chamber}$: 7 bar.
- p_{exit}: 1 bar (adapted to sea level)



Propulsive outputs:

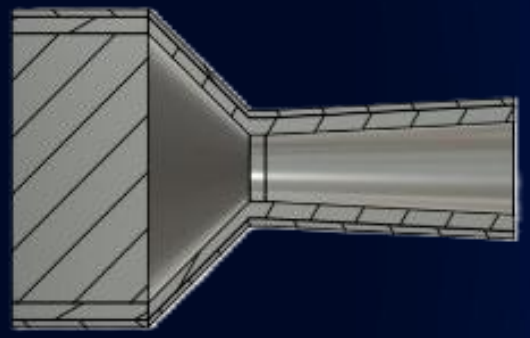
- Equivalent exhaust velocity (c): 1784.56 $^{m}/_{s}$
- Thrust coefficient (c_f): 1.07
- Characteristic velocity (c^*): 1666 $^{m}/_{s}$

Nozzle

Nozzle characteristic:

- Throat diameter: 15.9 mm
- Exit diameter: 22 mm
- Convergent angle: 45 deg
- Divergent angle: 3 deg
- Nozzle Length: 86.2 mm
- Thrust Velocity Coefficient: 1.07

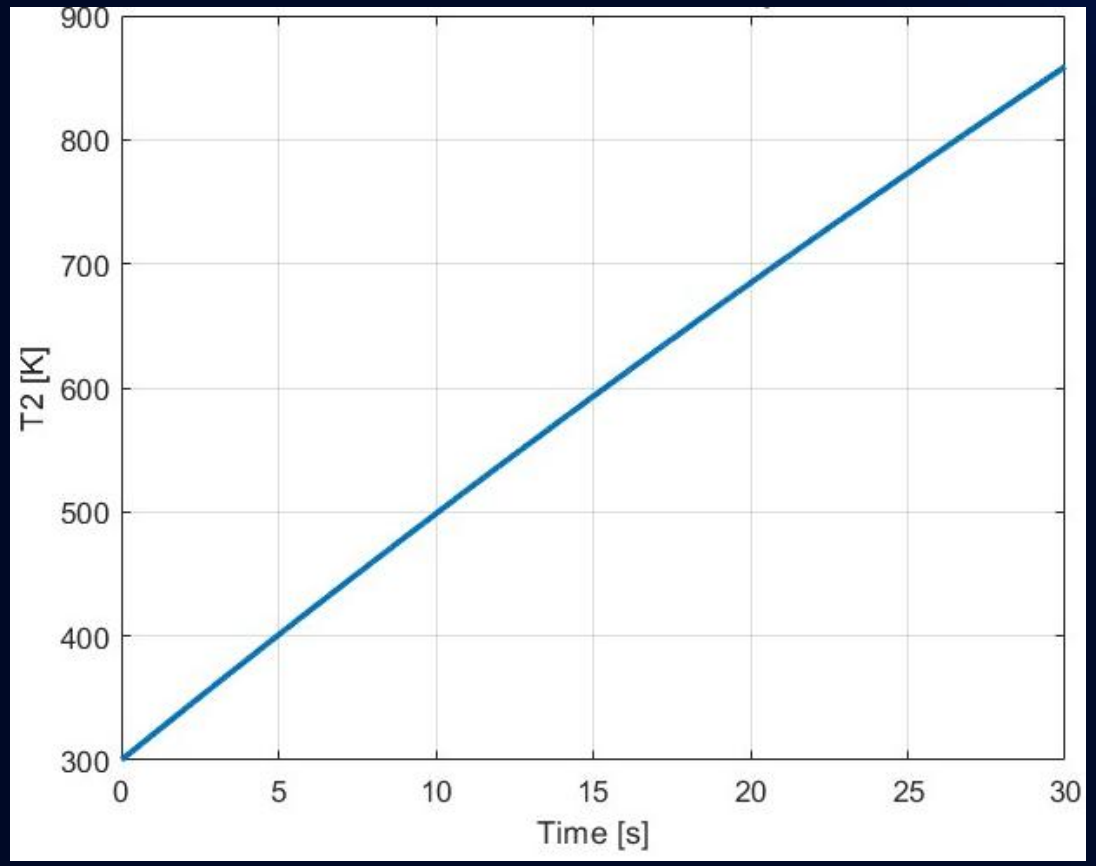




Steady-state cooling:

- Porous Graphite Liner: 4 mm
- Copper Heat Sink: 10 mm
- Throat final Temperature: 858.81 K

Nozzle



Heat -Sink Temperature in Throat over the time

Recap

Engine requirements:

- Thrust: 150 N
- Burning time: 30s

General charcteristics:

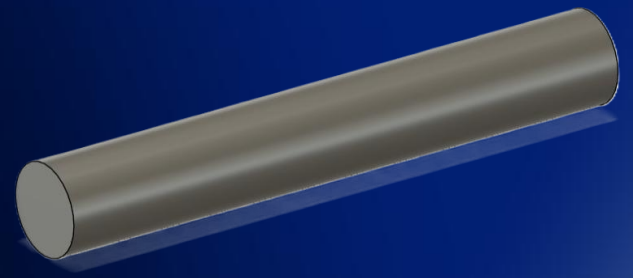
- Propellant: HTPB 1912
- Burning rate: 0.0141 $^{m}/_{s}$
- $\dot{m}$ = 0.0841 $^{kg}/_{s}$
- $p_{chamber}$: 7 bar
- p_{exit}: 1bar (adapted to sea level)
- Exit diameter: 22 mm

Input from CEA (Chemical equilibrium application):

- Expansion ratio: 1.9
- Specific Impulse: 182.1s (Corrected with 0.9)

Grain characteristic:

- End-burning (or «cigarette grain»)
- Grain diameter: 64mm
- Grain length: 423.1m
- Popellant mass: 2522 kg



Propulsive outputs:

- Thrust velocity coefficient (c_f): 1.07;
- Characteristic velocity (c^*): 1666 $^{m}/_{s}$.

Nozzle characteristic:

- Throat diameter: 15.9 mm
- Exit diameter: 22 mm
- Convergent angle: 45 deg
- Divergent angle: 3 deg



AERODYNAMICS

Sustain structural conditions:

- Sustain mass 11.224 kg (safe factor 1.3)
- Sustain case diameter 80.6 mm
- Reference Area 5000 mm^2

Targets:

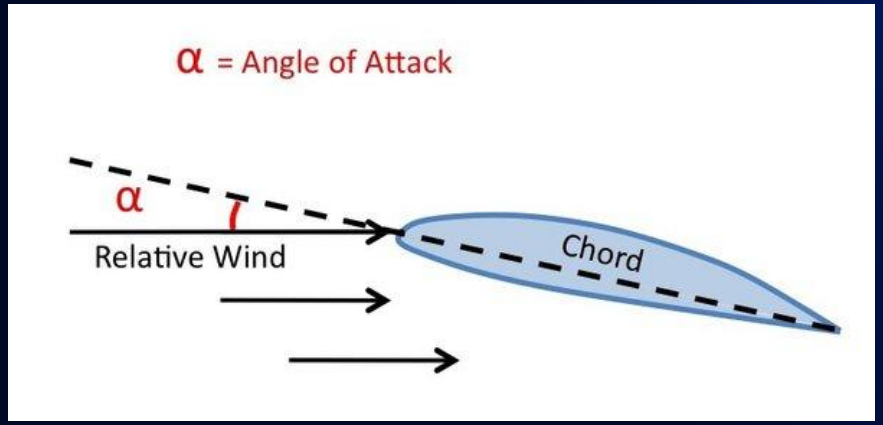
- n = 6
- F_z = 858.401 N



AERODYNAMICS

Flight parameters:

- $V_\infty = 250\ ^m/_s$
- Mach = 0.73
- $\alpha_{max} = 35°$

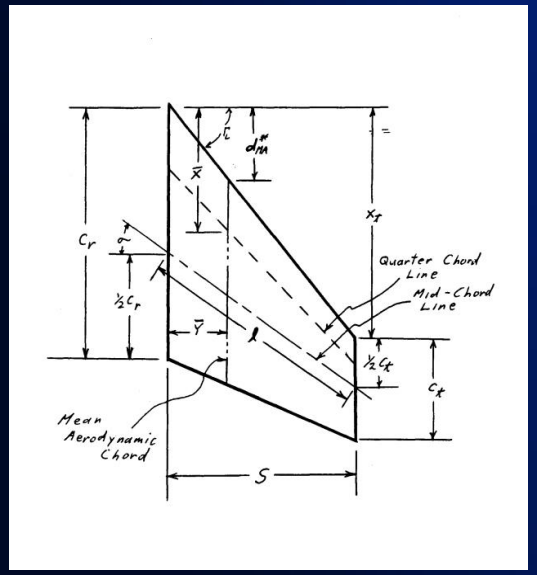


Profile considerations:

Profile	C_L	C_D	E_{max}	α_{Stall}
NACA 0015	0.6431	0.00921	77.9 (α =9°)	16.75°
NACA 0012	0.6940	0.00972	75.6 (α =7.5°)	15.5°
NACA 0010	0.6336	0.01033	70.2 (α =9.75°)	14.5°
NACA 0008	0.6671	0.0102	68.3 (α =6.75°)	12°
NACA 0006	0.6462	0.01479	56.6 (α =4.5°)	9°
NACA 63015A	0.6411	0.00968	72 (α =4.75°)	16.25°
NACA 63012A	0.6391	0.01033	64 (α =7°)	13.25°
NACA 63A010	0.6733	0.01131	59.7 (α =5.75°)	10.5°
NACA 64008A	0.6403	0.0159	49.9 (α =4.25°)	7°

Geometrical parameters:

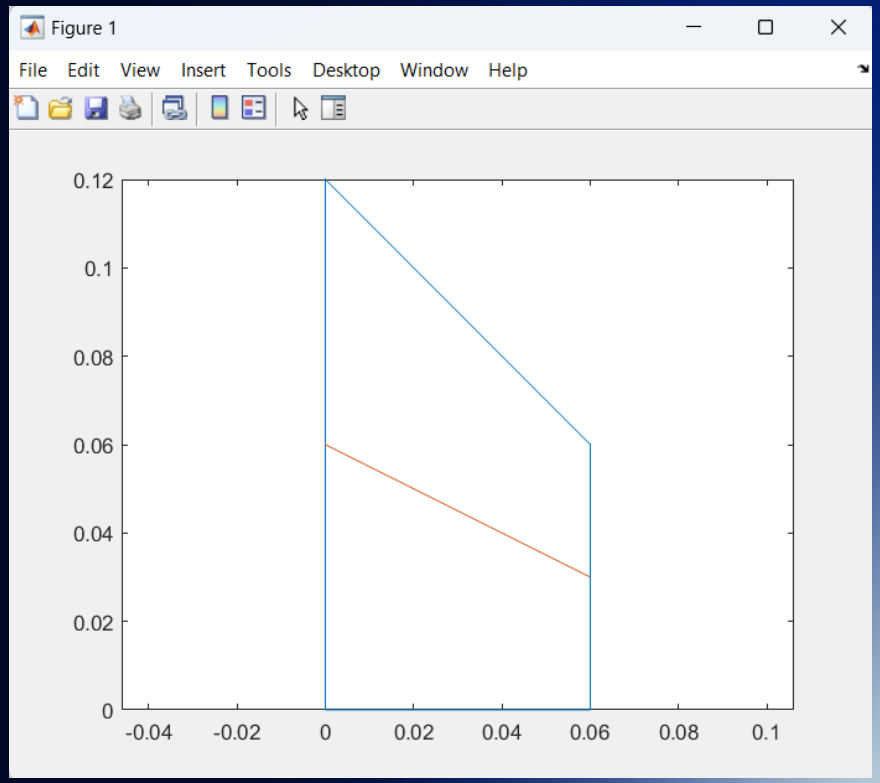
- C_{root} = 120 mm
- C_{tip} = 60 mm
- S = 60 mm
- Γ = 0.78 rad (45 °)
- $\Gamma_{c/4}$ = 0.64 rad (36.87 °)




```
Cr: 0.0800 m, Ct: 0.0400 m, S: 0.0400 m, d: 0.0800 m, n: 3.83
Cr: 0.0820 m, Ct: 0.0410 m, S: 0.0410 m, d: 0.0800 m, n: 3.94
Cr: 0.0840 m, Ct: 0.0420 m, S: 0.0420 m, d: 0.0800 m, n: 4.04
Cr: 0.0860 m, Ct: 0.0430 m, S: 0.0430 m, d: 0.0800 m, n: 4.15
Cr: 0.0880 m, Ct: 0.0440 m, S: 0.0440 m, d: 0.0800 m, n: 4.25
Cr: 0.0900 m, Ct: 0.0450 m, S: 0.0450 m, d: 0.0800 m, n: 4.36
Cr: 0.0920 m, Ct: 0.0460 m, S: 0.0460 m, d: 0.0800 m, n: 4.47
Cr: 0.0940 m, Ct: 0.0470 m, S: 0.0470 m, d: 0.0800 m, n: 4.59
Cr: 0.0960 m, Ct: 0.0480 m, S: 0.0480 m, d: 0.0800 m, n: 4.70
Cr: 0.0980 m, Ct: 0.0490 m, S: 0.0490 m, d: 0.0800 m, n: 4.82
Cr: 0.1000 m, Ct: 0.0500 m, S: 0.0500 m, d: 0.0800 m, n: 4.94
Cr: 0.1020 m, Ct: 0.0510 m, S: 0.0510 m, d: 0.0800 m, n: 5.06
Cr: 0.1040 m, Ct: 0.0520 m, S: 0.0520 m, d: 0.0800 m, n: 5.19
Cr: 0.1060 m, Ct: 0.0530 m, S: 0.0530 m, d: 0.0800 m, n: 5.31
Cr: 0.1080 m, Ct: 0.0540 m, S: 0.0540 m, d: 0.0800 m, n: 5.44
Cr: 0.1100 m, Ct: 0.0550 m, S: 0.0550 m, d: 0.0800 m, n: 5.57
Cr: 0.1120 m, Ct: 0.0560 m, S: 0.0560 m, d: 0.0800 m, n: 5.70
Cr: 0.1140 m, Ct: 0.0570 m, S: 0.0570 m, d: 0.0800 m, n: 5.83
Cr: 0.1160 m, Ct: 0.0580 m, S: 0.0580 m, d: 0.0800 m, n: 5.97
Cr: 0.1180 m, Ct: 0.0590 m, S: 0.0590 m, d: 0.0800 m, n: 6.11
Final Cr: 0.1200 m, Final Ct: 0.0600 m, Final S: 0.0600 m, Final d: 0.0800 m, Final n: 6.11
```

Aerodynamic parameters:

- $C_{Z,\alpha,total} = 7.43\ rad^{-1}$ (0.130 $\deg^{-1}$)
- n = 6
- $F_Z = 874.204$ N



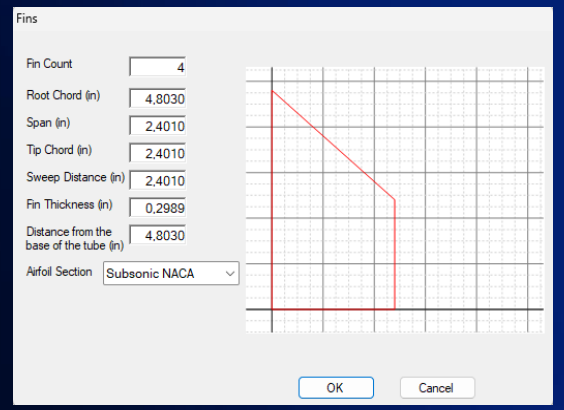
RASAERO

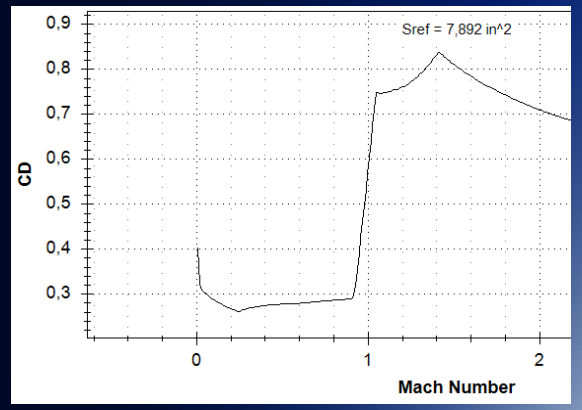
Stability parameters:

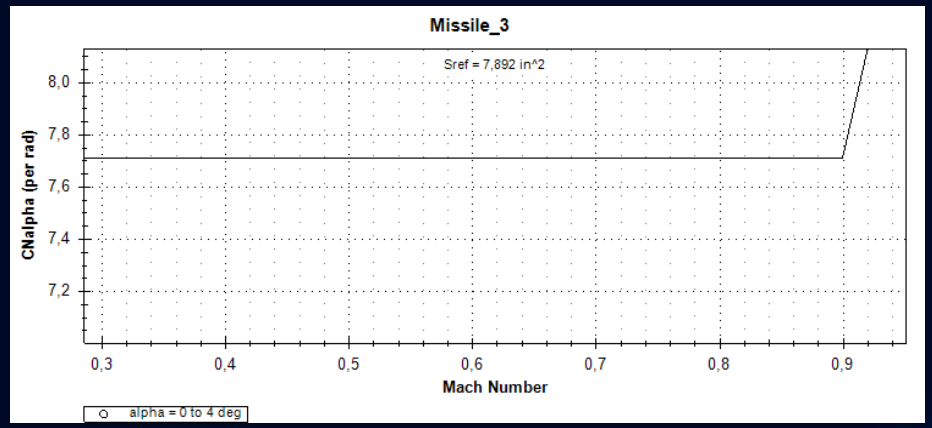
- Elliptical nose lenght: 200 mm
- Cp position: 687.3 mm
- Cg position (initial): 476.45 mm
- Cg position (end burn): 418.75 mm

RASAero data:

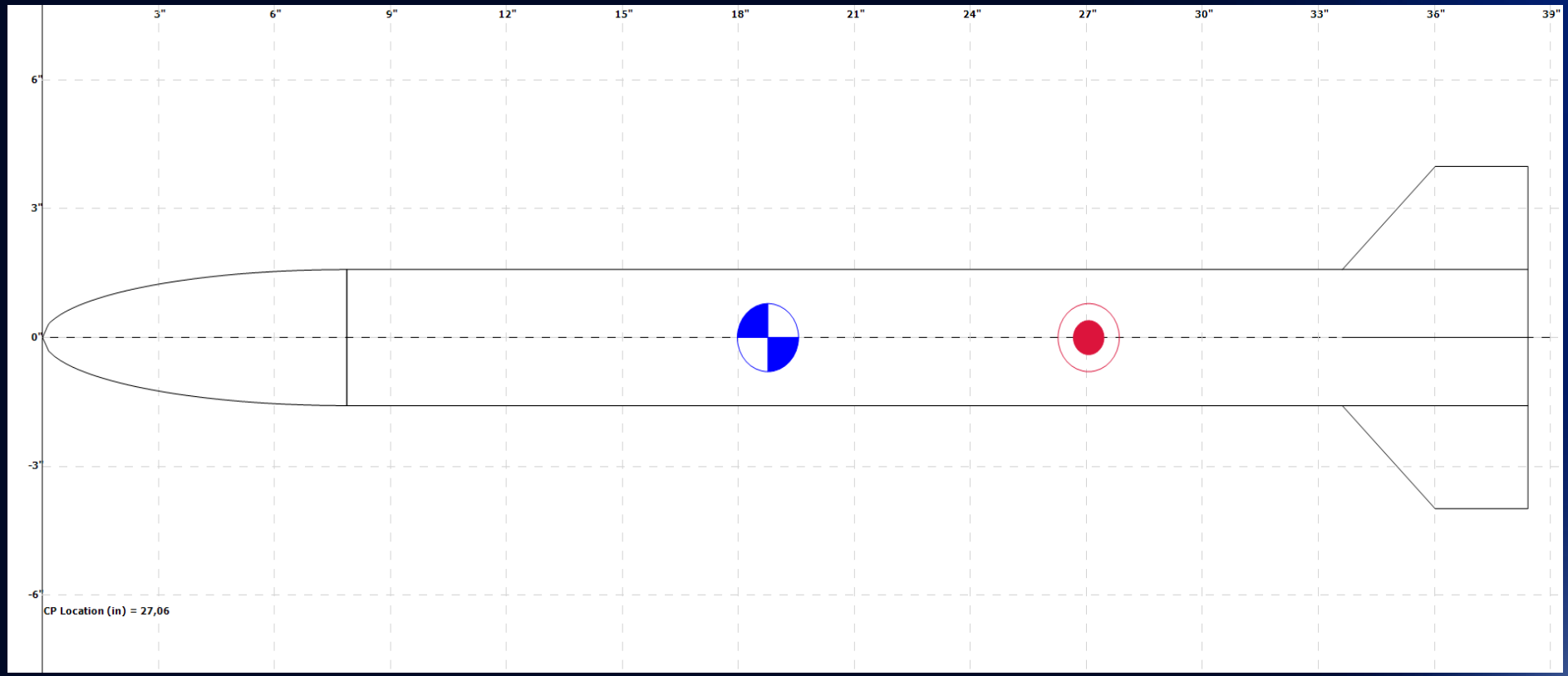
- C_D (subsonic): 0.29
- $C_{z,\alpha}$: 7.609 rad^{-1} = 0.1328 deg^{-1}
- $C_{M,\alpha}$ (initial) = $C_{Z_{alpha}} \cdot \frac{(C_{g_{position}} - C_{p_{position}})}{d}$: -0.345 deg^{-1}





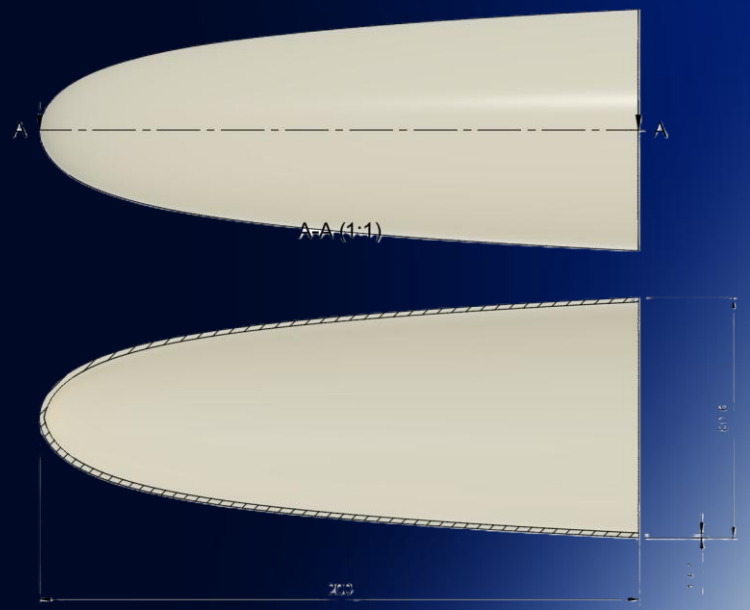


RASAERO



Nose parameters:

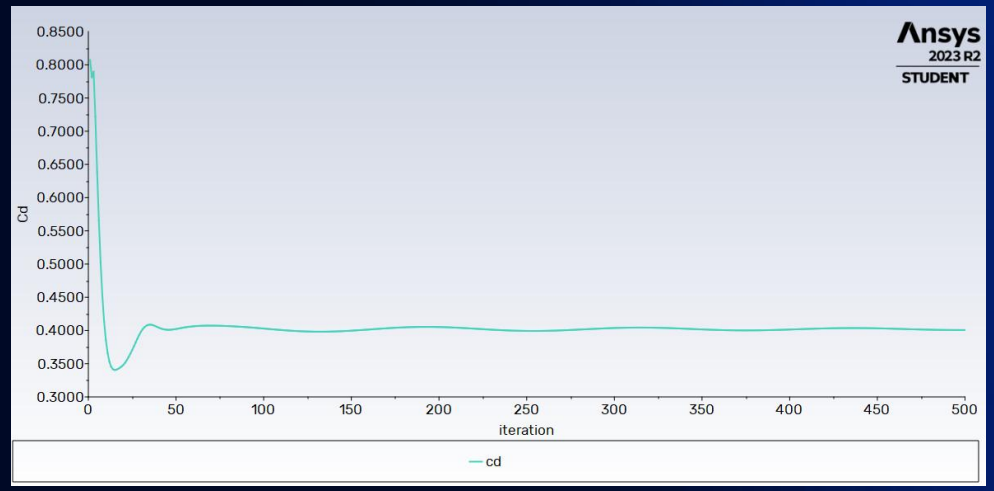
- $C_{Z\alpha_{Nose}} = 2\ rad^{-1}\ (0.0349\ deg^{-1})$
- Major semi-axis measures 200 mm
- Minor semi-axis measures 40 mm

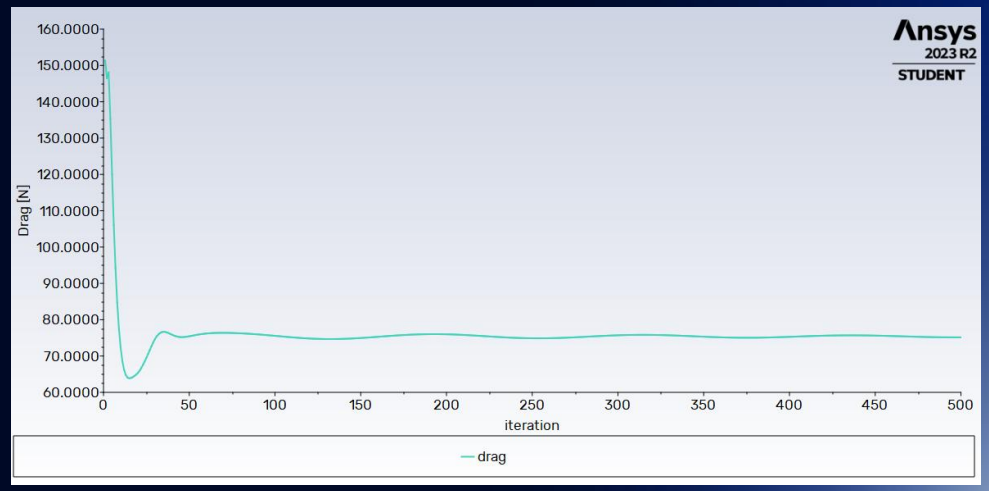


CFD Simulation

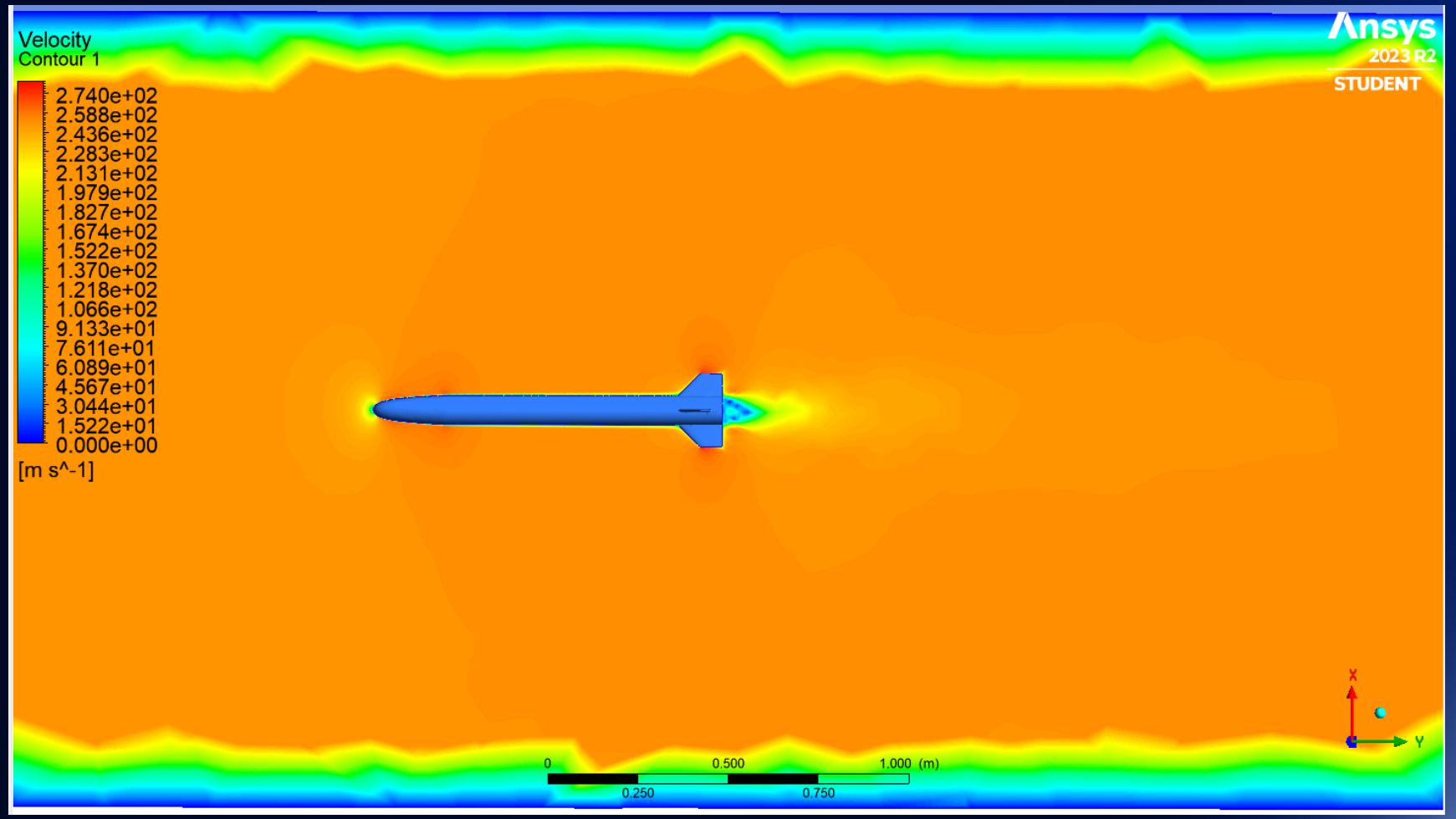
General considerations:

- D = 75 N
- C_D = 0.4

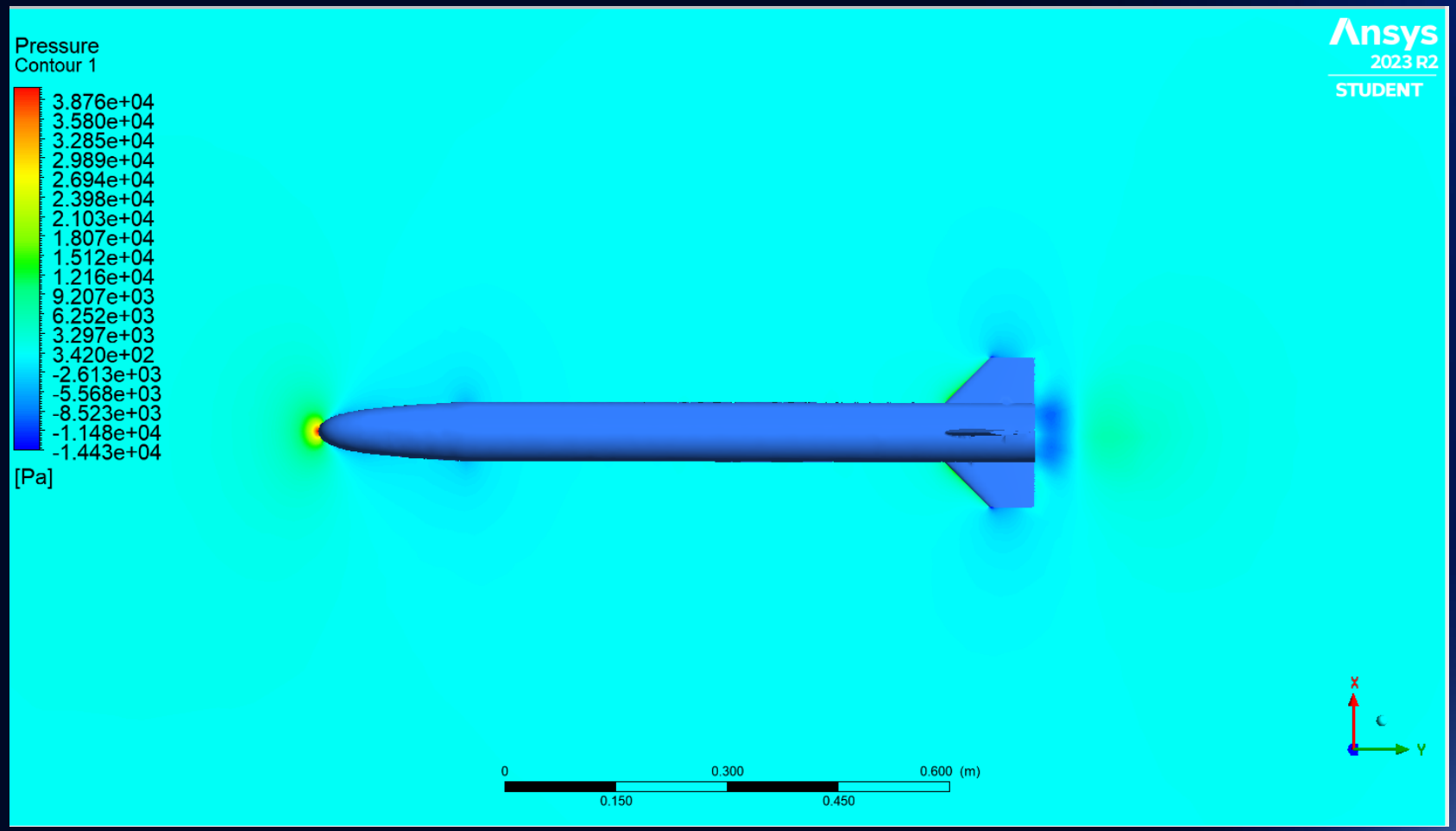




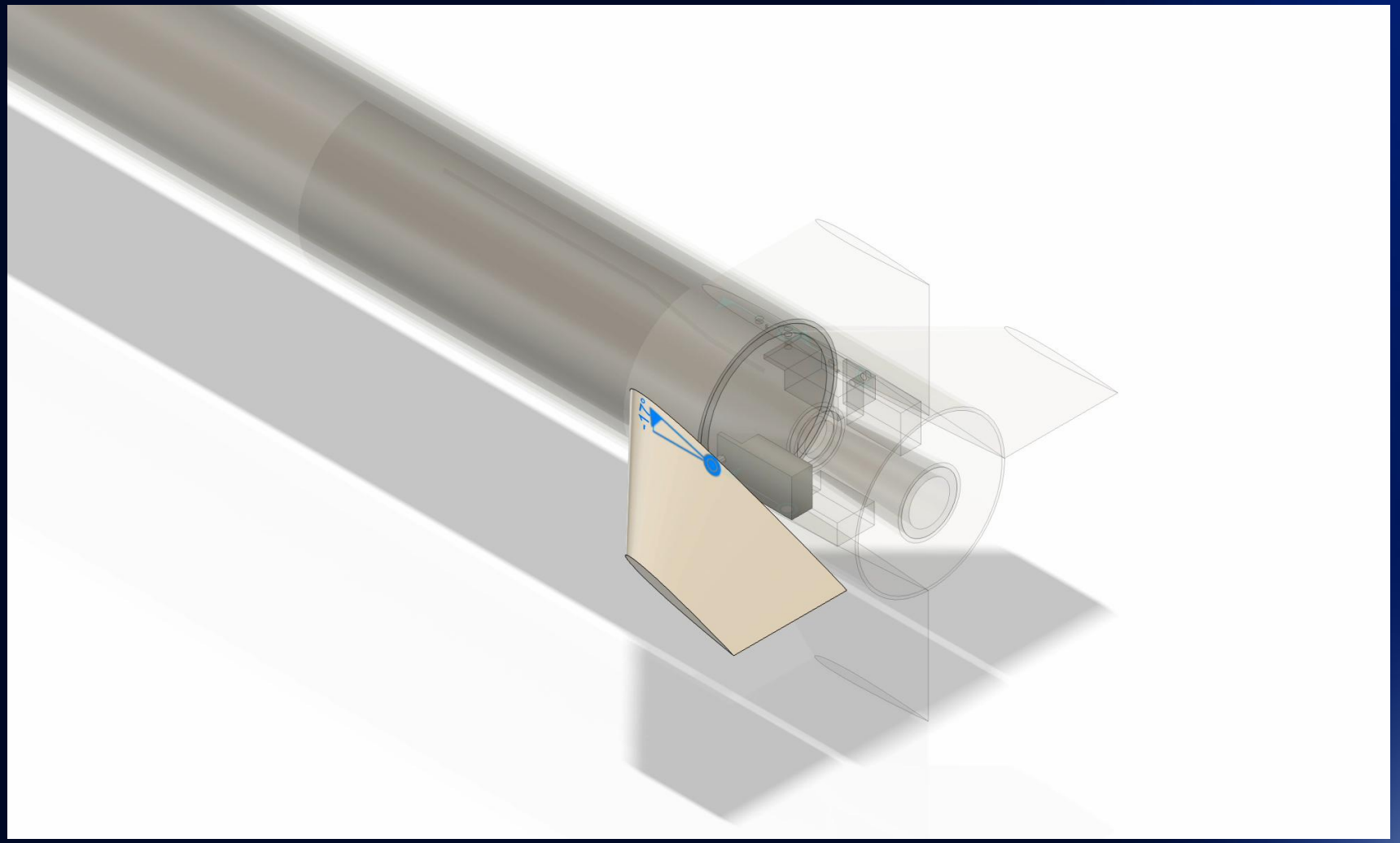
CFD Simulation



CFD Simulation



AERODYNAMICS



STRUCTURAL ANALYSIS

Outer Shell



Material: Alloy 7075, characteristics:

- Young Modulus 70 GPa
- Poisson modulus 0.3
- $\sigma_{cr} = \frac{P_{cr}}{A}; P_{cr} = \frac{\pi^2 EI}{(l_0 MS)^2}$ = first Eulerian Instability load
- $l_0 = \mu l$ = free bending length with $\mu = 2$
- $\tau_{cr} = \frac{1}{\sqrt{3}}\sigma_{\text{cr}}$

Hypothesis:

- Slender beam $\lambda_{\text{lim}} < 100$
- PSE can be applied
- MS = $\sqrt{5}$
- $\sigma_{fatigue} < 100MPa$
- Negligible torque moment

Data:

- Outer diameter 80.6 mm
- Length 1400 mm
- Arm Length 736 mm

Forces

- Compression 3626N
- Shear $nW_{Tot} = 705\ N$
- Bending Moment 520000Nmm

Sizing:

- Compression: $s_c = R \cdot \sigma_{cr} \left(\frac{E}{\sqrt{3(1-\nu^2)}} \right)^{-1} = 0.15\ mm$
- Shear, Jourasky's formula: $\tau = -\frac{T\,S}{I*t}$, where $S = -R_{med}^2 s \sin\theta \rightarrow s_s = 0.35\ mm$
- Bending, Navier's formula: $\sigma_{max} = \frac{M_f}{w_f} = \frac{M_f}{\frac{\pi\left(D^4-(D-2t_m)^4\right)}{32\,D}} \rightarrow s_m = 1.4\ mm \cong 1.5\ mm$

The formula leads to a thickness $s = 1.5\ mm$

Refers to this thickness the von Mises's equivalent stress is $\sigma_{eq} = 18\ MPa$

Outer Shell FEM

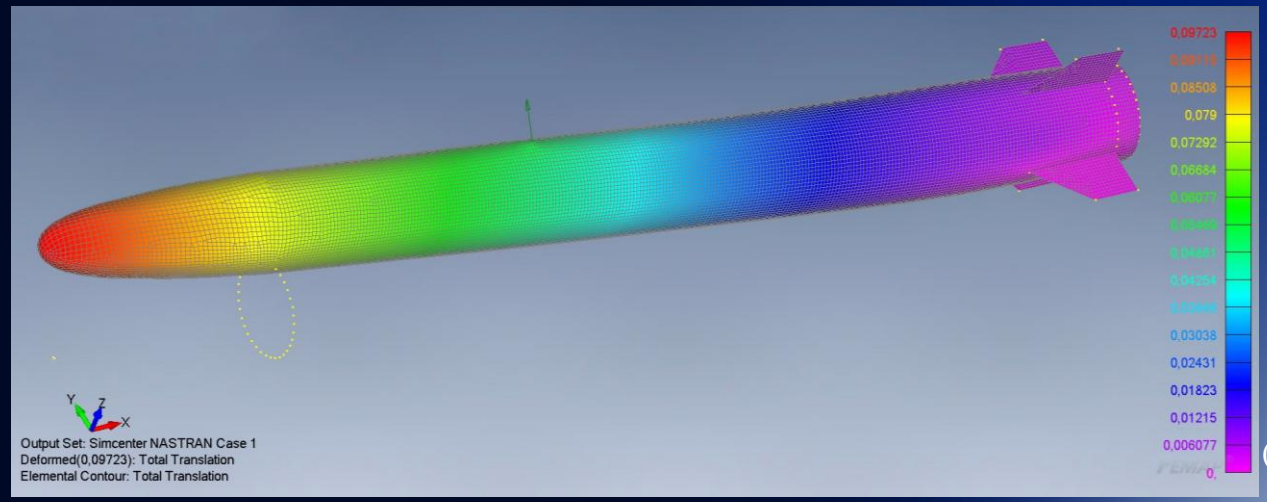
Type of Analysis:

- Maximum Deformation: 0.1 mm
- Von mises stress: 22 MPa

It's considered the entire external strcuture:

- Outer Shell: Alloy 7075
- Separation Ring: Steel AISI 4340
- Nose: Radome structure in pyroceramic
- Wings: Alloy 7075

The missle is modeled as a simply supported beam



(*mm*)

Grain Case & Nozzle



Material AISI 4340, characteristics:

- Young Modulus 175 GPa
- Poisson modulus 0.3
- Compression limit stress 240 Mpa
- Shear Limit stress 156MPa
- Expansion coefficient $6.6E-6$

Hypotesis:

- Thermal Gradient 600 K
- MS = $\sqrt{5}$
- Pressure 7 bar

Barlow's equation combained with thermal stress:

$$\sigma_y = MS \cdot \left(\frac{p \cdot r}{t} + E\alpha\nabla T\right) \rightarrow \begin{cases} t_{case} = 1.8\ mm \\ t_{nozzle} = 1.5\ mm \end{cases}$$

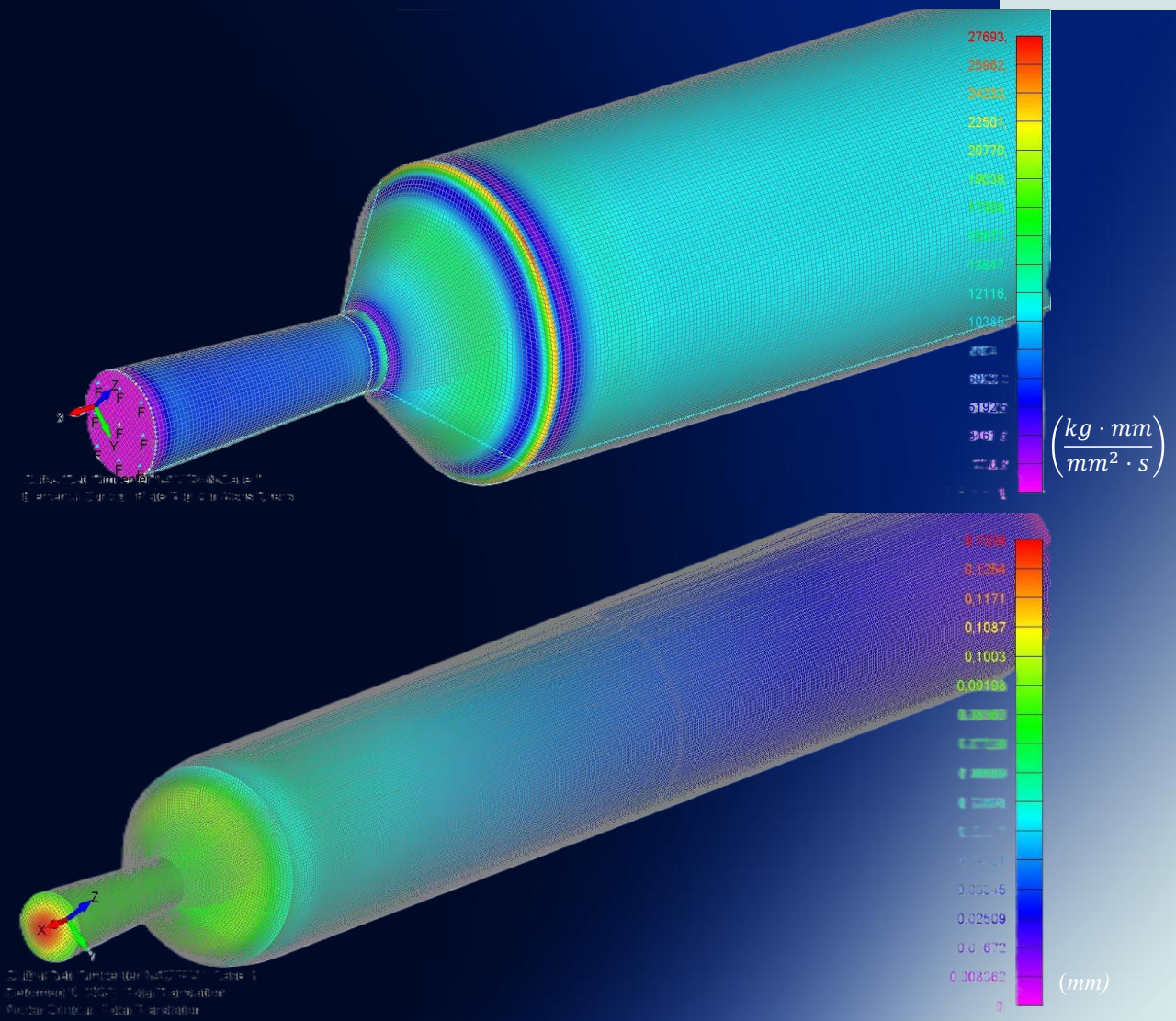
Chamber FEM

Analysis Results:

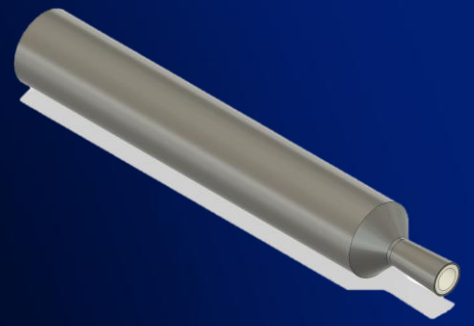
- Maximum Deformation: 0.1 mm
- Pressure stress: 18-19 MPa

Type of Analysis:

- Double Fixed
- Single Fixed



Thermal Management: Liner



Material porus graphite, characteristics:

- Conducivity 0.538 $^{W}/_{K \cdot m}$

Other characteristics:

- $h_{air} = 150\ ^{W}/_{K \cdot m^2}$
- $h_{gas} = 50000\ ^{W}/_{K \cdot m^2}$, estimated with Bartz's formula
- $k_{steel} = 17\ ^{W}/_{K \cdot m}$
- $k_{alloy} = 200\ ^{W}/_{K \cdot m}$

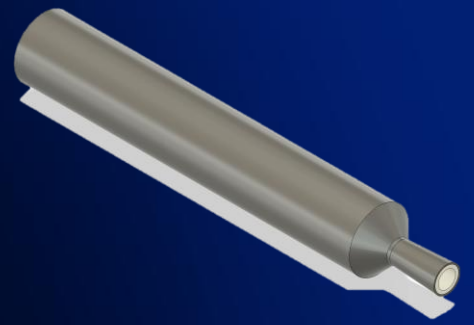
Hypotesis:

- Inner Layer temperature 3000 K
- Outer Layer temperature 855 K

It's increased t_{liner} until the $\Delta T > 600$:

$$R = \frac{1}{\left(\frac{1}{h_{air}} + \frac{1}{h_{gas}} + \frac{t_{case}}{k_{steel}} + \frac{t_{liner}}{k_{ceramic}} + \frac{t_{alloy}}{k_{alloy}}\right)} \rightarrow t_{liner} = 3.9\ mm$$

Thermal Managment: Heat Sink



Material copper, characteristics:

- Conducivity 360 $^{W}/_{K \cdot m}$

Iterations leads to a Thickness of 10 mm. The relative thermal conductance is k/δ = 36000 $^{W}/_{K \cdot m^2}$ >> the graphite one.

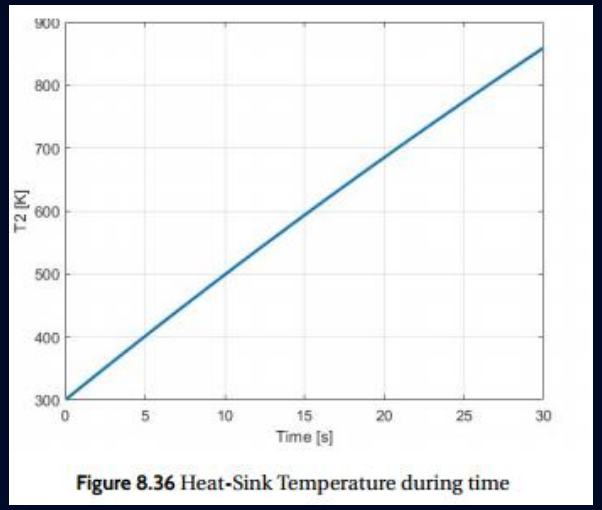


Figure 8.36 Heat-Sink Temperature during time

Hypotesis:

- Heat Sink initial temperature 300 K
- Copper Melting temperature 1360 K
- Adiabatic wall temperature 3440 K
- Firing Time 30 s

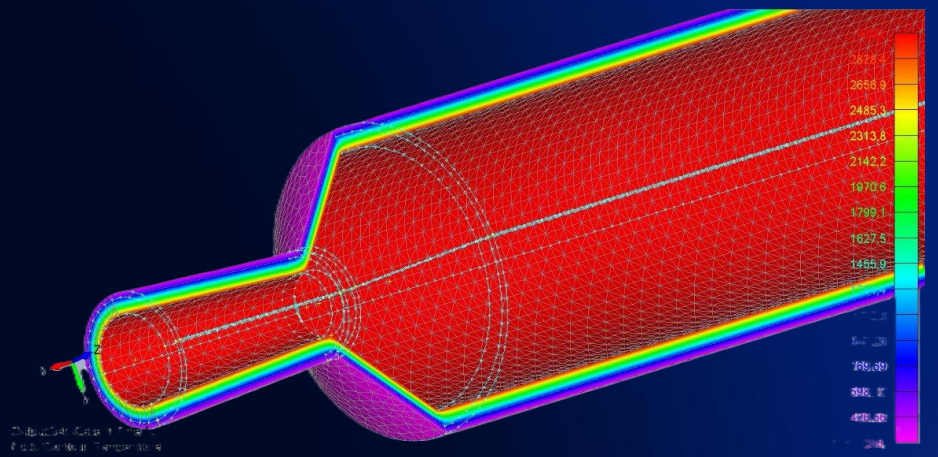
Heat Sink final temperature 858 K < the melting one: the sizing is correct

Chamber FEM 3D

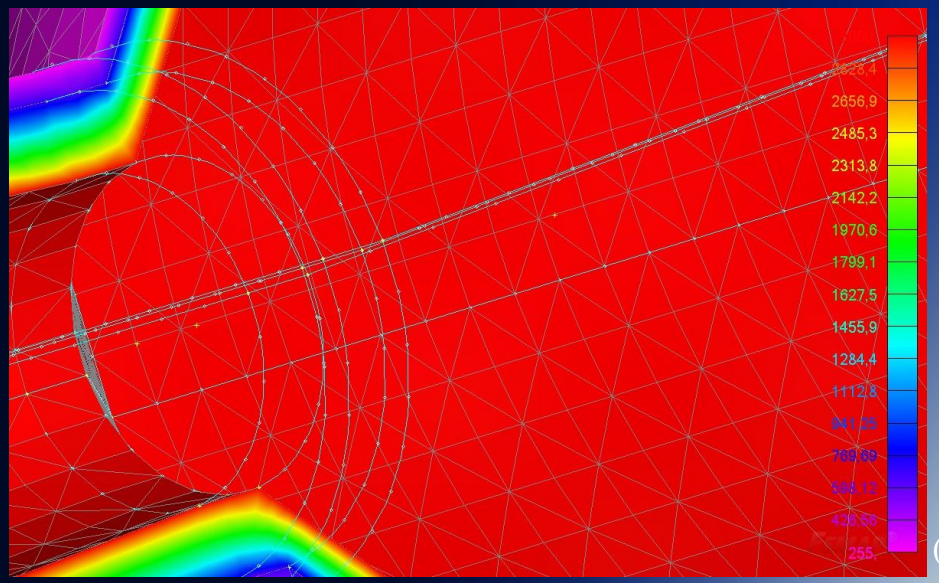
Type of Analysis:
- Temperature Evolution
- Thermal stresses

Precision:
- Cap tank and glued contact
- Convective gas flux
- Heat flux along every layer

Assumptions:
- Extarnal liner surface T = 3000 K
- Throat Internal liner surface T = 598 K
- Remain Nozzle internal liner surface T = 700 K
- Case internal liner surface T = 855 K
- External steel surface T = 255 K



(K)



(K)

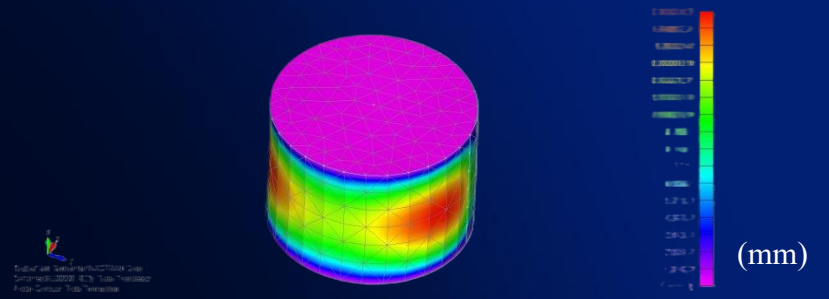
Thermal breakers

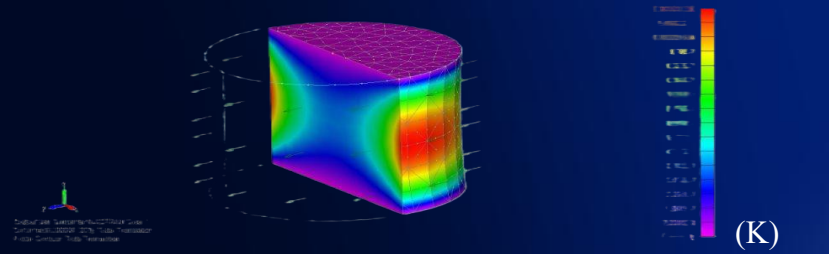
Material: porous graphite
The surfaces are glued thanks to a polyammide glue

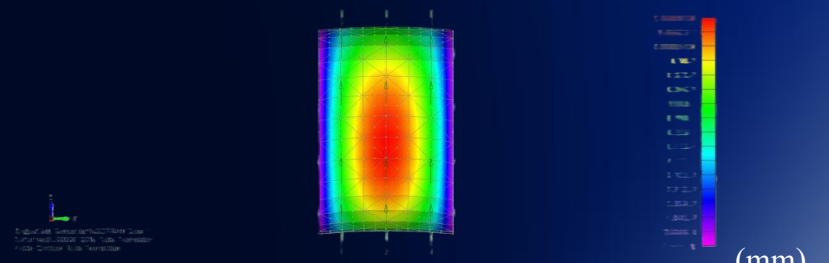
Sizing
- Trust force and shear force
- Thermal stresses

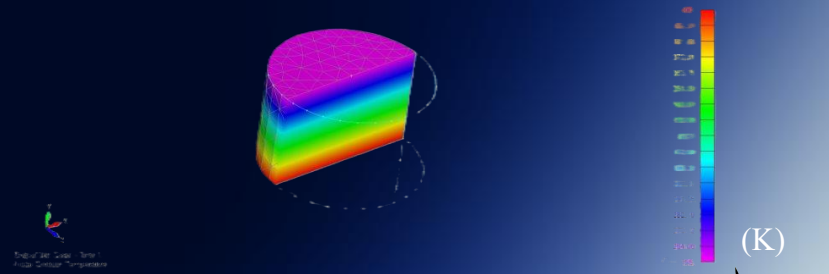
Two tipes of TB:
- Horizontal: both forces are considered;
- Vertical: only the trust is considered, cause the shear is absorbed by the structures (Outer Shell and Case)

Results
- H = 1.2 mm, wires space
- D = 1 mm Compromise between strength and ease of fabrication









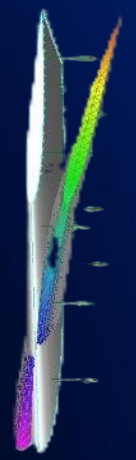
Wings FEM

Material: alloy 7075-T6

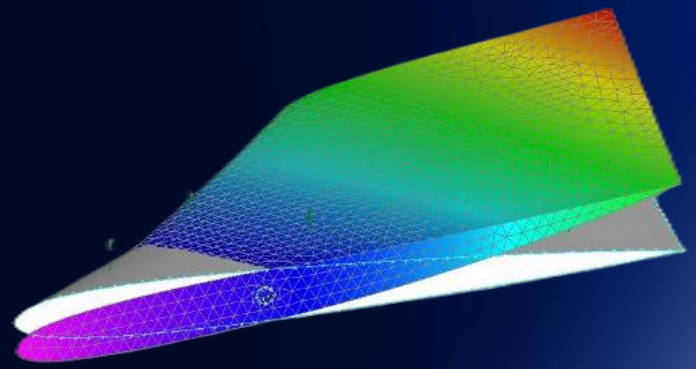
Criteria: f < 5% S; verify cause f = 0.05 mm

Assumptions:

- Constraint: Hinge hole
- Force apply on the surface
- The gravity and Feed rate accelerations were considered



(mm)

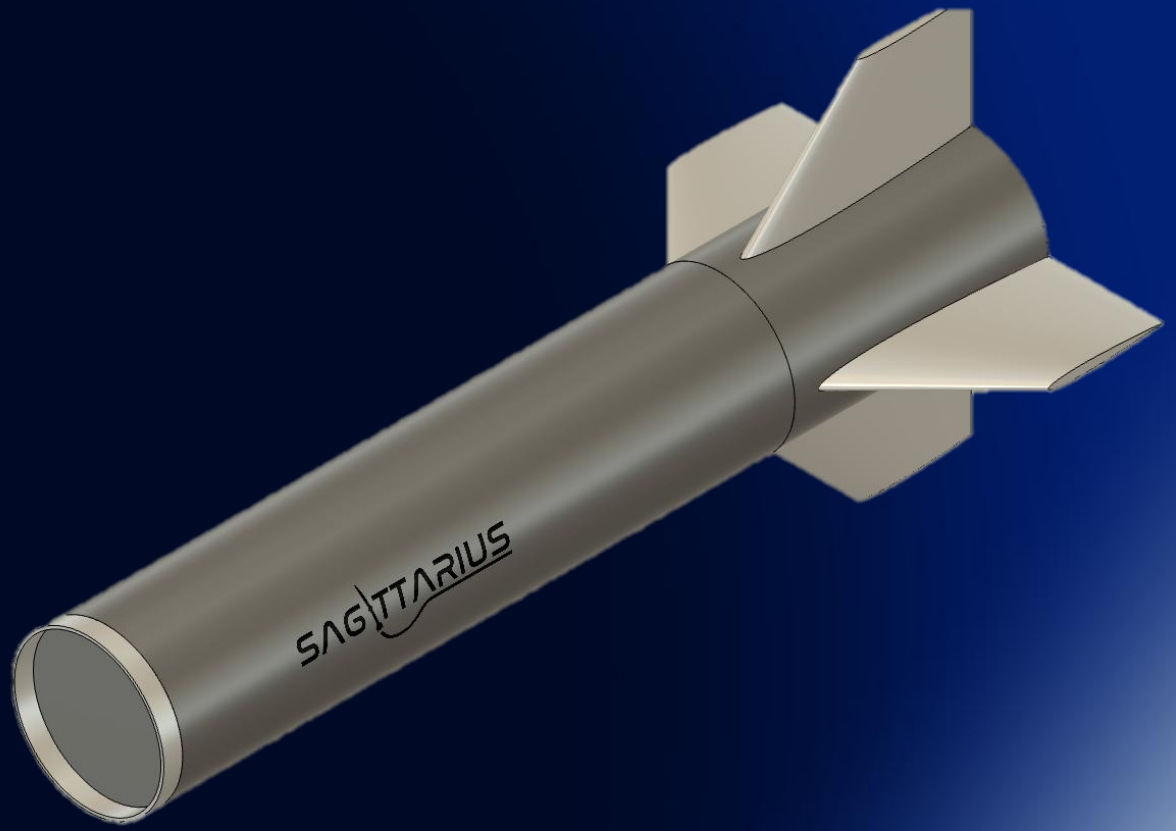


(mm)

FIRST STAGE

OVERVIEW

- Length: 456 mm
- Caliber: 80.6 mm
- Number of fins: 4
- Mass with full propellant loaded: 4811 g
- Mass without propellant: 3523 g
- Thrust: 3626 N



Grain Geometry

- The geometry chosen for the grain is a star-shaped geometry
- The reason behind this choice is that this geometry leads to a Thrust profile that gives maximum Thrust in the launch phase, that is preferable for a booster
- A star-shaped geometry minimize the erosive burning
- A star shaped geometry holds the combustion far from the wall, so there is no need to add a liner like in the Sustain



Grain Sizing

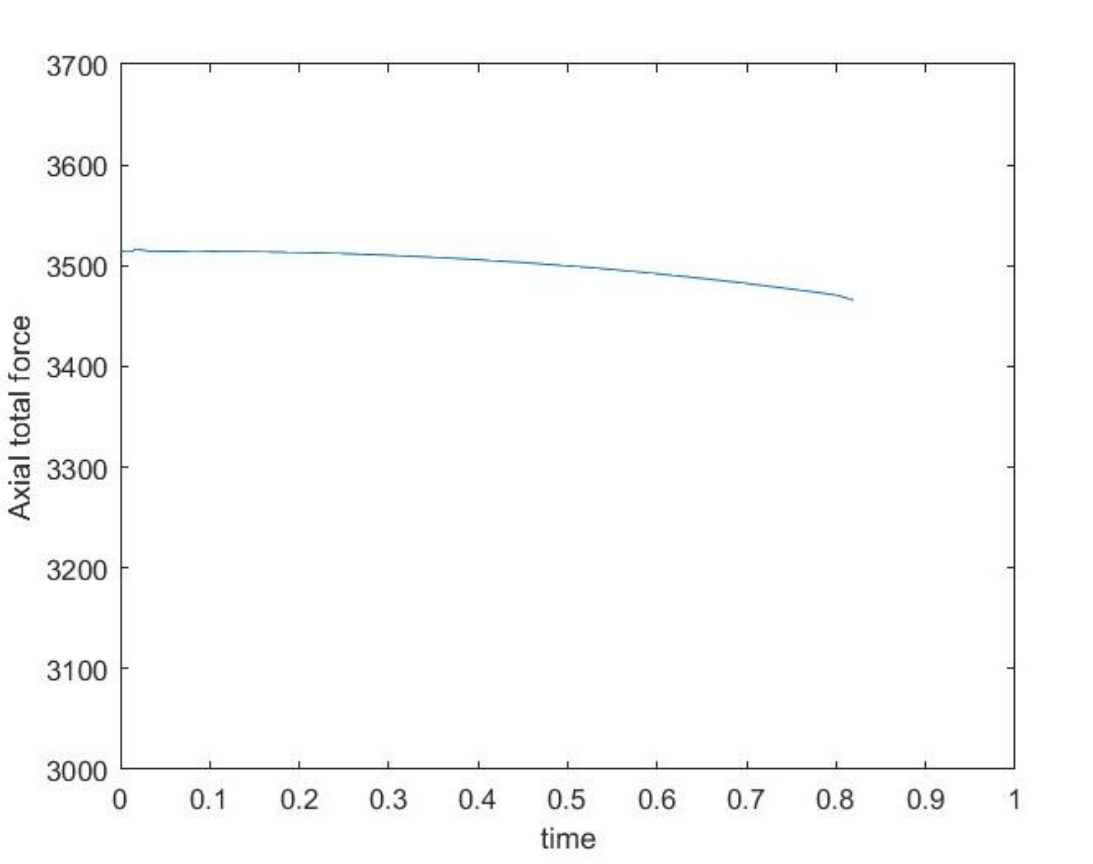
Characteristics:

- Propellant: HTPB 1912
- $V_{grain} = \frac{m_p}{\rho} = (\frac{\pi}{4} \cdot d_{grain}^2 \cdot L_{grain}) - (A_{door} \cdot L_{grain})$
- $A_{wet} = \frac{\dot{m}}{\rho \cdot \dot{r}} = (0.9 \cdot d_{grain} \cdot L_{grain} \cdot \pi)$

V_{grain} is the total volume of the propellant, A_{door} is the section of the star, while A_{wet} is the surface of the grain in contact with the flame during combustion, and it slightly varies over time.

- Case diameter: 76.2 mm
- Case length: 322.1 mm

Thrust Profile



- Thrust (T) = $\dot{m} \cdot c$
- Constant Thrust at 3626 N
- Small decrease of the net axial force because of the Drag (D)

Input From CEA

- Expansion ratio: 4.75
- Specific Impulse: 200.08 s (corrected with 0.9 factor)
- Thrust coefficient (c_f): 1.507
- Characteristic velocity (c^*): 1532 $^m/_s$
- Equivalent exhaust velocity (c): 2308.72 $^m/_s$

Propellant Flow Rate

The calculation of the propellant flow rate is crucial to determine the performances of the booster. This calculation has been made thanks to the following parameters:

- $p_{chamber}$ (p_0)
- Critical area (A_{cr})
- Gamma function $(f(\gamma))$
- Specific gas constant (R)
- Stagnation Temperature (T_0)

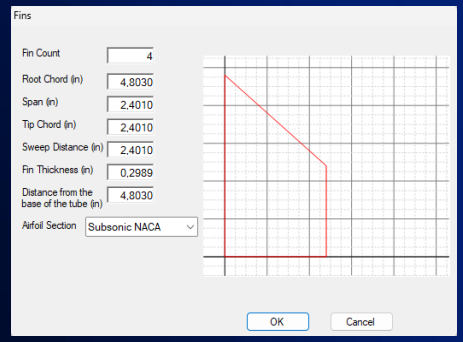
$$\dot{m} = \frac{p_0 \cdot A_{cr} \cdot f(\gamma)}{\sqrt{R \cdot T_0}} = 1570.6 \ {}^{g}/_{s}$$

NOZZLE

- Inlet diameter: 76.2 mm
- Throat diameter: 32 mm
- Exit diameter: 70 mm
- Throat length: 4 mm
- Nozzle length: 133.9 mm
- Divergent angle: 10°
- Convergent angle: 45°
- The converging angle is assumed to be 45°, to avoid issues related to excessive thermal heating
- The divergent angle is very small to avoid performance losses due to non-axial outflow; however, it cannot be too small because the flow in a very long duct would experience losses due to the boundary layer.



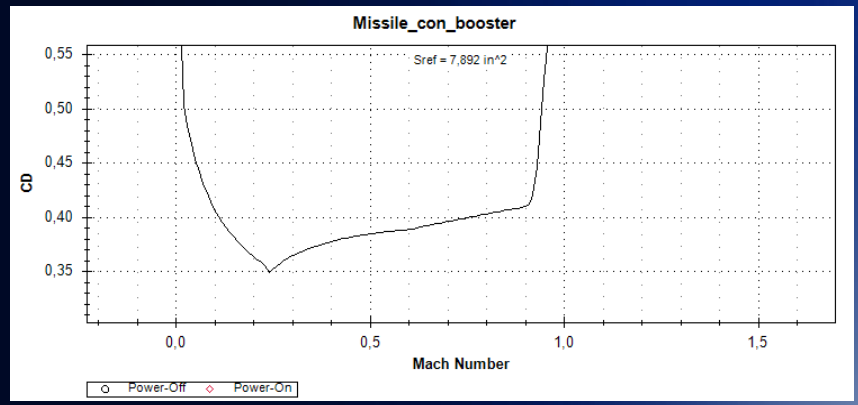
AERODYNAMICS



Wing characteristic, same of II stage.

General characteristics (I + II stage):

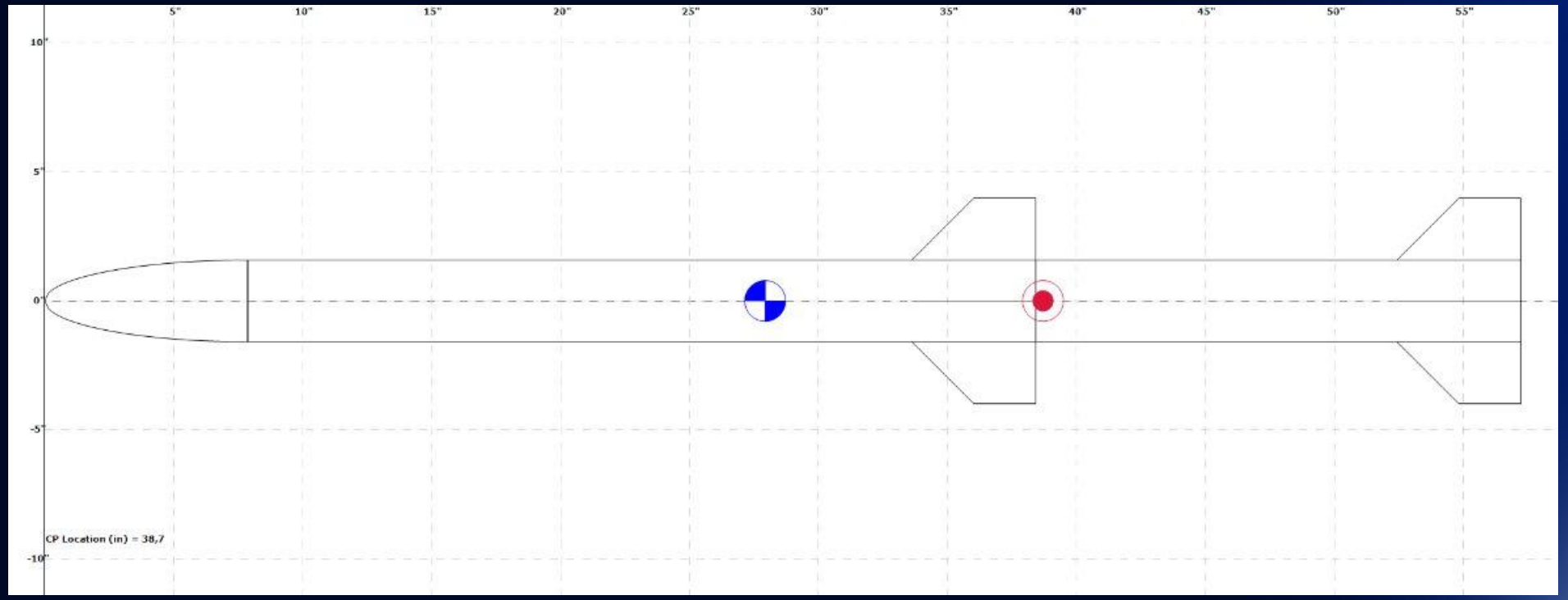
- C_p = 983.00 mm
- C_g (*initial*) = 709.41 mm
- C_g (*final*) = 625.5 mm



RASAero data:

- C_D = 0.4
- $C_{Z,\alpha}$= 13.4 rad^{-1}= 0.234 deg^{-1}

AERODYNAMICS



STRUCTURAL ANALYSIS

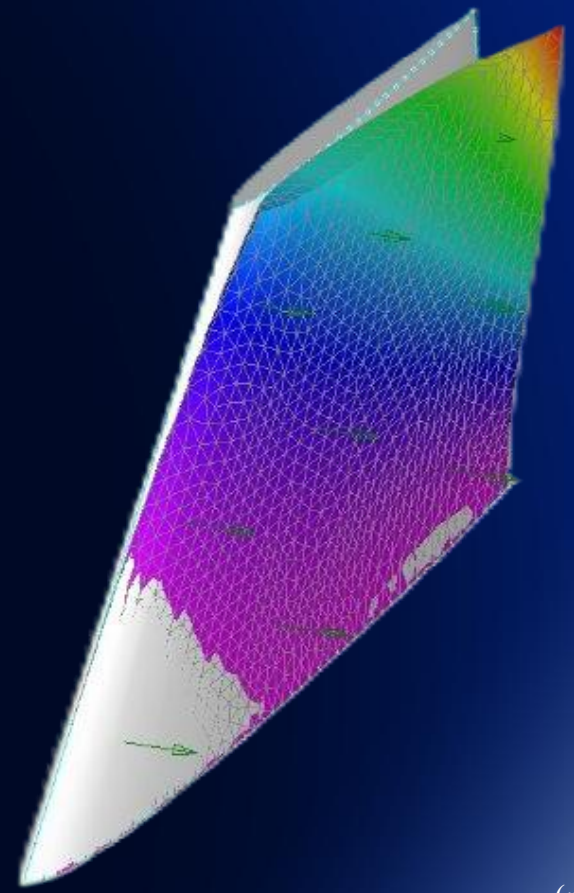
STRUCTURAL ANALYSIS

Characteristic:

- No Outer Shell
- Grain Case thickness 2.2 mm
- Nozzle thickness 2.2 mm
- No thermal cooling

Hypotesis:

- p =30 bar
- Short time use
- Same approach used for second stage
- Focus on FEM constrained wings with the same criteria seen for second stage; verify



(mm)

OFF-DESIGN SCENARIOS

Target conditions:

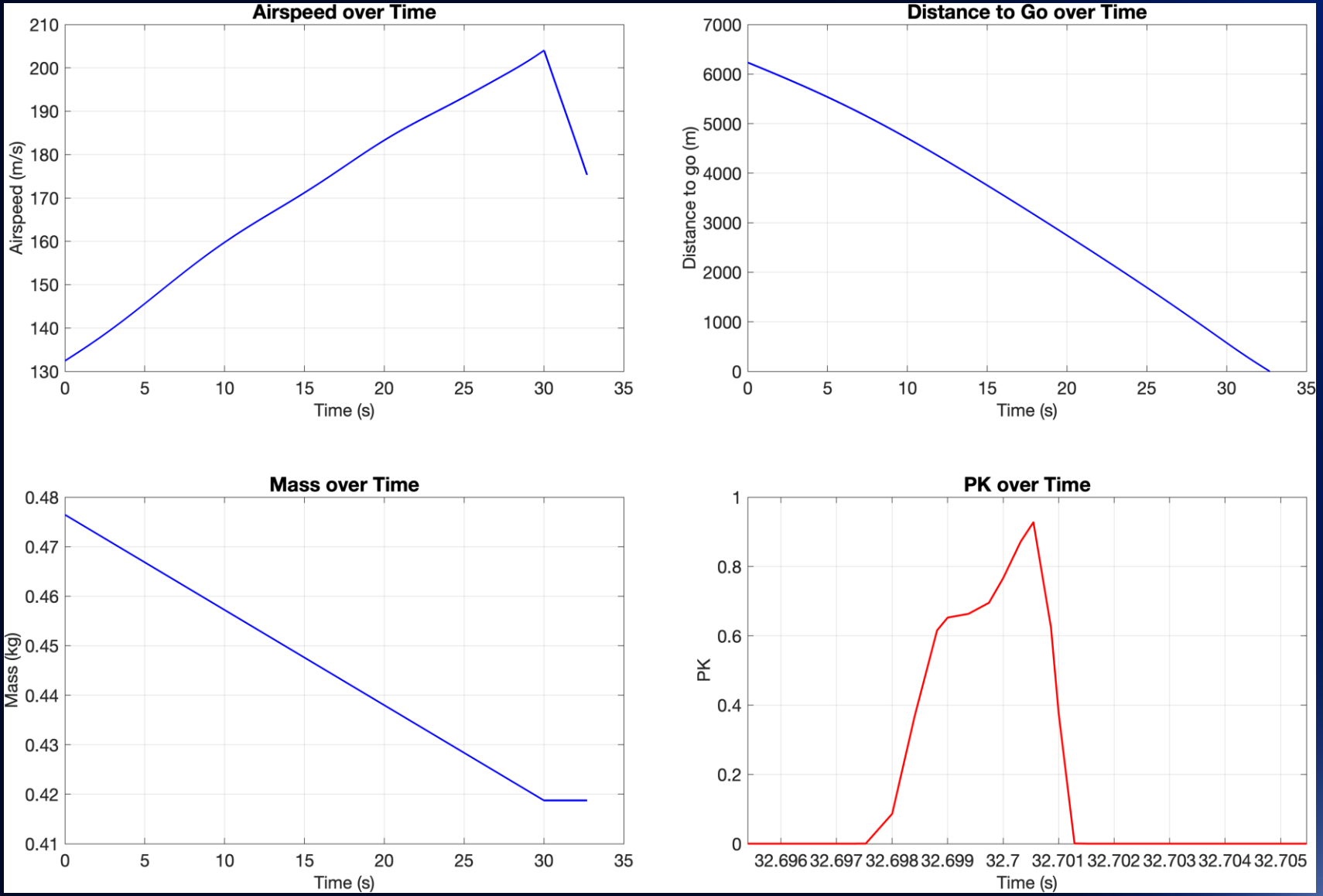
X(0) = 3000 m

Y(0) = 5500 m

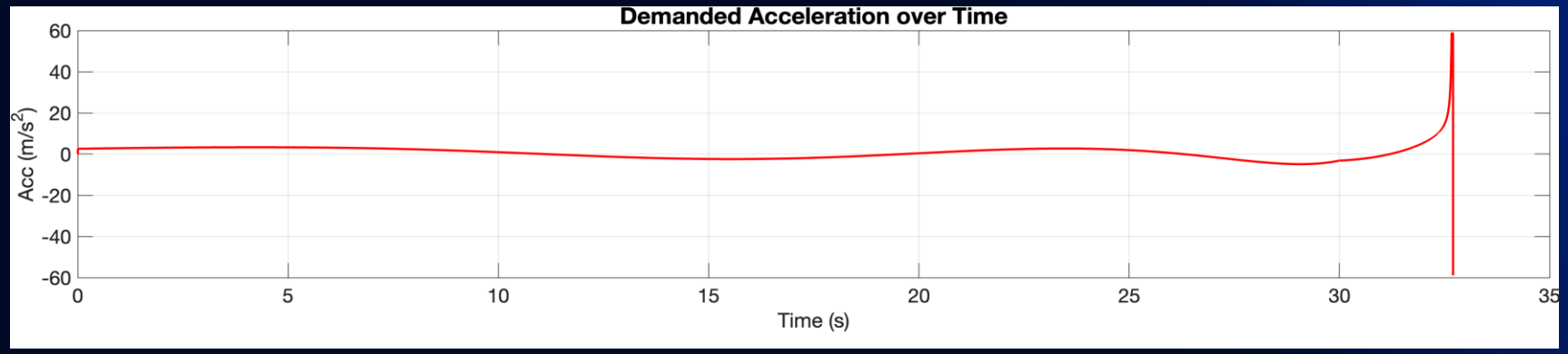
v = 60 $^{m}/_{s}$

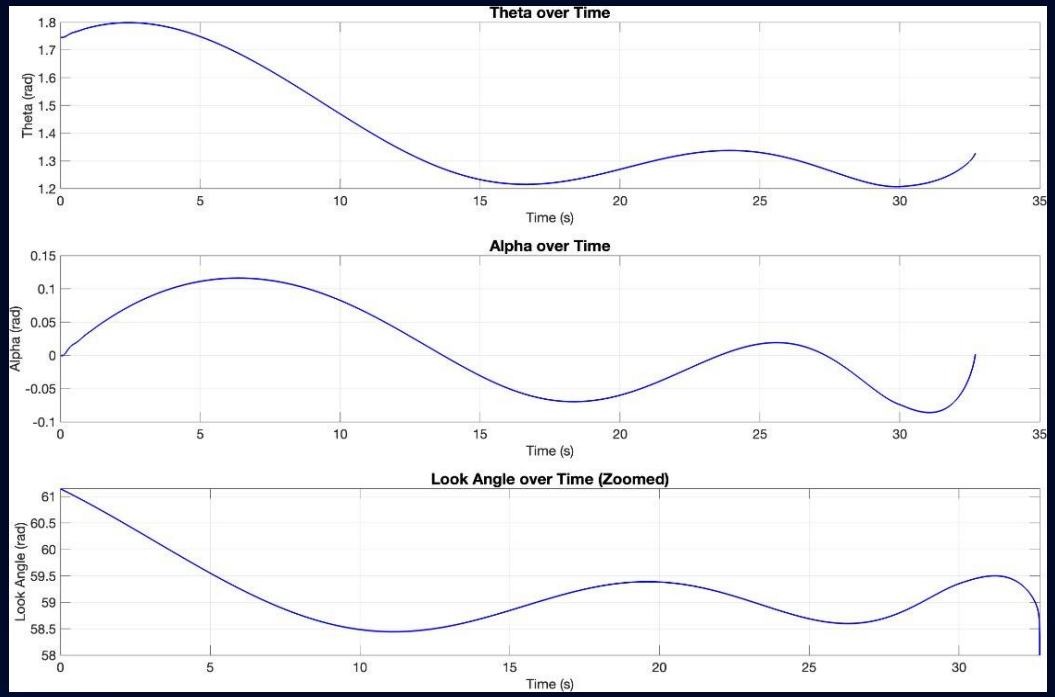


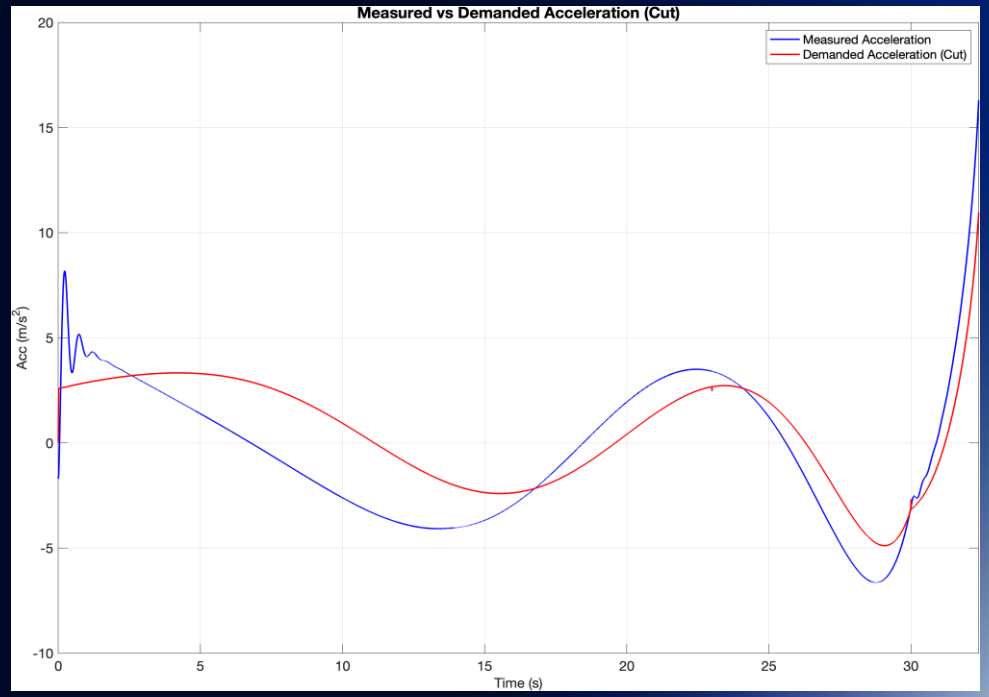
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







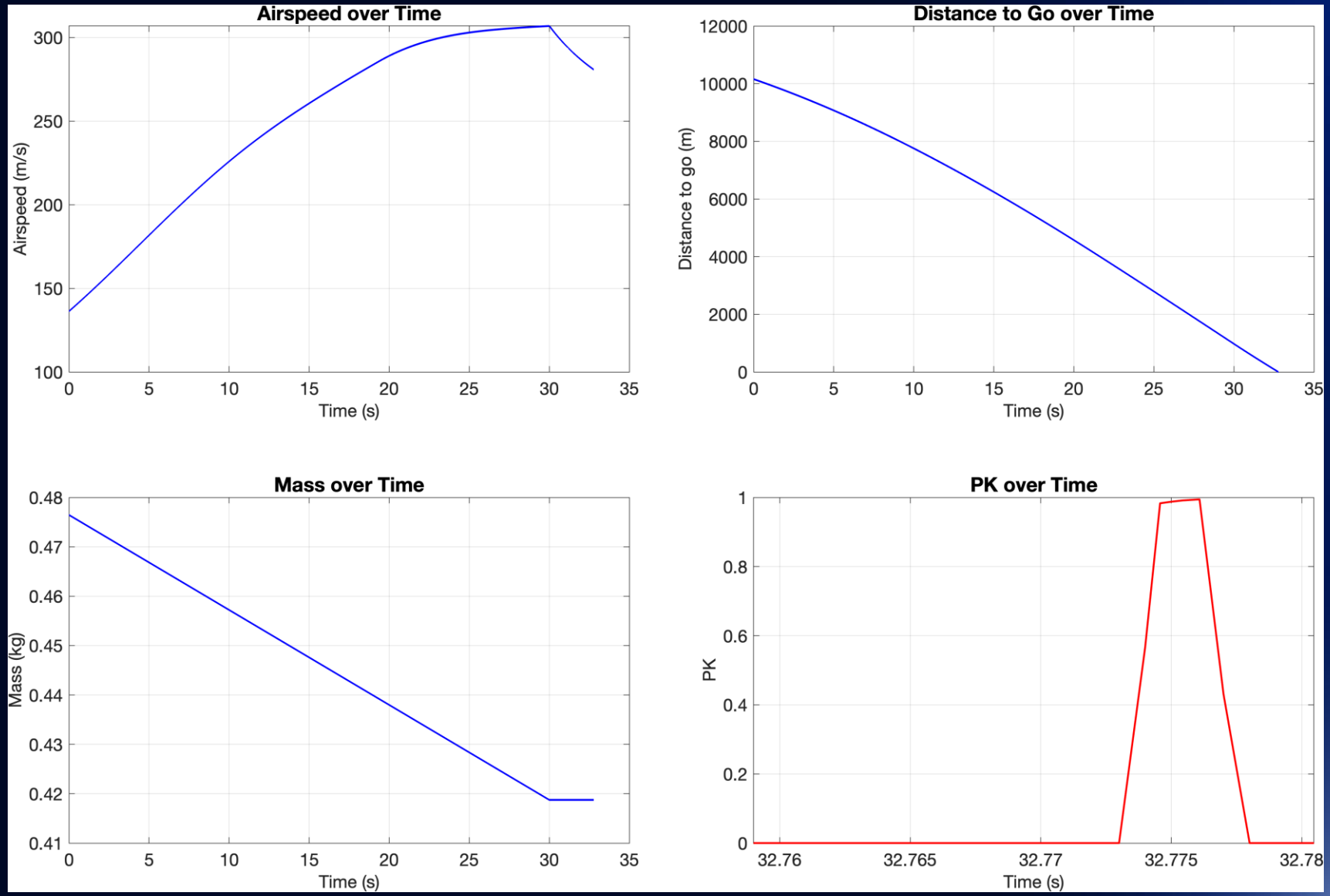
Simulation Results

Target conditions:

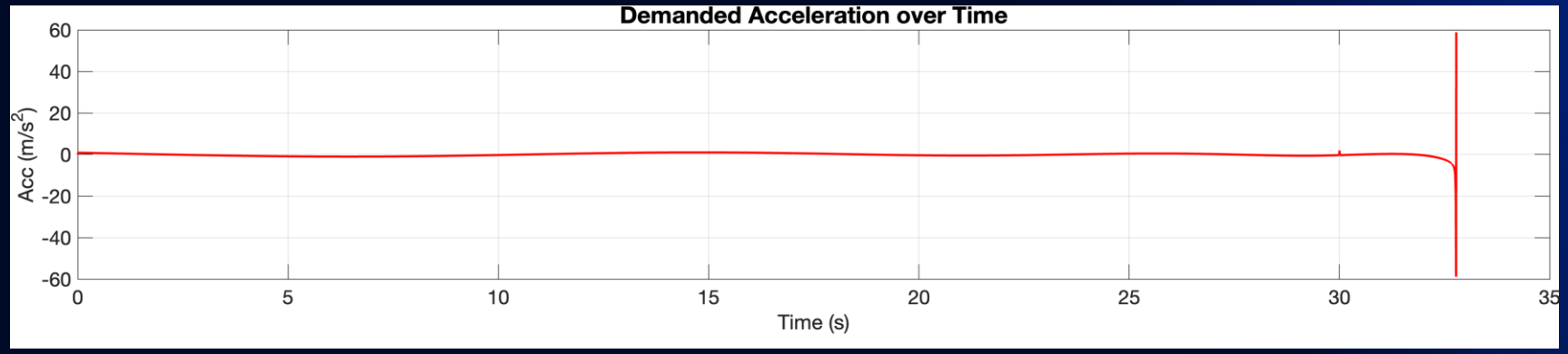
X(0) = 10000 m
Y(0) = 2000 m
v = 60 $^{m}/_{s}$

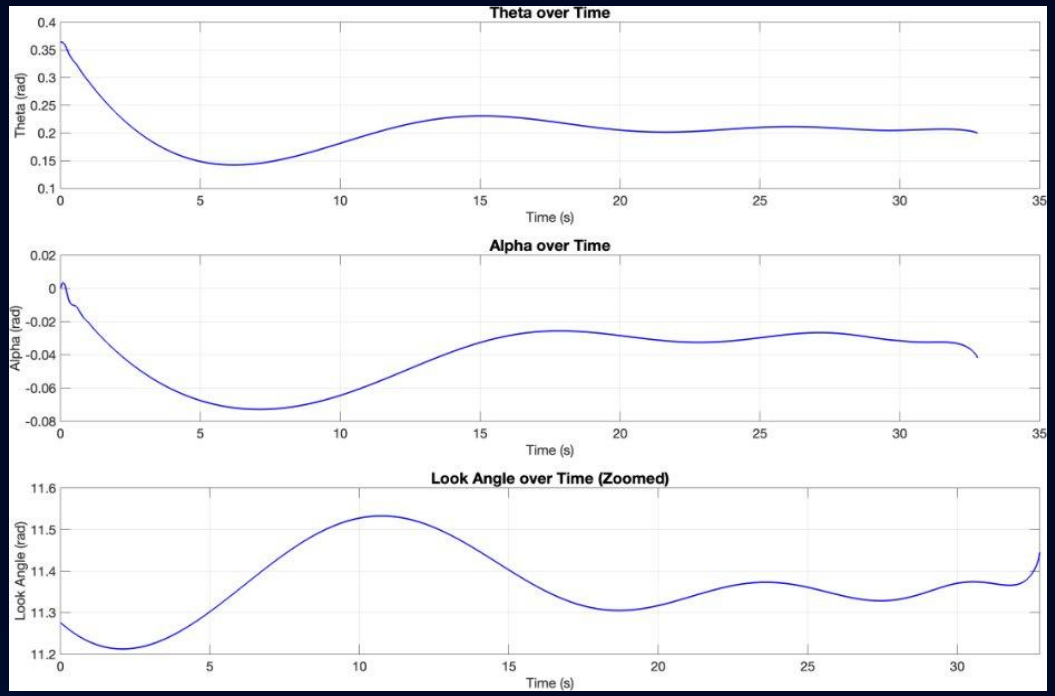


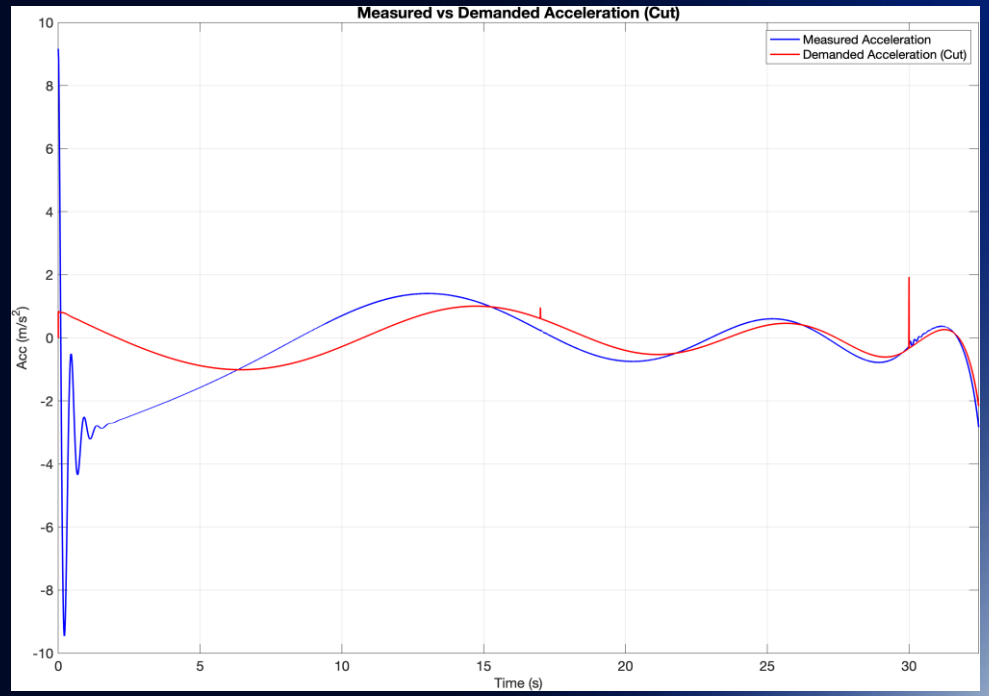
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







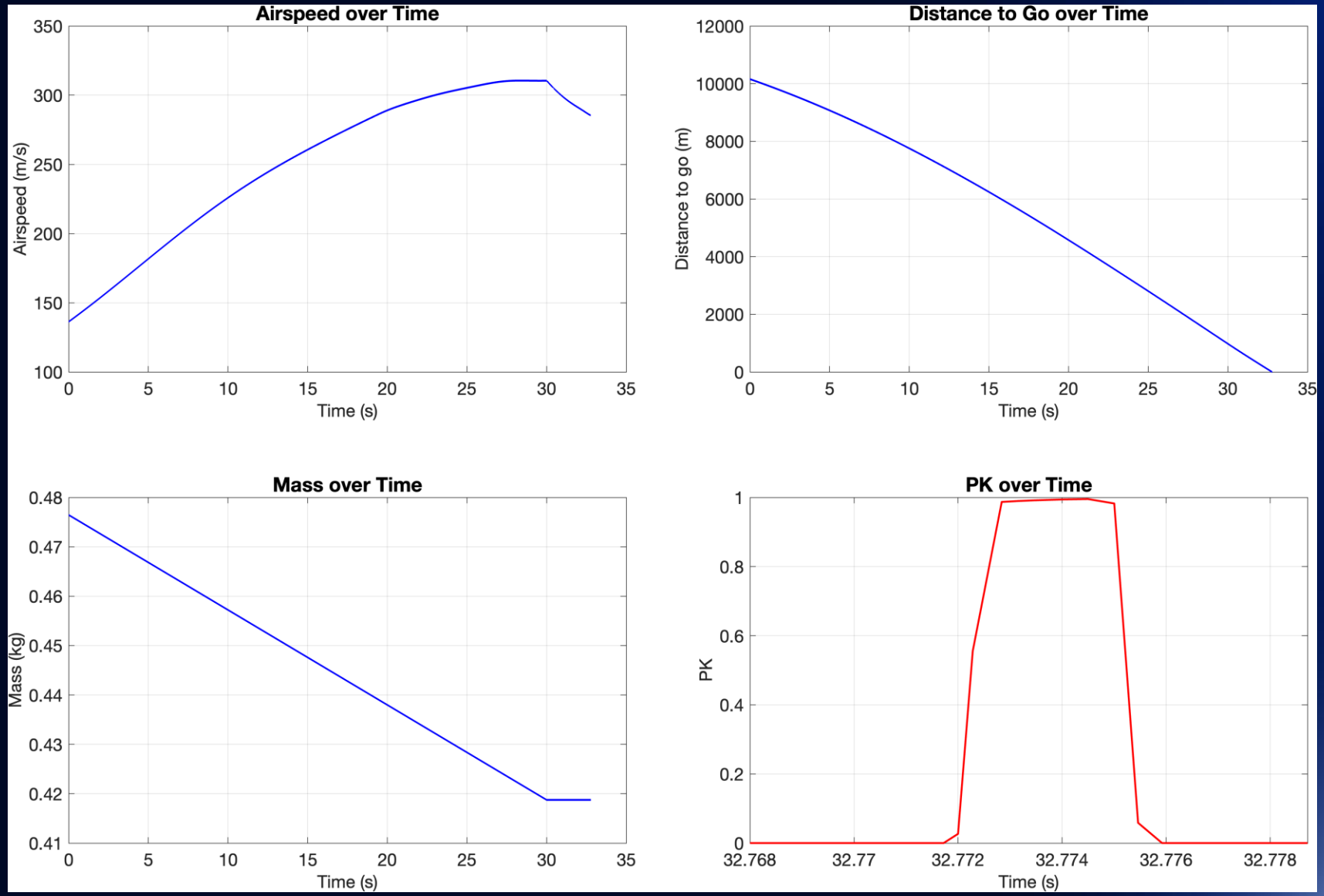
Simulation Results

Target conditions:

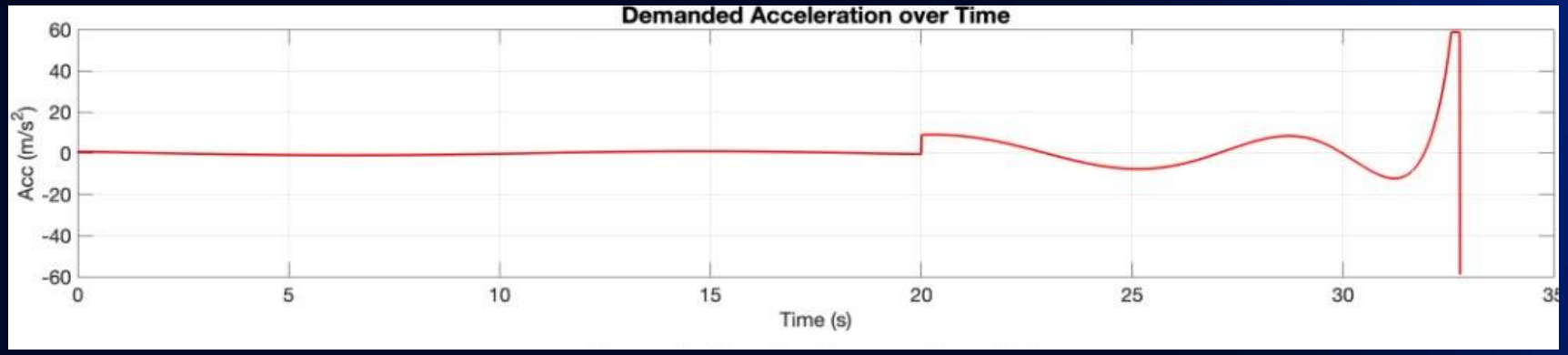
X(0) = 10000 m
Y(0) = 2000 m
v = 60 $^{m}/_{s}$
30 degrees target dive at t = 20s

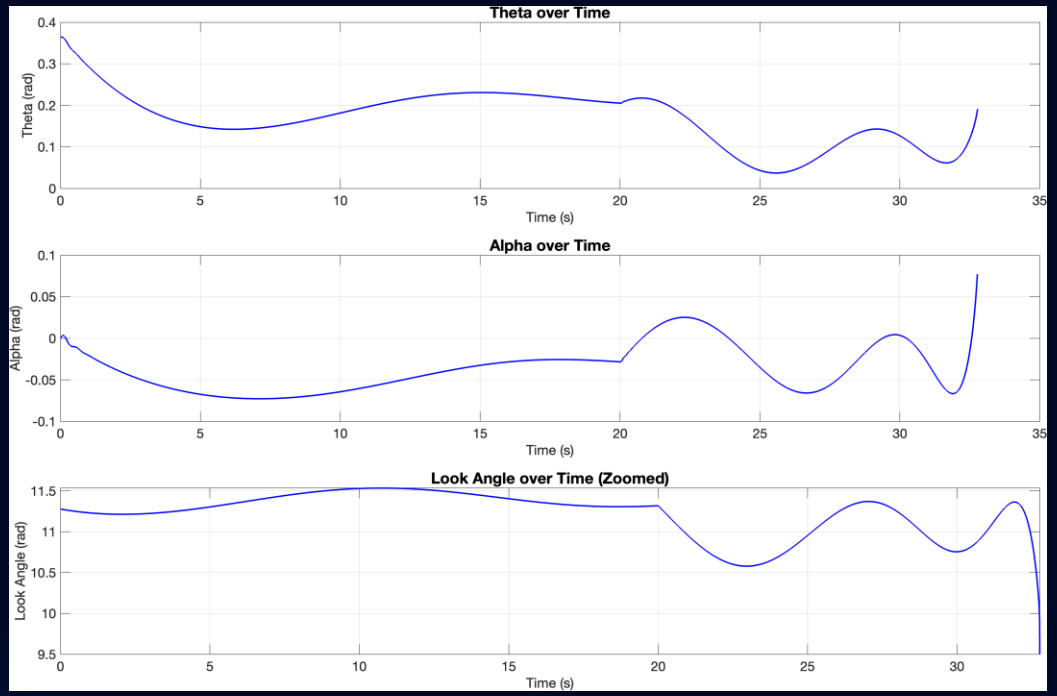


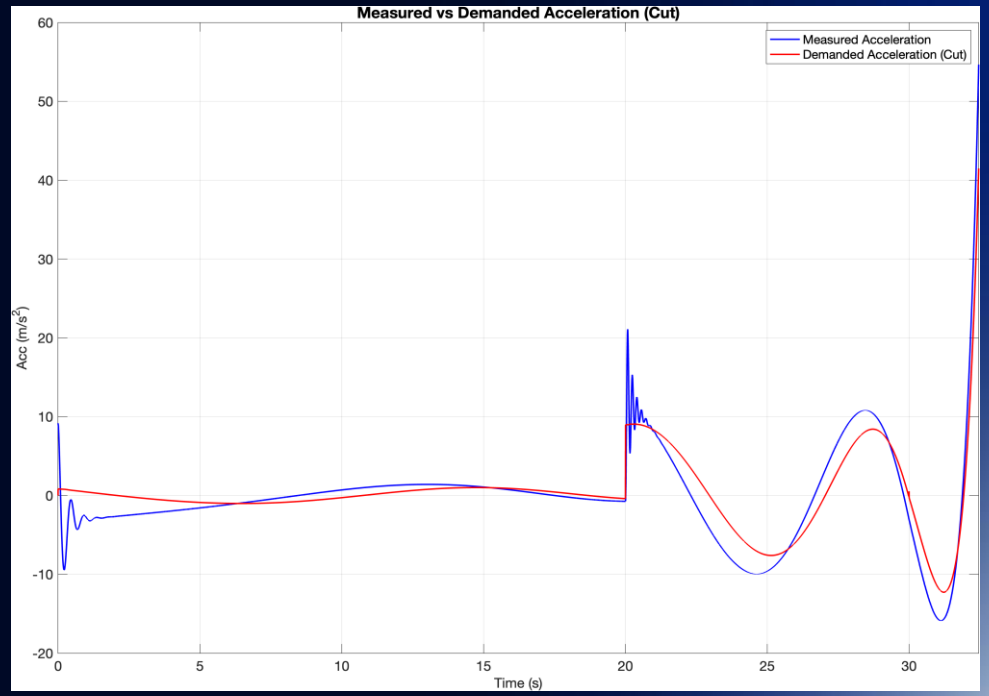
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







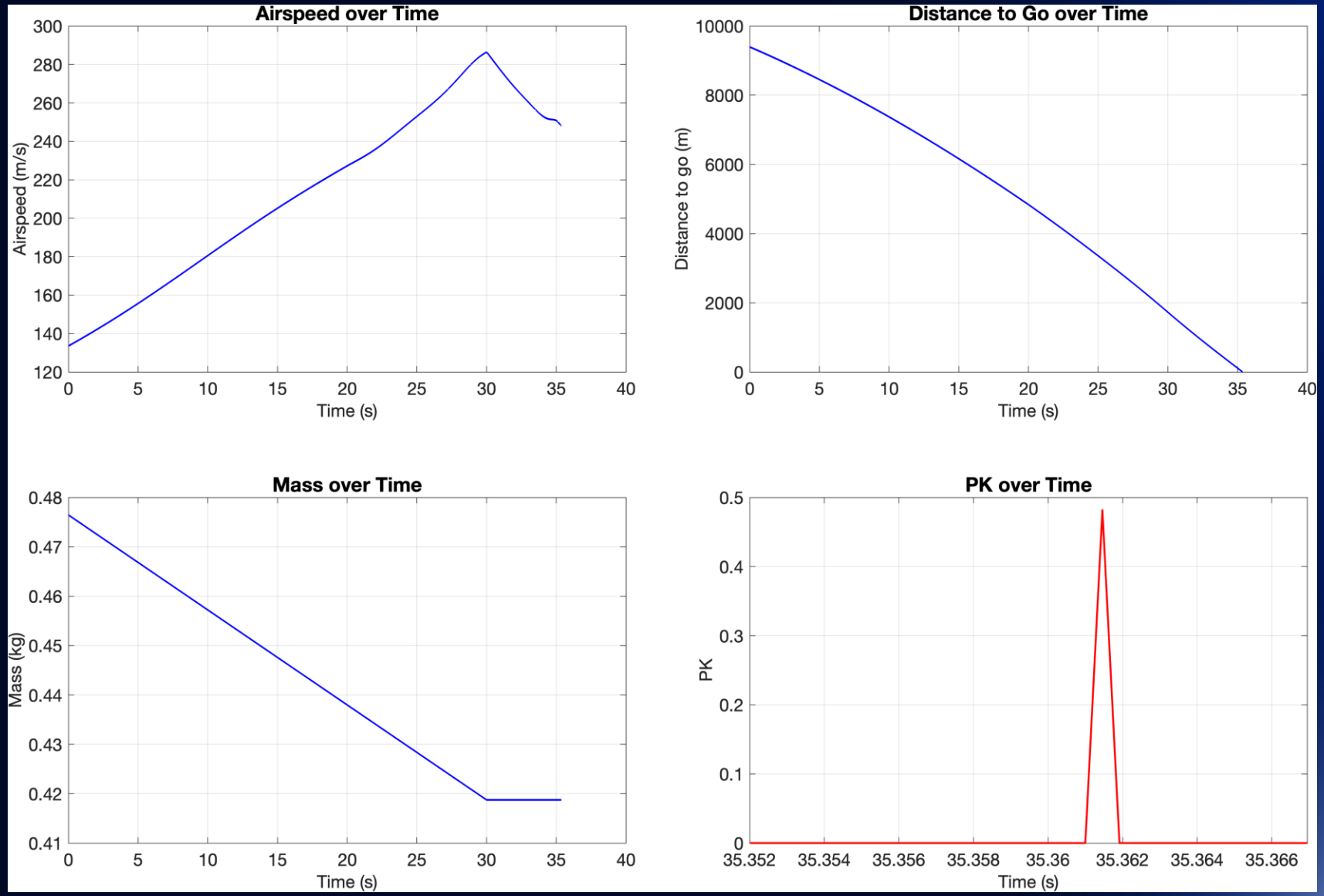
Simulation Results

Target conditions:

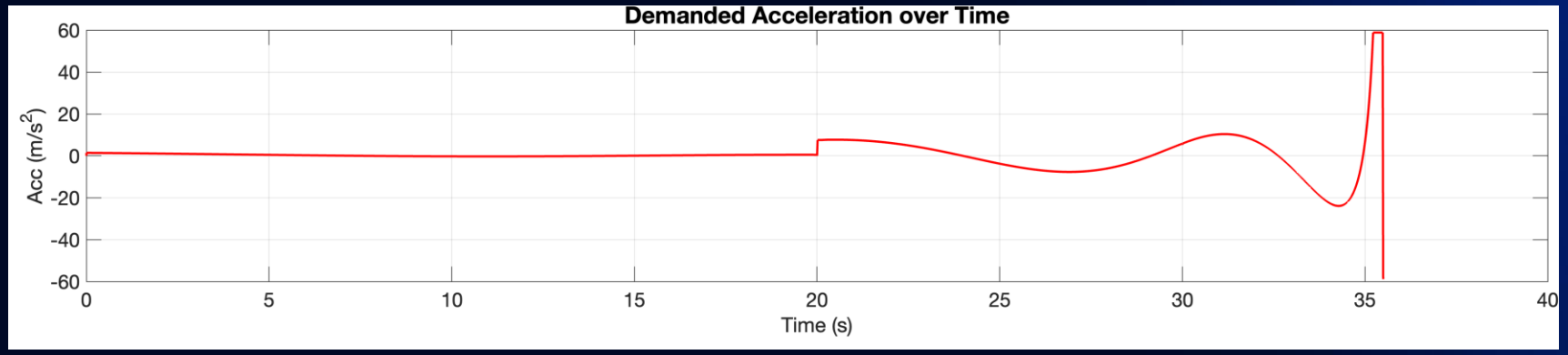
X(0) = 8000 m
Y(0) = 5000 m
v = 60 $^{m}/_{s}$
45 degrees target dive at t = 20s

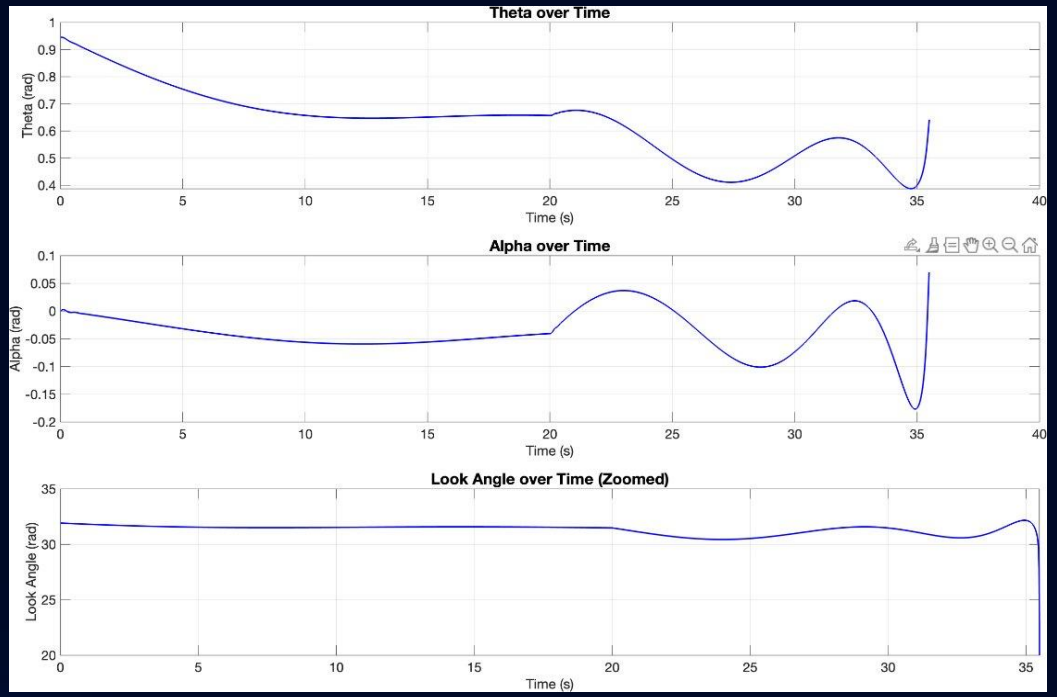


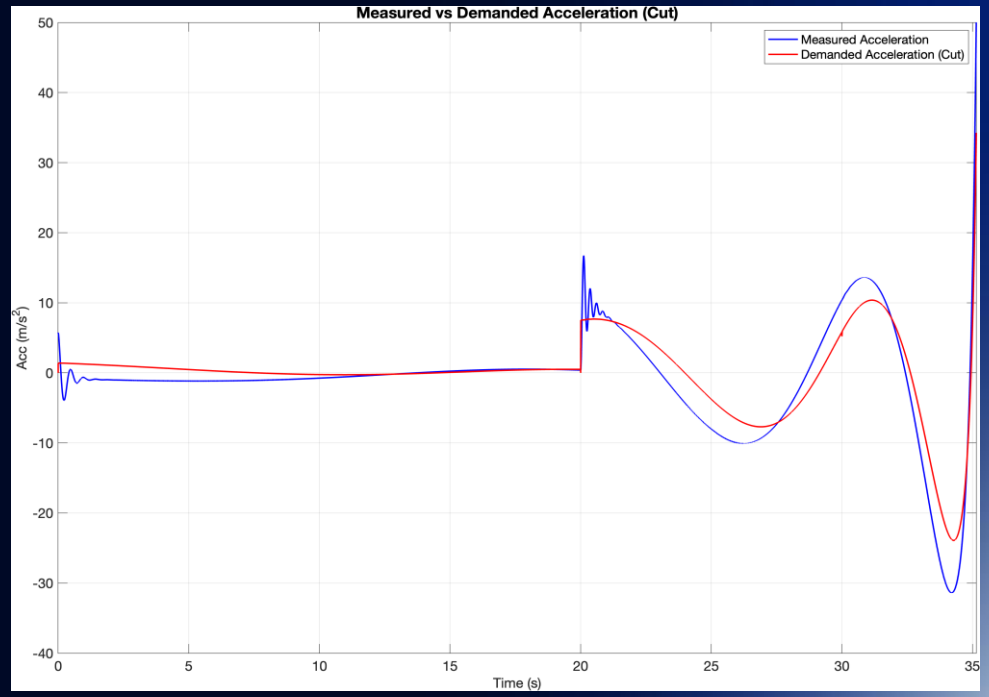
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







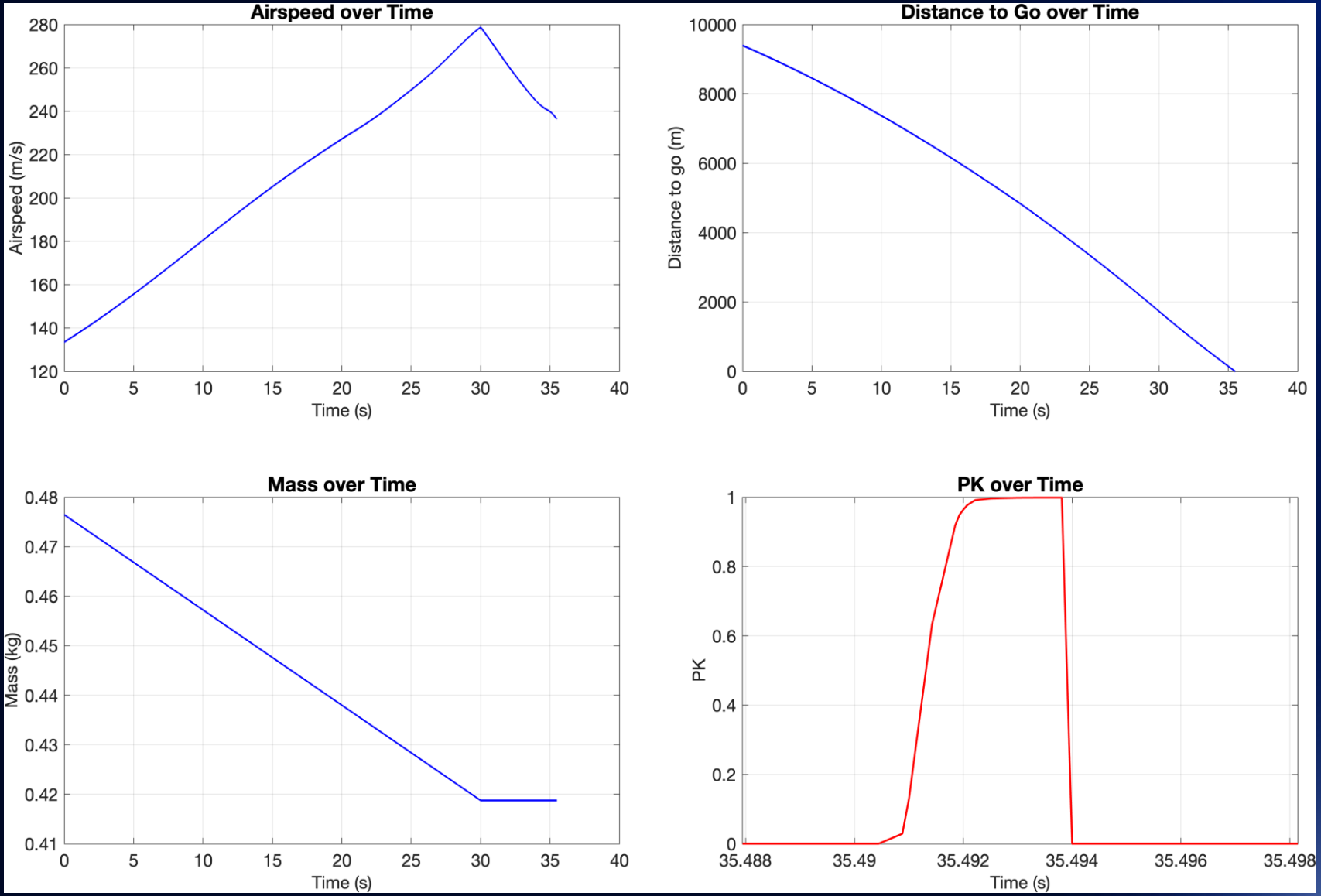
Simulation Results



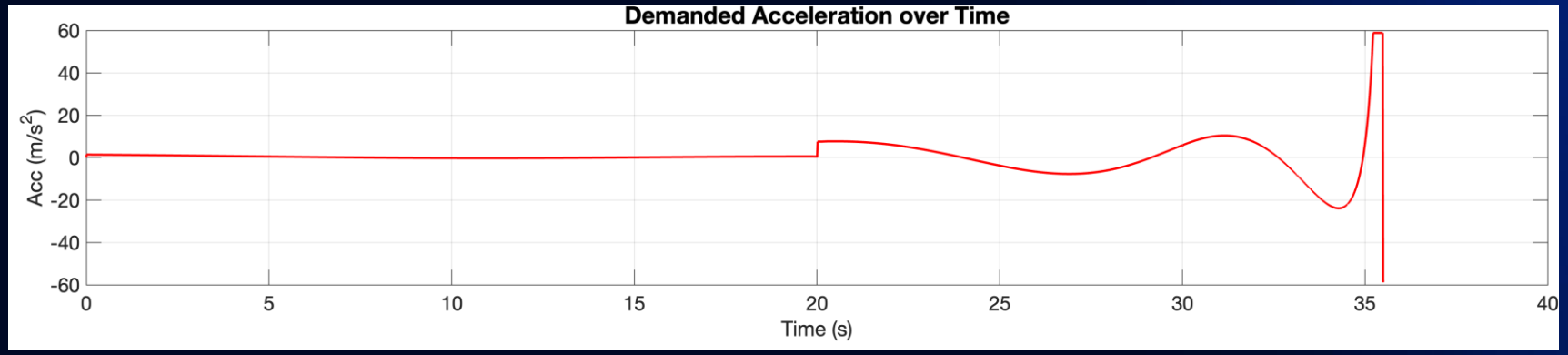
Target conditions:

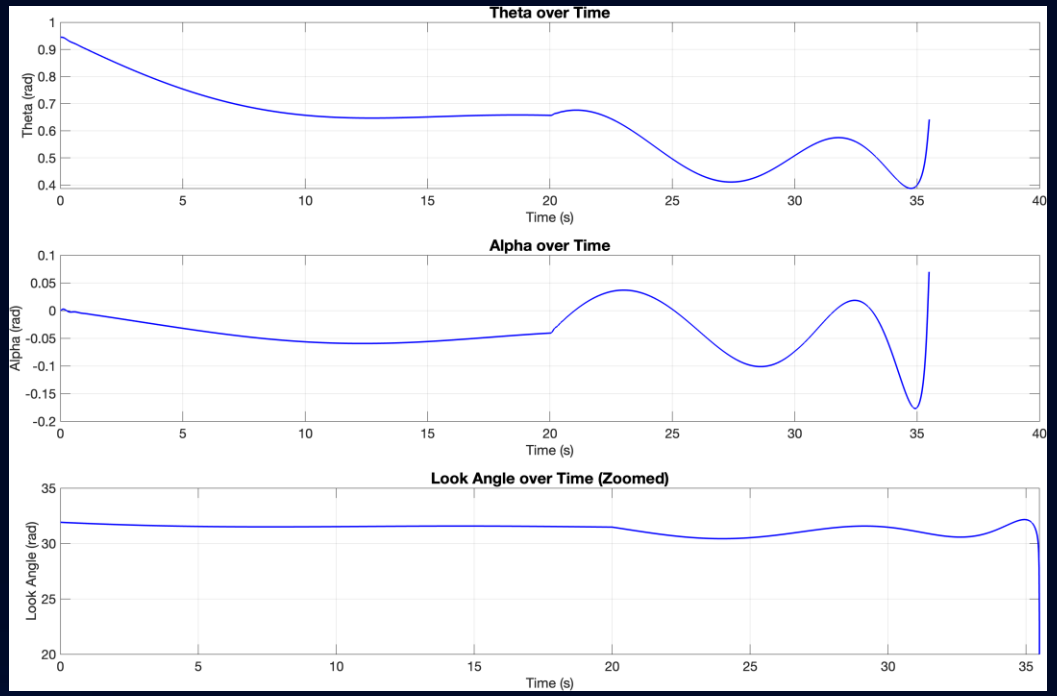
X(0) = 8000 m
Y(0) = 5000 m
v = 60 $^{m}/_{s}$
30 degrees target dive at t = 20s

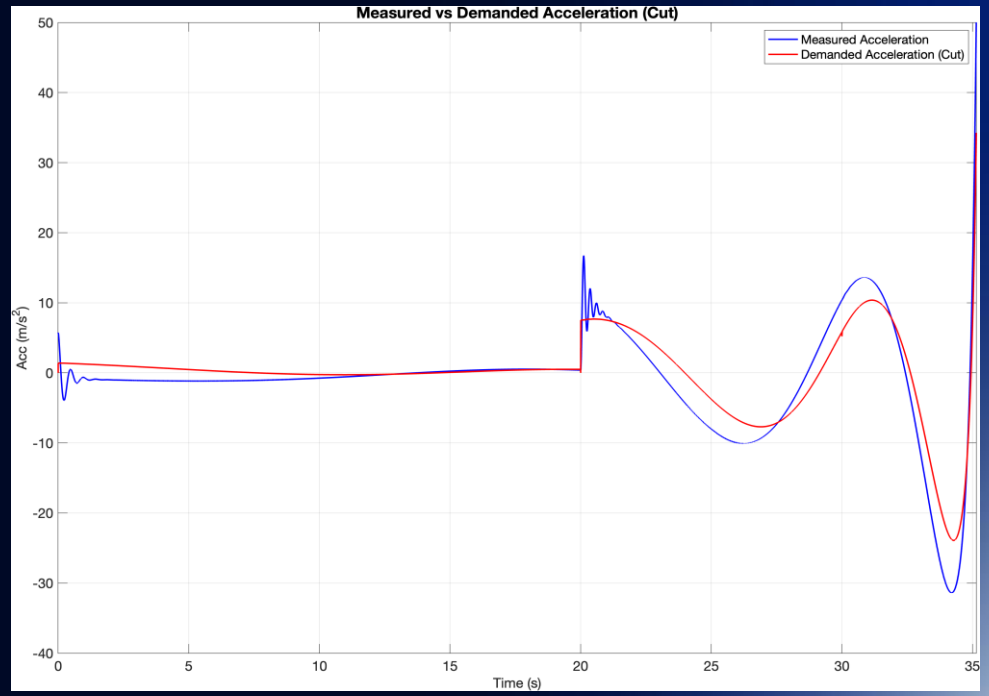
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







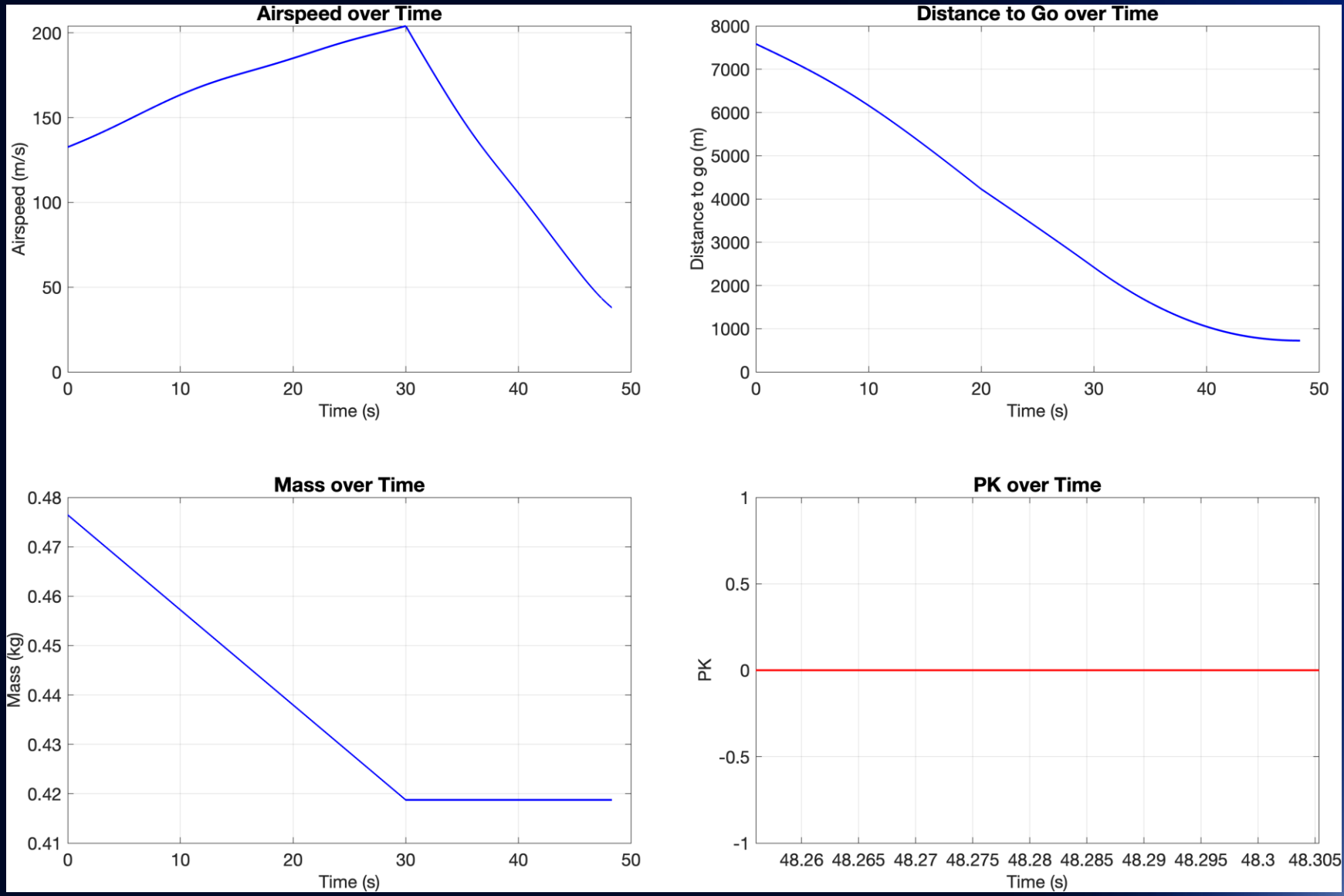
Simulation Results

Target conditions:

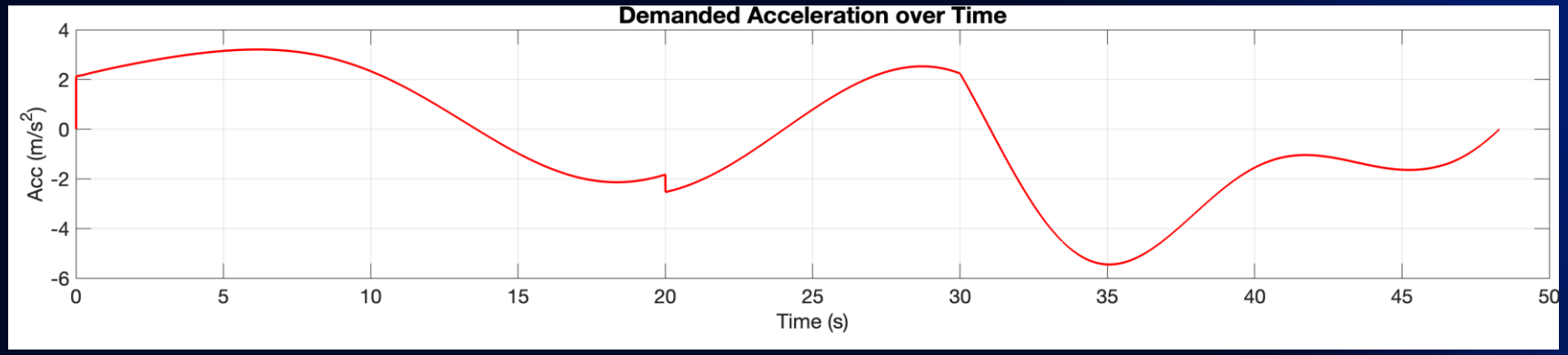
X(0) = 3000 m
Y(0) = 7000 m
v = 60 $^{m}/_{s}$
30 degrees target surge at t = 20s

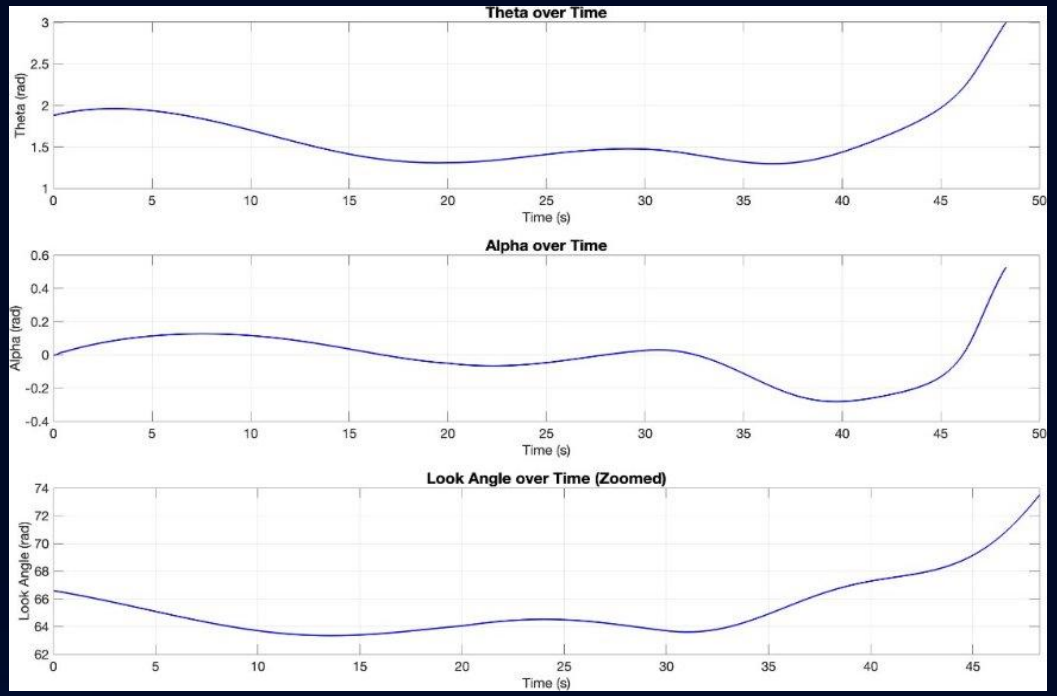


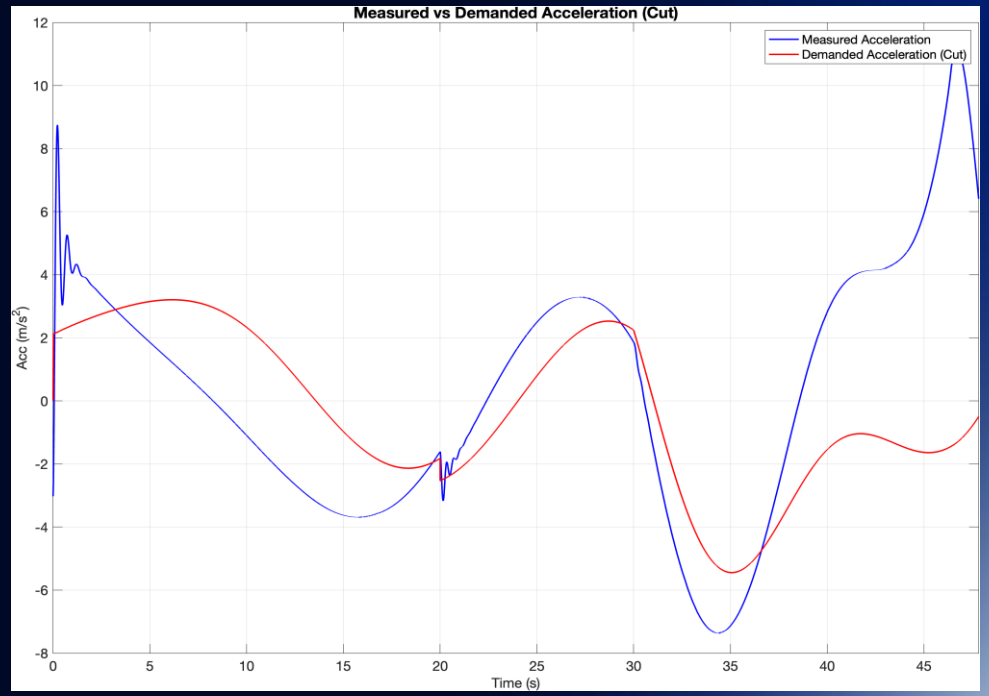
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







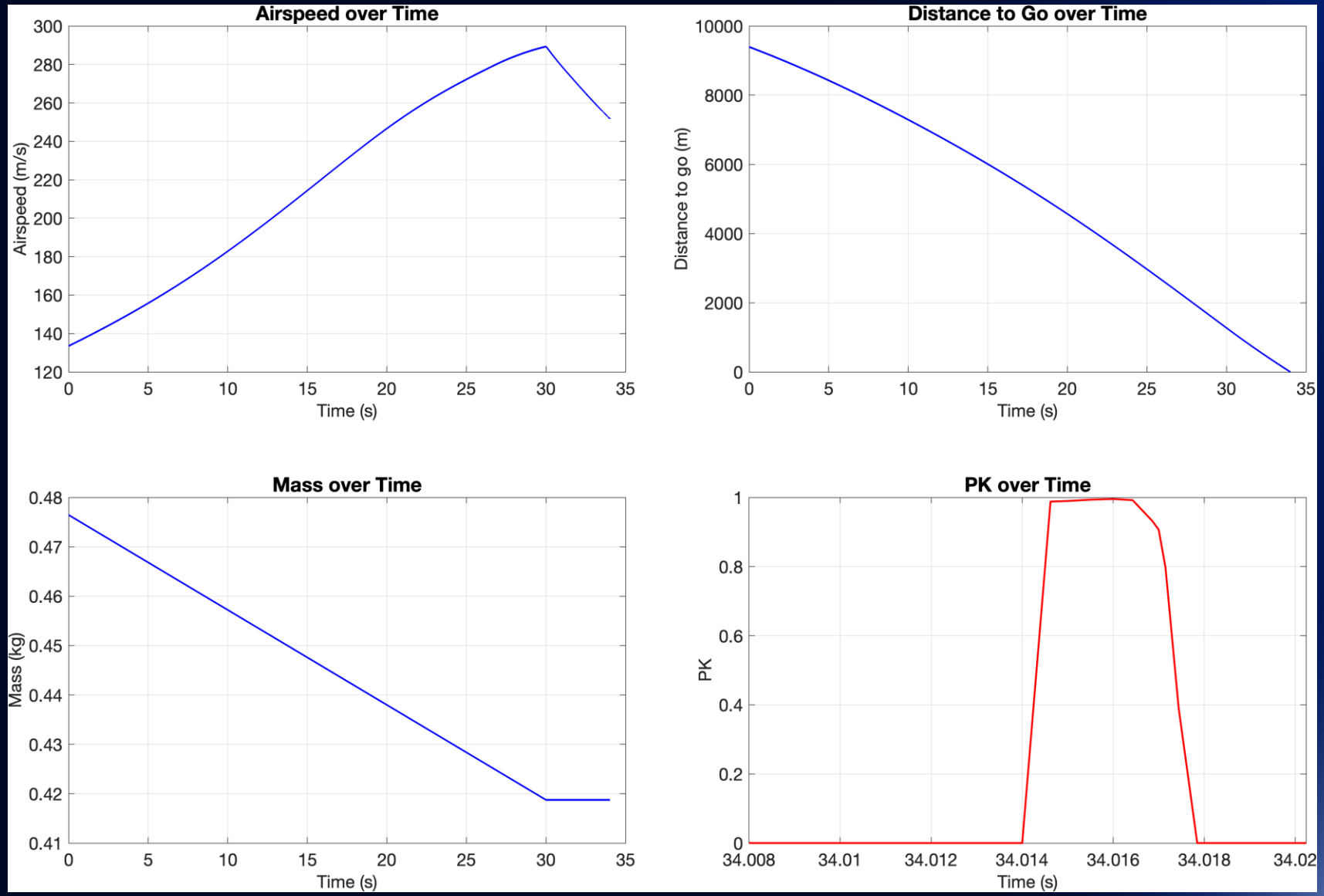
Simulation Results

Target conditions:

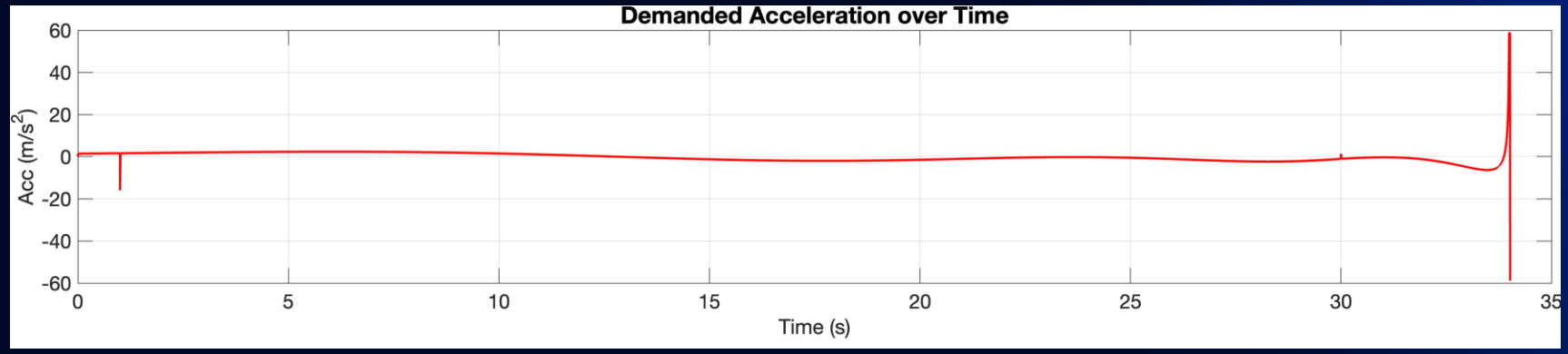
X(0) = 8000 m
Y(0) = 5000 m
v = 60 $^{m}/_{s}$
Sinusoidal target trajectory
- $f = 0.1\ s^{-1}$
- $A = 0.8\ rad$

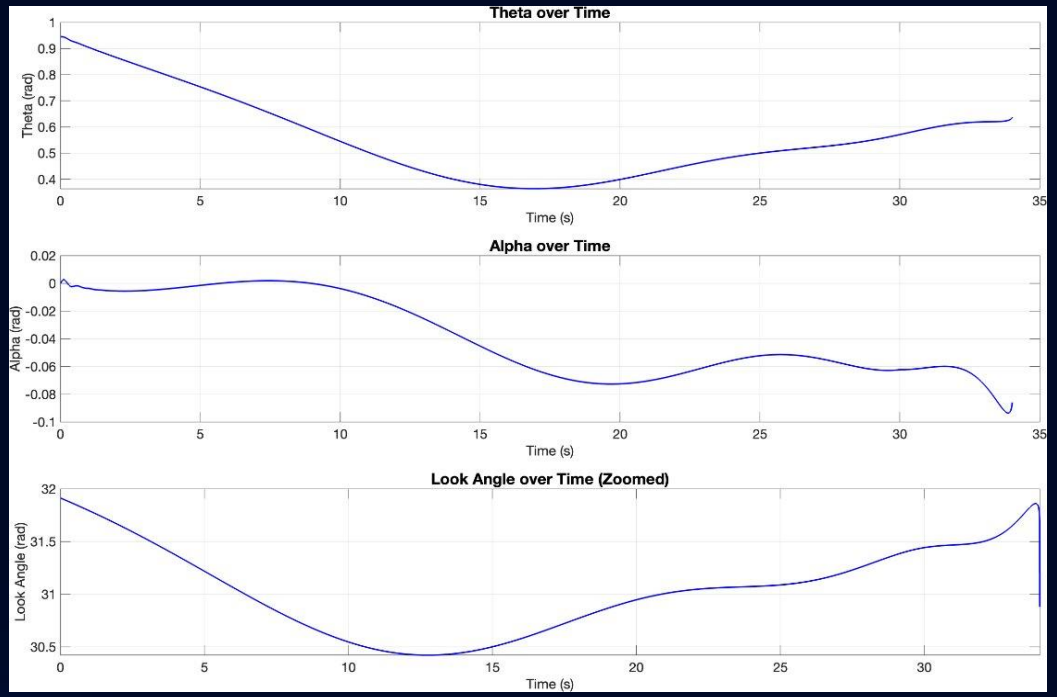


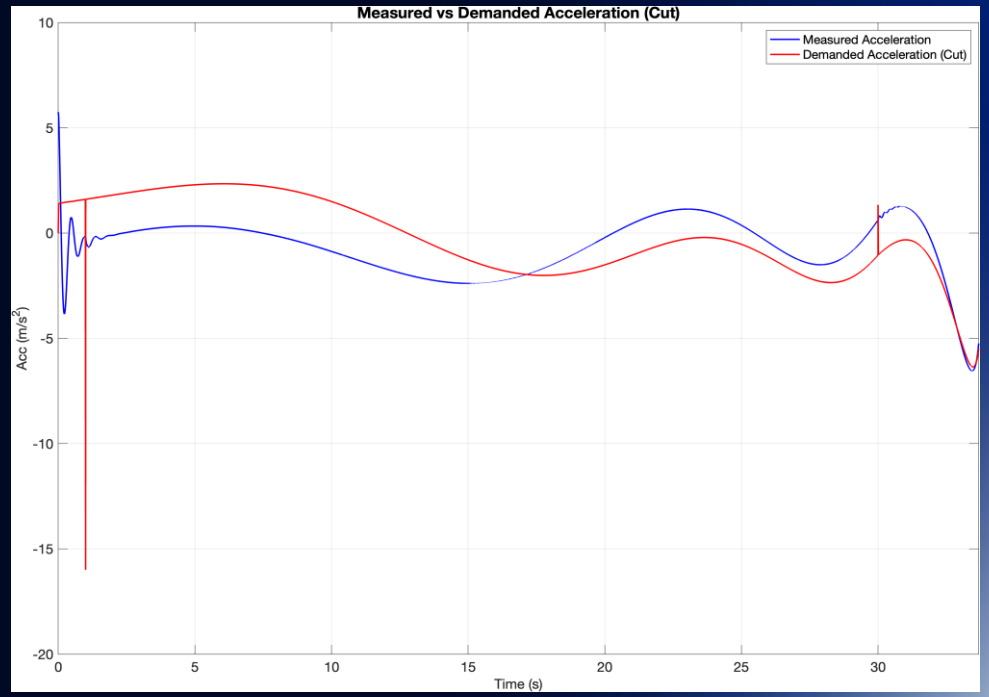
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







Simulation Results

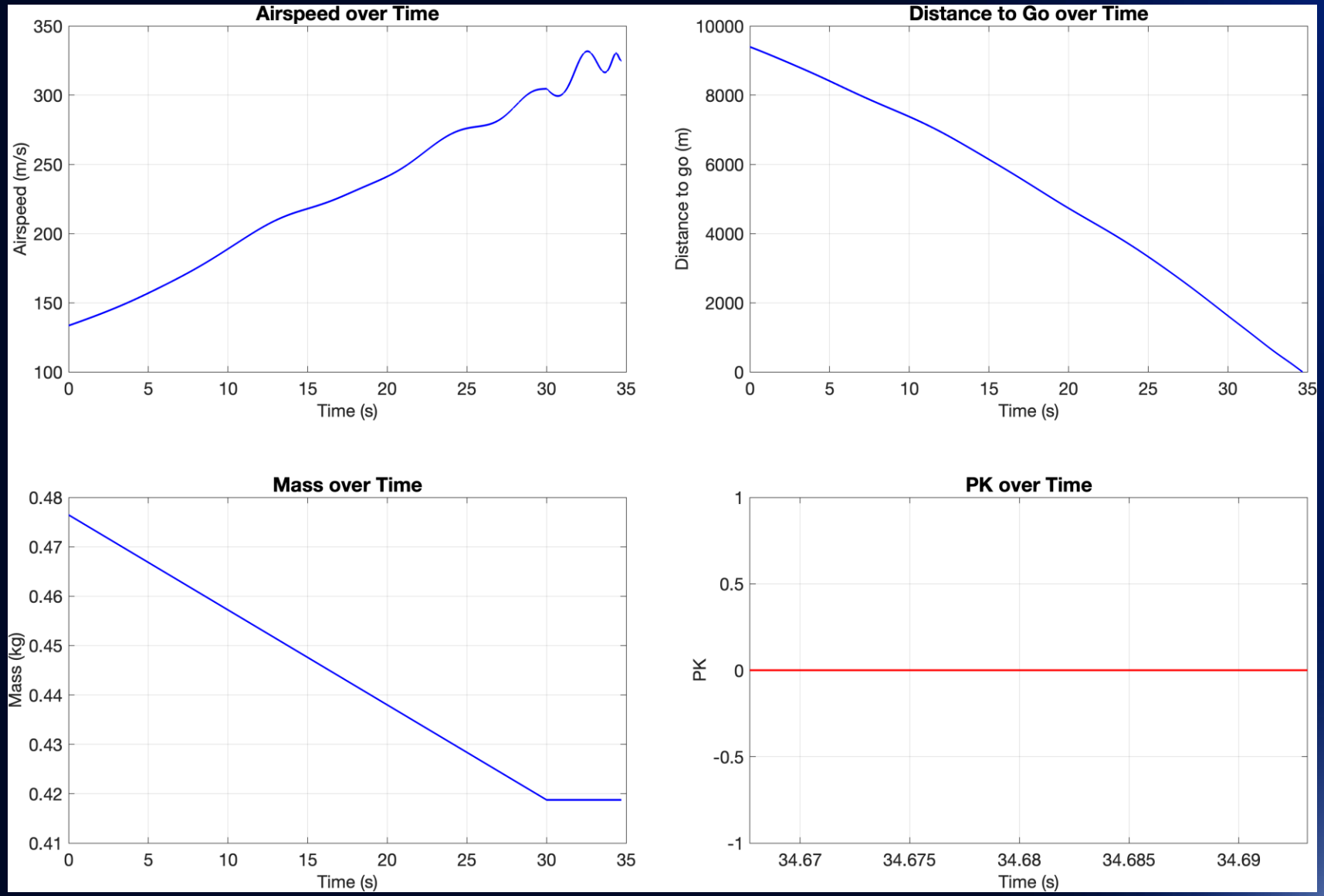


Target conditions:

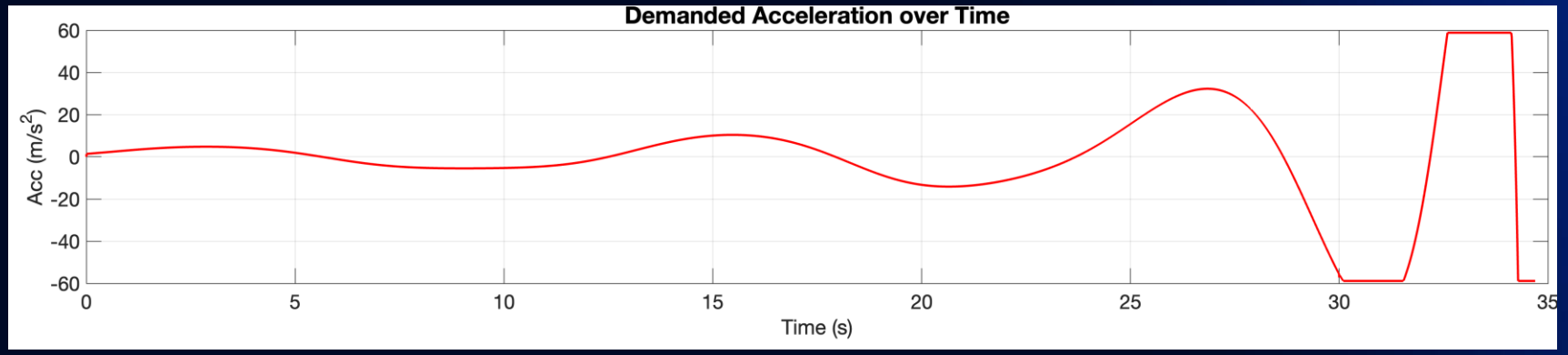
X(0) = 8000 m
Y(0) = 5000 m
v = 60 $^{m}/_{s}$
Sinusoidal target trajectory

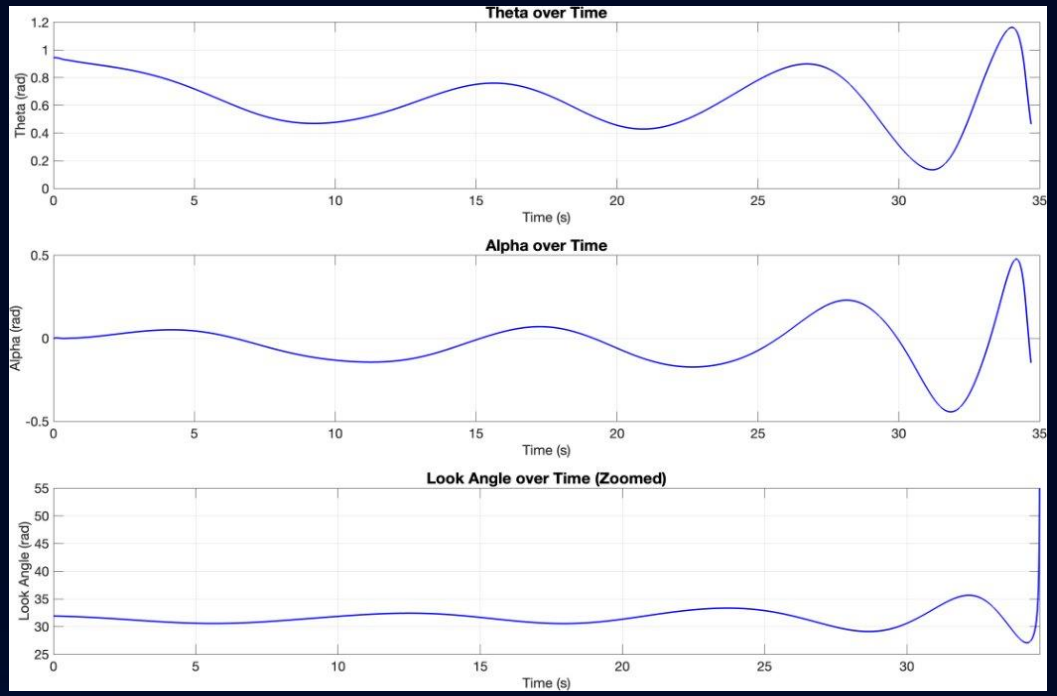
- $f = 0.5\ s^{-1}$
- $A = 0.8\ rad$

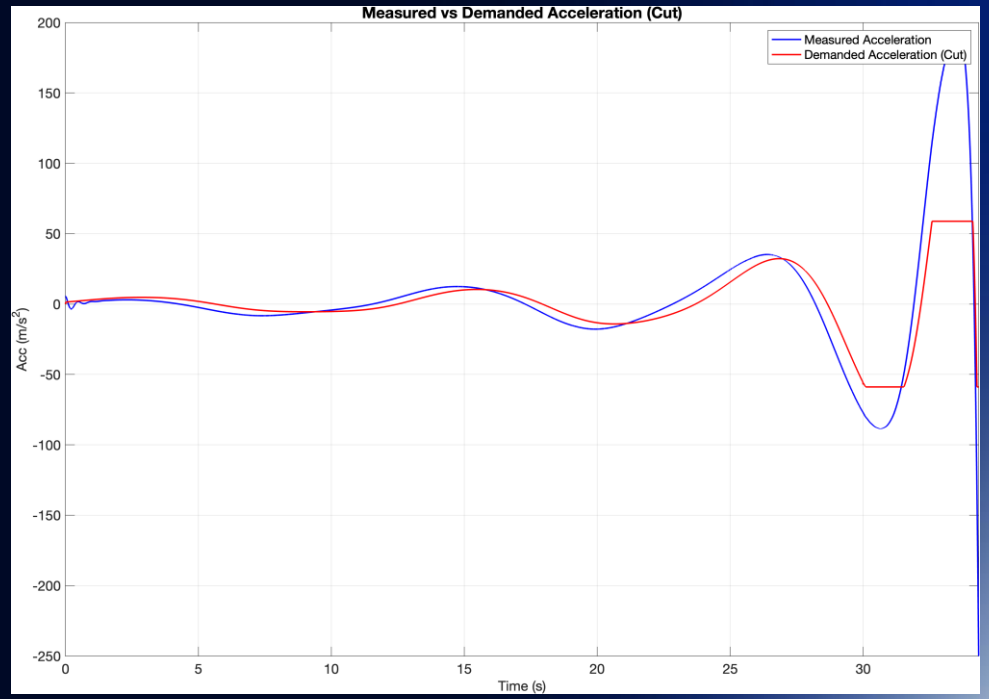
OFF-DESIGN SCENARIOS



OFF-DESIGN SCENARIOS







Thanks for the attention.

Any question?